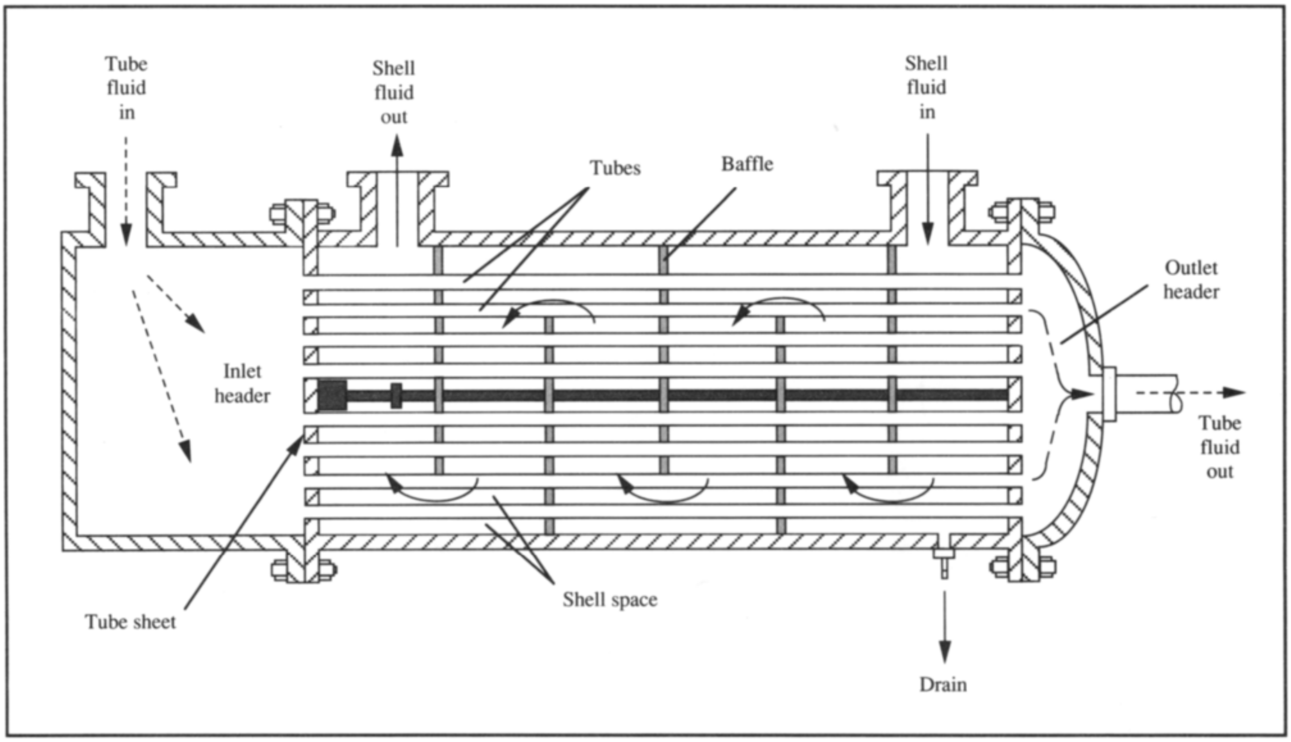
Consider the device of Figure 8.6 for exchange of sensible heat from one fluid to another. The heat-transfer system is divided into two sections: a *tube bundle* containing pipes through which one fluid flows, and a *shell* or cavity where the other fluid flows. Hot or cold fluid may be put into either the tubes or the shell. In a single-pass exchanger, the shell and tube fluids pass down the length of the equipment only once. The fluid which is to travel in the tubes enters at the inlet header. The header is divided from the rest of the apparatus by a *tube sheet*. Open tubes are fitted into the tube sheet; fluid in the header cannot enter the main cavity of the exchanger but must pass into the tubes. The tube-side fluid leaves the exchanger through another header at the outlet. Shell-side fluid enters the internal cavity of the exchanger and flows around the outsides of the tubes in a direction which is largely countercurrent to the tube fluid. Heat is exchanged across the tube walls from hot fluid to cold fluid. As shown in Figure 8.6, *baffles* are often installed in the shell to decrease the cross-sectional area for flow and divert the shell fluid so it flows mainly across rather than parallel to the tubes. Both these effects promote turbulence in the shell fluid which improves the rate of heat transfer. As well as directing the flow of shell fluid, baffles also support the tube bundle and keep the tubes from sagging.

The length of tubes in a single-pass heat exchanger determines the surface area available for heat transfer, and therefore the rate at which heat can be exchanged. However, there are practical and economic limits to the maximum length of single-pass tubes; if greater heat-transfer capacity is required *multiple-pass heat exchangers* are employed. Heat exchangers containing more that one tube pass are used routinely; Figure 8.7 shows the structure of a heat exchanger with one shell pass and a double tube pass. In this device, the header for the tube fluid is divided into two sections. Fluid entering the header is channelled into the lower half of the tubes and flows to the other end of the exchanger. The header at the other end diverts the fluid into the upper tubes; the tube-side fluid therefore leaves the exchanger at the same end it entered. On the shell side, the configuration is the same as for the single-pass structure of Figure 8.6; fluid enters one end of the shell and flows around several baffles to the other end.

In a double-tube pass exchanger, flow of tube and shell fluids is mainly countercurrent for one tube pass and mainly cocurrent for the other; however, because of the action of the baffles, cross-flow of shell-side fluid normal to the tubes also

Figure 8.7 Double tube-pass heat exchanger.



occurs. Temperature curves for the exchanger depend on the location of the shell-side entry nozzle; this is illustrated in Figure 8.8 for hot fluid flowing in the shell. In the temperature profile of Figure 8.8(b), a *temperature cross* occurs where, at some point in the exchanger, the temperature of the cold fluid equals the temperature of the hot fluid. This situation should be avoided because, after the cross, the cold fluid is actually cooled rather than heated. The solution is an increased number of shell passes or, more practically, provision of another heat exchanger in series with the first.

Heat exchangers with multiple shell-passes can also be used. However, in comparison with multiple-tube pass equipment, multiple-shell exchangers are complex in construction and are normally applied only in very large installations.

8.2 Mechanisms of Heat Transfer

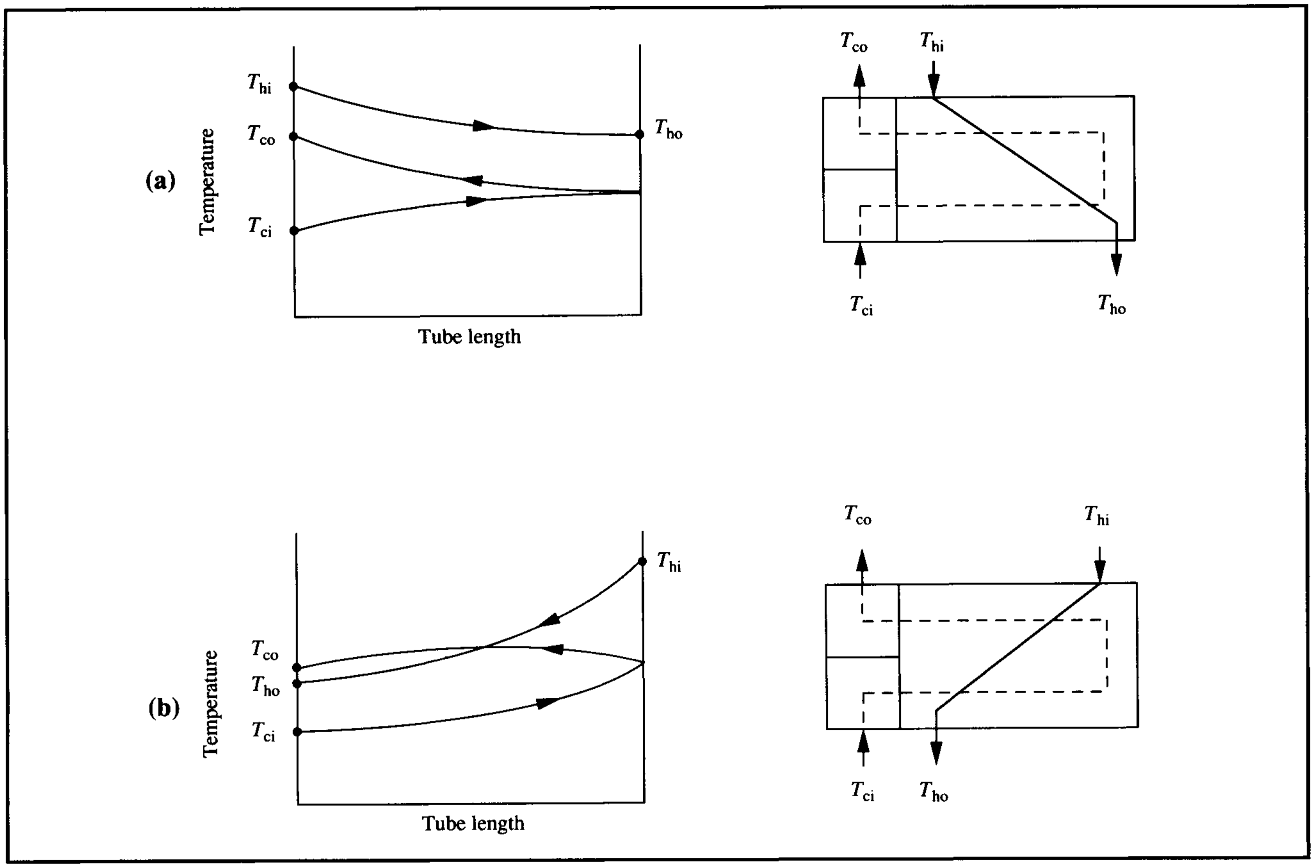
Heat transfer occurs by one or more of the following three mechanisms.

(i) *Conduction.* Heat conduction occurs by transfer of vibrational energy between molecules, or movement of free electrons. Conduction is particularly important with metals and occurs without observable movement of matter.

(ii) *Convection.* Convection requires movement on a macroscopic scale; it is therefore confined to gases and liquids. *Natural convection* occurs when temperature gradients in the system generate localised density differences which result in flow currents. In *forced convection,* flow currents are set in motion by an external agent such as a stirrer or pump and are independent of density gradients. Higher rates of heat transfer are possible with forced convection compared with natural convection.

(iii) *Radiation.* Energy is radiated from all materials in the form of waves; when this radiation is absorbed by matter it appears as heat. Because radiation is important at much higher temperatures than those normally encountered in biological processing, it will not be mentioned further.

Figure 8.8 Temperature changes for a double tube-pass heat exchanger. (From J.M. Coulson and J.F. Richardson, 1977, *Chemical Engineering,* 3rd edn, Pergamon Press, Oxford.)



8.3 Conduction

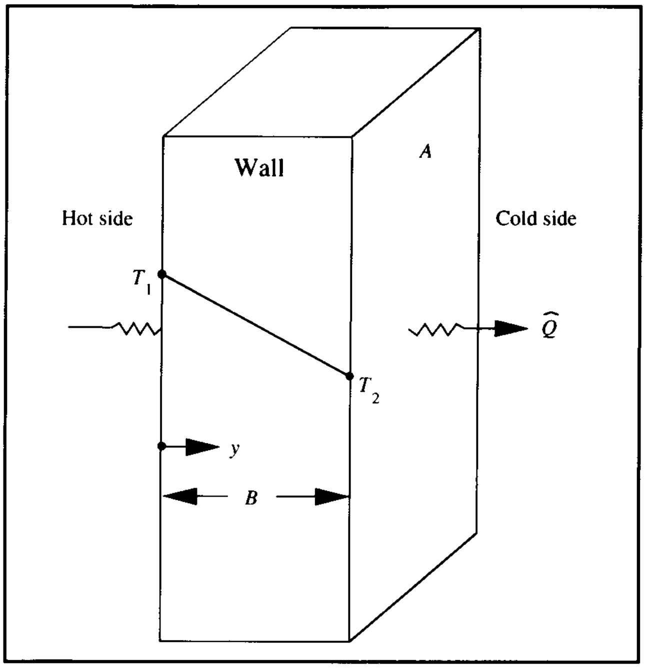
In most heat-transfer equipment, heat is exchanged between fluids separated by a solid wall. Heat transfer through the wall occurs by conduction. In this section we consider equations describing rate of conduction as a function of operating variables.

Conduction of heat through a homogeneous solid wall is depicted in Figure 8.9. The wall has thickness B; on one side of the wall the temperature is T_1, on the other side the temperature is T_2. The area of wall exposed to each temperature is A. The rate of heat conduction through the wall is given by *Fourier's law*:

$$\hat{Q} = -kA\frac{\mathrm{d}T}{\mathrm{d}y} \tag{8.1}$$

where $\hat{Q}$ is rate of heat transfer, k is the *thermal conductivity* of the wall, A is surface area perpendicular to the direction of heat flow, T is temperature, and y is distance measured normal to A. $\mathrm{d}T/\mathrm{d}y$ is the *temperature gradient*, or change of temperature with distance through the wall. The negative sign in Eq. (8.1) indicates that heat always flows from hot to cold irrespective of whether $\mathrm{d}T/\mathrm{d}y$ is positive or negative. To illustrate this, when

Figure 8.9 Heat conduction through a flat wall.



the temperature gradient is negative relative to co-ordinate y (as shown in Figure 8.9), heat flows in the positive y-direction; conversely, if the gradient were positive (i.e. $T_1 < T_2$), heat would flow in the negative y-direction.

Fourier's law can also be expressed in terms of ***heat flux***, $\hat{q}$. Heat flux is defined as the rate of heat transfer per unit area normal to the direction of heat flow. Therefore, from Eq. (8.1):

$$\hat{q} = -k\frac{\mathrm{d}T}{\mathrm{d}y}. \tag{8.2}$$

Rate of heat transfer $\hat{Q}$ is also known as *power*. The SI unit for $\hat{Q}$ is the watt (W); in imperial units $\hat{Q}$ is measured in Btu h^{-1}. Corresponding units of $\hat{q}$ are W m^{-2} and Btu h^{-1} ft^{-2}.

Thermal conductivity is a transport property of materials; values can be found in handbooks. The dimensions of k are $\mathrm{LMT}^{-3}\Theta^{-1}$; units include W m^{-1} K^{-1} and Btu h^{-1} ft^{-1} $°\mathrm{F}^{-1}$. The magnitude of k in Eqs (8.1) and (8.2) reflects the ease with which heat is conducted; the higher the value of k the faster is the heat transfer. Table 8.1 lists thermal conductivities for some common materials; metals generally have higher thermal conductivities than other substances. Solids with low k values are used as *insulators* to minimise the rate of conduction, for example, around steam pipes or in buildings. Thermal conductivity varies somewhat with temperature; however for small ranges of temperature, k can be considered constant.

8.3.1 Analogy Between Heat and Momentum Transfer

An analogy exists between the equations for heat and momentum transfer. Newton's law of viscosity given by Eq. (7.6):

$$\tau = -\mu\frac{\mathrm{d}v}{\mathrm{d}y} \tag{7.6}$$

has the same mathematical form as Eq. (8.2). In heat transfer, the temperature gradient $\mathrm{d}T/\mathrm{d}y$ is the driving force for heat flow; in momentum transfer the driving force is the velocity gradient $\mathrm{d}v/\mathrm{d}y$. Both heat flux and momentum flux are directly proportional to the driving force, and the proportionality constant, μ or k, is a physical property of the material. As we shall see in Chapter 9, the analogy between heat and momentum transfer can be extended to include mass transfer.

Table 8.1 Thermal conductivities

(From J.M. Coulson and J.F. Richardson, 1977, Chemical Engineering, *vol. 1, 3rd edn, Pergamon Press, Oxford)*

Material	*Temperature* (K)	k ($W\,m^{-1}\,K^{-1}$)	k ($Btu\,h^{-1}\,ft^{-1}\,{}^{o}F^{-1}$)
Solids: Metals			
Aluminium	573	230	133
Bronze	–	189	109
Copper	373	377	218
Iron (wrought)	291	61	35
Iron (cast)	326	48	27.6
Lead	373	33	19
Stainless steel	293	16	9.2
Steel (1% C)	291	45	26
Solids: Non-metals			
Asbestos	273	0.16	0.09
Bricks (building)	293	0.69	0.4
Cotton wool	303	0.050	0.029
Glass	303	1.09	0.63
Rubber (hard)	273	0.15	0.087
Cork	303	0.043	0.025
Glass wool	–	0.041	0.024
Liquids			
Acetic acid (50%)	293	0.35	0.20
Ethanol (80%)	293	0.24	0.137
Glycerol (40%)	293	0.45	0.26
Water	303	0.62	0.356
Water	333	0.66	0.381
Gases			
Air	273	0.024	0.014
Air	373	0.031	0.018
Carbon dioxide	273	0.015	0.0085
Oxygen	273	0.024	0.0141
Water vapour	373	0.025	0.0145

8.3.2 Steady-State Conduction

Consider again the conduction of heat through the wall shown in Figure 8.9. At steady state there can be neither accumulation nor depletion of heat within the wall; this means that the rate of heat flow $\hat{Q}$ must be the same at each point in the wall. If k is largely independent of temperature and A is also constant, the only variables in Eq. (8.1) are temperature T and distance y. We can therefore integrate Eq. (8.1) to obtain an expression for rate of conduction as a function of the temperature difference across the wall.

Separating variables in Eq. (8.1) gives:

$$\hat{Q}\,\mathrm{d}y = -kA\,\mathrm{d}T. \tag{8.3}$$

Both sides of Eq. (8.3) can be integrated after taking the constants $\hat{Q}$, k and A outside of the integral signs:

$$\hat{Q}\int dy = -kA\int dT. \tag{8.4}$$

From the rules of integration given in Appendix D:

$$\hat{Q}\,y = -k\,A\,T + K \tag{8.5}$$

where K is the integration constant. K is evaluated by applying a single boundary condition; in this case we can use the boundary condition: $T = T_1$ at $y = 0$. Substituting this information into Eq. (8.5) gives:

$$K = k\,A\,T_1. \tag{8.6}$$

Thus eliminating K from Eq. (8.5) gives:

$$\hat{Q}\,y = -kA(T_1 - T) \tag{8.7}$$

or

$$\hat{Q} = \frac{kA}{y}(T_1 - T). \tag{8.8}$$

Because $\hat{Q}$ at steady state is the same at all points in the wall, Eq. (8.8) holds for all values of y including at $y = B$ where $T = T_2$. Substituting these values into Eq. (8.8) gives the expression:

$$\hat{Q} = \frac{kA}{B}(T_1 - T_2) \tag{8.9}$$

or

$$\hat{Q} = \frac{kA}{B}\Delta T. \tag{8.10}$$

Eq. (8.10) allows us to calculate $\hat{Q}$ if we know the heat-transfer area A and the total temperature drop across the slab ΔT. Eq. (8.10) can also be written in the form:

$$\hat{Q} = \frac{\Delta T}{R_w} \tag{8.11}$$

where R_w is the *thermal resistance* to heat transfer offered by the wall:

$$R_w = \frac{B}{kA}. \tag{8.12}$$

In heat transfer, the ΔT responsible for flow of heat is known as the *temperature-difference driving force.* Eq. (8.11) is an example of the general rate principle which equates rate of transfer to the ratio of driving force and resistance. Eq. (8.12) can be interpreted as follows: the wall would pose more of a resistance to heat transfer if its thickness were increased; on the other hand, resistance is reduced if the surface area is increased, or the material in the wall is replaced with a substance of higher thermal conductivity.

8.3.3 Combining Thermal Resistances in Series

When a system contains several different heat-transfer resistances in series, ***the overall resistance is equal to the sum of the individual resistances.*** For example, if the wall shown in Figure 8.9 were constructed of several layers of different material, each layer would represent a separate resistance to heat transfer. Consider the three-layer system illustrated in Figure 8.10 with surface area A, layer thicknesses B_1, B_2 and B_3, thermal conductivities k_1, k_2 and k_3, and temperature drops across the layers ΔT_1, ΔT_2 and ΔT_3. If the layers are in perfect thermal contact so that there is no temperature drop across the interfaces, the temperature change across the entire structure is:

$$\Delta T = \Delta T_1 + \Delta T_2 + \Delta T_3. \tag{8.13}$$

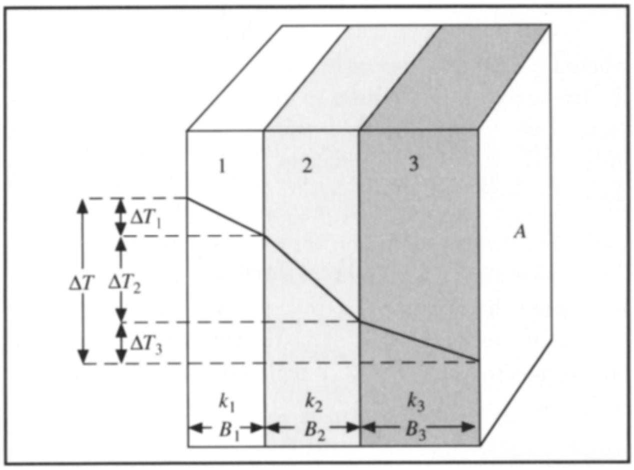
Rate of heat conduction in this system is given by Eq. (8.11), with the overall resistance R_w equal to the sum of the individual resistances:

$$\hat{Q} = \frac{\Delta T}{R_w} = \frac{\Delta T}{(R_1 + R_2 + R_3)} \tag{8.14}$$

where R_1, R_2 and R_3 are the thermal resistances of the individual layers:

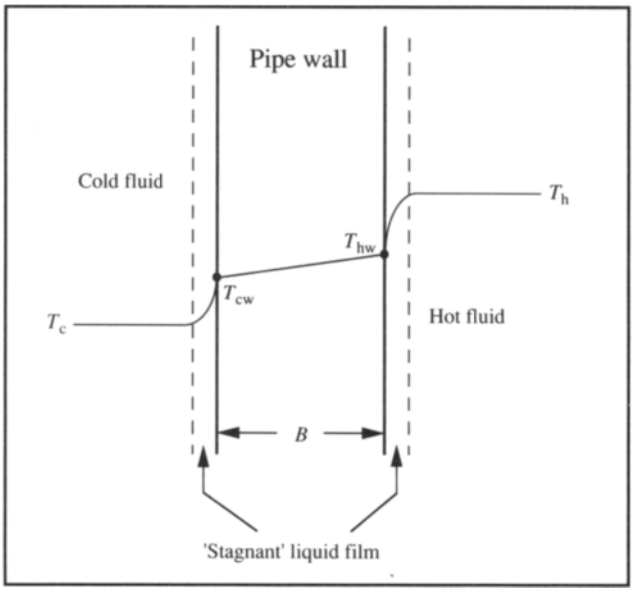
$$R_1 = \frac{B_1}{k_1 A} \qquad R_2 = \frac{B_2}{k_2 A} \qquad \text{and} \qquad R_3 = \frac{B_3}{k_3 A}. \tag{8.15}$$

Figure 8.10 Heat conduction through three resistances in series.



Eq. (8.14) represents the important principle of *additivity of resistances*. We shall use this principle later for analysis of convective heat transfer in pipes and stirred vessels.

Figure 8.11 Heat transfer between fluids separated by a solid wall.



8.4 Heat Transfer Between Fluids

Convection and conduction both play important roles in heat transfer in fluids. In agitated, single-phase systems, convective heat transfer in the bulk fluid is linked directly to mixing and turbulence and is generally quite rapid. However, in the heat-exchange devices described in Section 8.1, additional resistances are encountered.

8.4.1 Thermal Boundary Layers

Figure 8.11 depicts the heat-transfer situation at any point on the pipe wall of a heat exchanger. Hot and cold fluids flow on either side of the wall; we assume that both fluids are in turbulent flow. The bulk temperature of the hot fluid away from the wall is T_h; T_c is the bulk temperature of the cold fluid. T_{hw} and T_{cw} are the respective temperatures of hot and cold fluids at the wall.

As explained in Section 7.2.3, when fluid contacts a solid, a fluid boundary layer develops at the surface as a result of viscous drag. Therefore, fluids such as those represented in Figure 8.11 consist of a turbulent core which accounts for the bulk of the fluid, and a thin sublayer or film near the wall where the velocity is relatively low. In the turbulent part of the fluid, rapidly moving eddies transfer heat quickly so that any temperature gradients in the bulk fluid can be neglected. The film of liquid at the wall is called the *thermal boundary layer* or *stagnant film*, although the fluid in it is not actually stationary. This viscous sublayer has an important effect on the rate of heat transfer. Most of the resistance to heat transfer to or from the fluid is contained in the film; the reason for this is that heat flow through it must occur mainly by conduction rather than convection because of the reduced velocity of the fluid. The width of the film indicated by the broken lines in Figure 8.11 is the approximate distance from the wall where the temperature reaches the bulk-fluid temperature, either T_h or T_c. The thickness of the thermal boundary layer in most heat-transfer situations is less than the hydrodynamic boundary layer described in Section 7.2.3. In other words, as we move away from the wall, the temperature normally reaches that of the bulk fluid before the velocity reaches that of the bulk flow stream.

8.4.2 Individual Heat-Transfer Coefficients

Heat exchanged between the fluids in Figure 8.11 encounters three major resistances in series: the hot-fluid film resistance at the wall, resistance due to the wall itself, and the cold-fluid film resistance. Equations for rate of conduction through the wall have already been developed in Section 8.3.2. Rate of heat transfer through each thermal boundary layer is given by an equation somewhat analogous to Eq. (8.10) for steady-state conduction:

$$\hat{Q} = hA\Delta T \tag{8.16}$$

where h is the *individual heat-transfer coefficient*, A is the area for heat transfer normal to the direction of heat flow, and ΔT is the temperature difference between the wall and the bulk stream. $\Delta T = T_h - T_{hw}$ for the hot-fluid film; $\Delta T = T_{cw} - T_c$ for the cold-fluid film. Eq. (8.16) does not contain a separate term for the thickness of the boundary layer; this thickness is difficult to measure and depends strongly on the prevailing flow conditions. Instead, the effect of film thickness is included in the value of h so that, unlike thermal conductivity, values for h cannot be found in handbooks. The heat-transfer coefficient h is an empirical parameter incorporating the effects of system geometry, flow conditions and fluid properties. Because it involves fluid flow, convective heat transfer is a more complex process than conduction. Consequently, there is little theoretical basis for calculation of h; h must be determined experimentally or evaluated using published correlations based on experimental data. Suitable correlations for heat-transfer coefficients are presented in Section 8.5.3. SI units for h are W m^{-2} K^{-1}; in the imperial system h is expressed as Btu h^{-1} ft^{-2} $°F^{-1}$. Magnitudes of h vary greatly; some typical values are listed in Table 8.2.

Rate of heat transfer $\hat{Q}$ in each fluid boundary layer can be written as the ratio of the driving force ΔT and the resistance. Therefore, from Eq. (8.16) the two resistances on either side of the pipe wall are as follows:

$$R_h = \frac{1}{h_h A} \tag{8.17}$$

and

$$R_c = \frac{1}{h_c A} \tag{8.18}$$

where R_h is the resistance to heat transfer in the hot fluid, R_c is the resistance to heat transfer in the cold fluid, h_h is the individual heat-transfer coefficient for the hot fluid, h_c is the individual heat-transfer coefficient for the cold fluid and A is the surface area for heat transfer.

8.4.3 Overall Heat-Transfer Coefficient

Use of Eq. (8.16) to calculate rate of heat transfer in each boundary layer requires knowledge of ΔT for each fluid; this is usually difficult because of lack of information about T_{hw} and T_{cw}. It is easier and more accurate to measure the bulk temperatures of fluids rather than wall temperatures. This problem is removed by introduction of the *overall heat-transfer coefficient*, U, for the total heat-flow process through both fluids and the wall. U is defined by the equation:

$$\hat{Q} = UA\Delta T = UA(T_h - T_c). \tag{8.19}$$

The units of U are the same as h, e.g. W m^{-2} K^{-1} or Btu h^{-1} ft^{-2} $°F^{-1}$. Eq. (8.19) written in terms of the ratio of driving force (ΔT) and resistance yields an expression for the total resistance to heat flow, R_T:

$$R_T = \frac{1}{UA}\,. \tag{8.20}$$

In Section 8.3.3 it was noted that when there are thermal

Table 8.2 Individual heat-transfer coefficients

(From W.H. McAdams, 1954, Heat Transmission, *3rd edn, McGraw-Hill, New York)*

Process	*Range of values of h* (W m^{-2} K^{-1})	(Btu ft^{-2} h^{-1} $°F^{-1}$)
Condensing steam	6000–115 000	1000–20 000
Boiling water	1700–50 000	300–9000
Condensing organic vapour	1100–2200	200–400
Heating or cooling water	300–17 000	50–3000
Heating or cooling oil	60–1700	10–300
Superheating steam	30–110	5–20
Heating or cooling air	1–60	0.2–10

resistances in series, the total resistance is the sum of the individual resistances. Applying this now to the situation of heat exchange between fluids, R_T is equal to the sum of R_h, R_w and R_c:

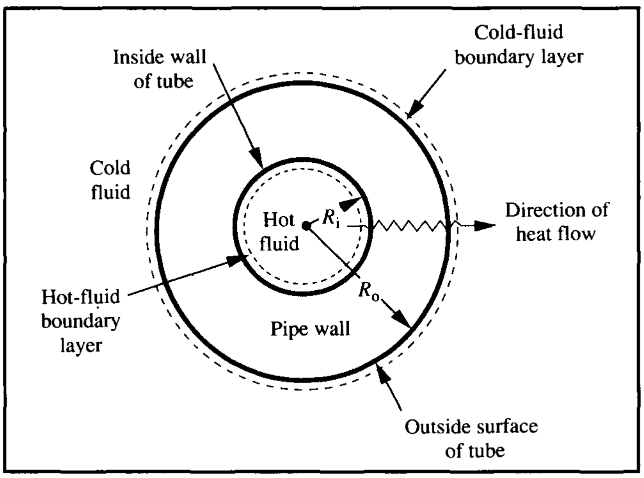
$$R_T = R_h + R_w + R_c. \tag{8.21}$$

Combining Eqs (8.12), (8.17), (8.18), (8.20) and (8.21) gives:

$$\frac{1}{UA} = \frac{1}{h_h A} + \frac{B}{kA} + \frac{1}{h_c A}. \tag{8.22}$$

In Eq. (8.22), the surface area A appears in each term. When fluids are separated by a flat wall, the surface area for heat transfer through each boundary layer and the wall is the same, so that A can be cancelled from the equation. However, a minor complication arises with cylindrical geometry such as pipes. Let us assume that hot fluid is flowing inside a pipe while cold fluid flows outside, as shown in Figure 8.12. Because the inside diameter of the pipe is smaller than the outside diameter, the surface areas for heat transfer between the fluid and the pipe wall are different for the two fluids. The surface area of a cylinder is equal to its circumference multiplied by length, i.e. $A = 2\pi RL$ where R is the radius of the cylinder and L is its length. Therefore, the heat-transfer area at the hot-fluid boundary layer inside the tube is $A_i = 2\pi R_i L$; the heat-transfer area at the cold-fluid film outside the tube is

Figure 8.12 Effect of pipe wall thickness on surface area for heat transfer.



$A_o = 2\pi R_o L$. The surface area available for conduction through the wall varies between A_i and A_o.

The variation of heat-transfer area in cylindrical systems depends on the thickness of the pipe wall; for thin walls the variation will be relatively small. In engineering design, these variations in surface area are incorporated into the equations for heat transfer. However, for the sake of simplicity, in this chapter we will ignore any differences in surface area; we will assume in effect that the pipes are thin-walled. Accordingly, for cylindrical as well as flat geometry, we can cancel A from Eq. (8.22), and write a simplified equation for U:

$$\frac{1}{U} = \frac{1}{h_h} + \frac{B}{k} + \frac{1}{h_c}. \tag{8.23}$$

The overall heat-transfer coefficient characterises the operating conditions used for heat transfer. Small U for a particular process means that the system has limited capacity for heat exchange; U can be improved by manipulating operating conditions such as fluid velocity in shell-and-tube equipment or stirrer speed in bioreactors. U is independent of A. Heat transfer in an exchanger with small U can also be improved by increasing the heat-transfer area and size of the unit; however increasing A raises the cost of the equipment. If U is large, the heat exchanger is well designed and is operating under conditions which enhance heat transfer.

8.4.4 Fouling Factors

Heat-transfer equipment in service does not remain clean. Dirt and scale deposit on one or both sides of the pipes, providing additional resistance to heat flow and reducing the overall heat-transfer coefficient. Resistances to heat transfer when fouling affects both sides of the heat-transfer surface are represented in Figure 8.13. Five resistances are present in series: the thermal boundary layer on the hot-fluid side, a fouling layer on the hot-fluid side, the pipe wall, a fouling layer on the cold-fluid side, and the cold-fluid boundary layer.

Each fouling layer has associated with it a heat-transfer coefficient; for scale and dirt the coefficient is called a *fouling factor*. Let h_{fh} be the fouling factor on the hot-fluid side, and h_{fc} be the fouling factor on the cold-fluid side. When these additional resistances are present, they must be included in the expression for the overall heat-transfer coefficient, U. Eq. (8.23) becomes:

$$\frac{1}{U} = \frac{1}{h_{fh}} + \frac{1}{h_h} + \frac{B}{k} + \frac{1}{h_c} + \frac{1}{h_{fc}}. \tag{8.24}$$

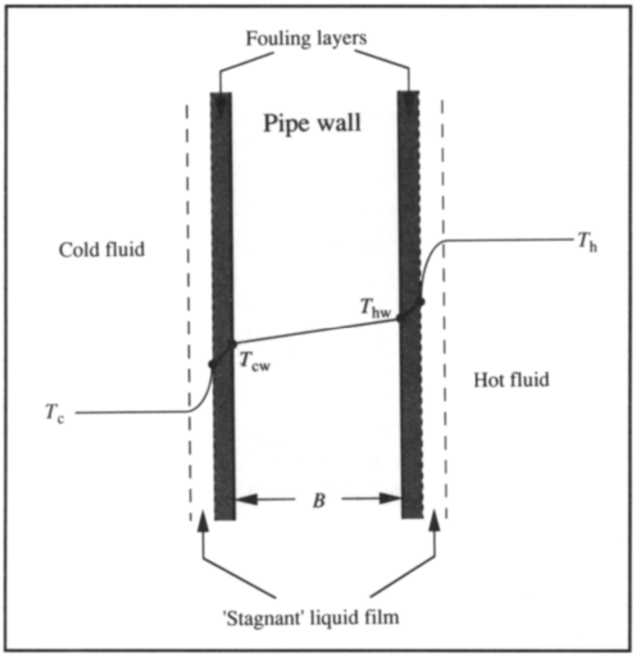


Figure 8.13 Resistances to heat transfer with fouling deposits on both surfaces.

Adding fouling factors in Eq. (8.24) increases $1/U$, thus decreasing the value of U.

Accurate estimation of fouling factors is very difficult. The chemical nature of the deposit and its thermal conductivity depend on the fluid in the tube and the temperature; fouling thickness can also vary between cleanings. Typical values of fouling factors for various fluids are listed in Table 8.3.

8.5 Design Equations For Heat-Transfer Systems

The basic equation for design of heat exchangers is Eq. (8.19). If $\hat{Q}$, U and ΔT are known, this equation allows us to calculate A. Specification of A is a major objective of heat-exchanger design; the surface area required dictates the configuration and size of the equipment and its cost. In the following sections, we will consider procedures for determining $\hat{Q}$, U and ΔT for use in Eq. (8.19).

8.5.1 Energy Balance

In heat-exchanger design, energy balances are applied to determine $\hat{Q}$ and all inlet and outlet temperatures used to specify

Table 8.3 Fouling factors for scale deposits

(Data from J.M. Coulson and J.F. Richardson, 1977, Chemical Engineering*, vol. 1, 3rd edn, Pergamon Press, Oxford)*

Source of deposit	*Fouling factor*	
	($W\,m^{-2}\,K^{-1}$)	($Btu\,ft^{-2}\,h^{-1}\,°F^{-1}$)
Water*		
Distilled	11 000	2 000
Sea	11 000	2 000
Clear river	4 800	800
Untreated cooling tower	1 700	300
Hard well	1 700	300
Steam		
Good quality, oil free	19 000	3 000
Liquids		
Treated brine	3 700	700
Organics	5 600	1 000
Fuel oils	1 000	200
Gases		
Air	2 000–4 000	300–700
Solvent vapour	7 000	1 300

* Velocity 1 m s^{-1}; temperature less than 320K

ΔT. These energy balances are based on general equations for flow systems derived in Chapters 5 and 6.

Let us first consider the equations for double-pipe or shell-and-tube heat-exchangers. From Eq. (6.10), under steady-state conditions ${dE}/{dt} = 0$ and in the absence of shaft work ($\hat{W}_s = 0$), the energy-balance equation is:

$$\hat{M}_i h_i - \hat{M}_o h_o - \hat{Q} = 0 \tag{8.25}$$

where $\hat{M}_i$ is mass flow rate in, $\hat{M}_o$ is mass flow rate out, h_i is specific enthalpy of the incoming stream, h_o is specific enthalpy of the outgoing stream and $\hat{Q}$ is rate of heat removal from the system. Unfortunately, the conventional symbols for individual heat-transfer coefficient and specific enthalpy are the same: h. In this section, h in Eqs (8.25)–(8.30) and (8.33) denotes specific enthalpy; otherwise in this chapter, h represents the individual heat-transfer coefficient.

Eq. (8.25) can be applied separately to each fluid in the heat exchanger. As the mass flow rate is the same at the inlet as at the outlet, for the hot fluid:

$$\hat{M}_h (h_{hi} - h_{ho}) - \hat{Q}_h = 0 \tag{8.26}$$

$$\hat{M}_h (h_{hi} - h_{ho}) = \hat{Q}_h \tag{8.27}$$

where subscript h denotes hot fluid and $\hat{Q}_h$ is the rate of heat transfer from that fluid. Equations similar to Eqs (8.26) and (8.27) can be derived for the cold fluid:

$$\hat{M}_c (h_{ci} - h_{co}) + \hat{Q}_c = 0 \tag{8.28}$$

or

$$\hat{M}_c (h_{co} - h_{ci}) = \hat{Q}_c \tag{8.29}$$

where subscript c refers to cold fluid. $\hat{Q}_c$ is the rate of heat flow into the cold fluid; therefore $\hat{Q}_c$ is added rather than subtracted in Eq. (8.28).

When there are no heat losses from the exchanger, all heat removed from the hot stream is taken up by the cold stream. We can therefore equate $\hat{Q}$ terms in Eqs (8.27) and (8.29): $\hat{Q}_h = \hat{Q}_c = \hat{Q}$. Therefore:

$$\hat{M}_h (h_{hi} - h_{ho}) = \hat{M}_c (h_{co} - h_{ci}) = \hat{Q}\,. \tag{8.30}$$

When sensible heat is exchanged between fluids, the enthalpy differences in Eq. (8.30) can be expressed in terms of the heat capacity C_p and the temperature change for each fluid. If we assume C_p is constant over the temperature range in the exchanger, Eq. (8.30) becomes:

$$\hat{M}_h C_{ph} (T_{hi} - T_{ho}) = \hat{M}_c C_{pc} (T_{co} - T_{ci}) = \hat{Q} \tag{8.31}$$

where C_{ph} is the heat capacity of the hot fluid, C_{pc} is the heat capacity of the cold fluid, T_{hi} is the inlet temperature of the hot fluid, T_{ho} is the outlet temperature of the hot fluid, T_{ci} is the inlet temperature of the cold fluid, and T_{co} is the outlet temperature of the cold fluid.

In heat-exchanger design, Eq. (8.31) is used to determine $\hat{Q}$ and the inlet and outlet conditions of the fluid streams. This is illustrated in Example 8.1.

Example 8.1 Heat exchanger

Hot, freshly-sterilised nutrient medium is cooled in a double-pipe heat exchanger before being used in a fermentation. Medium leaving the steriliser at 121°C enters the exchanger at a flow rate of 10 $m^3\ h^{-1}$; the desired outlet temperature is 30°C. Heat from the medium is used to raise the temperature of 25 $m^3\ h^{-1}$ water initially at 15°C. The system operates at steady state. Assume that nutrient medium has the properties of water.

(a) What rate of heat transfer is required?
(b) Calculate the final temperature of the cooling water as it leaves the heat exchanger.

Solution:
The density of water and medium is 1000 kg m^{-3}. Therefore:

$$\hat{M}_h = 10\ \mathrm{m^3\,h^{-1}} \cdot \left| \frac{1\ \mathrm{h}}{3600\ \mathrm{s}} \right| \cdot \left| \frac{1000\ \mathrm{kg}}{1\ \mathrm{m^3}} \right| = 2.78\ \mathrm{kg\,s^{-1}}$$

$$\hat{M}_c = 25\ \mathrm{m^3\,h^{-1}} \cdot \left| \frac{1\ \mathrm{h}}{3600\ \mathrm{s}} \right| \cdot \left| \frac{1000\ \mathrm{kg}}{1\ \mathrm{m^3}} \right| = 6.94\ \mathrm{kg\,s^{-1}}.$$

The heat capacity of water can be taken as 75.4 J $\mathrm{gmol^{-1}\,°C^{-1}}$ for most of the temperature range of interest (Table B.3). Therefore:

$$C_{ph} = C_{pc} = 75.4\ \mathrm{J\,gmol^{-1}\,°C^{-1}} \cdot \left| \frac{1\ \mathrm{gmol}}{18\ \mathrm{g}} \right| \cdot \left| \frac{1000\ \mathrm{kg}}{1\ \mathrm{kg}} \right|$$
$$= 4.19 \times 10^3\ \mathrm{J\,kg^{-1}\,°C^{-1}}.$$

(a) From Eq. (8.31) for the hot fluid:

$$\hat{Q} = (2.78\ \mathrm{kg\,s^{-1}})\,(4.19 \times 10^3\ \mathrm{J\,kg^{-1}\,°C^{-1}})\,(121 - 30)°\mathrm{C}$$
$$= 1.06 \times 10^6\ \mathrm{J\,s^{-1}} = 1060\ \mathrm{kW}.$$

(b) For the cold fluid, from Eq. (8.31):

$$T_{co} = T_{ci} + \frac{\hat{Q}}{\hat{M}_c C_{pc}}$$

$$T_{co} = 15°\mathrm{C} + \frac{1.06 \times 10^6\ \mathrm{J\,s^{-1}}}{(6.94\ \mathrm{kg\,s^{-1}})\,(4.19 \times 10^3\ \mathrm{J\,kg^{-1}\,°C})} = 51.5°\mathrm{C}.$$

The exit water temperature is 52°C.

Eq. (8.31) can also be applied to heat removal from a reactor for the purpose of temperature control. At steady state, the temperature of the hot fluid, e.g. fermentation broth, does not change; therefore the left-hand side of Eq. (8.31) is zero. If energy is absorbed by the cold fluid as sensible heat, the energy-balance equation becomes:

$$\hat{M}_c\, C_{pc}\,(T_{co} - T_{ci}) = \hat{Q}. \tag{8.32}$$

To use Eq. (8.32) for bioreactor design we must know $\hat{Q}$. $\hat{Q}$ is found by considering all significant heat sources and sinks in the system; an expression involving $\hat{Q}$ for fermentation systems was presented in Chapter 6 based on relationships derived in Chapter 5:

$$\frac{\mathrm{d}E}{\mathrm{d}t} = -\Delta\hat{H}_{rxn} - \hat{M}_v\,\Delta h_v - \hat{Q} + \hat{W}_s \tag{6.12}$$

where $\Delta\hat{H}_{rxn}$ is the rate of heat absorption or evolution due to metabolic reaction, $\hat{M}_v$ is the mass flow rate of evaporated liquid leaving the system, Δh_v is the latent heat of evaporation, and $\hat{W}_s$ is the rate of shaft work done on the system. For exothermic reactions $\Delta\hat{H}_{rxn}$ is negative, for endothermic reactions $\Delta\hat{H}_{rxn}$ is positive. In most fermentation systems the only source of shaft work is the stirrer; therefore $\hat{W}_s$ is the power P dissipated by the impeller. Methods for estimating P are described in Section 7.10. Eq. (6.12) represents a considerable simplification of the energy balance. It is applicable to systems in which heat of reaction dominates the energy balance so that contributions from sensible heat and heats of solution can be ignored. In large insulated fermenters, heat produced by metabolic activity is by far the dominant source of heat; energy dissipated by stirring may also be worth considering. The other heat sources and sinks are relatively minor and can generally be neglected.

At steady state $\mathrm{d}E/\mathrm{d}t = 0$ and Eq. (6.12) becomes:

$$\hat{Q} = -\Delta\hat{H}_{rxn} - \hat{M}_v \Delta h_v + \hat{W}_s. \tag{8.33}$$

Application of Eq. (8.33) to determine $\hat{Q}$ is illustrated in Examples 5.7 and 5.8. Once $\hat{Q}$ has been estimated, Eq. (8.32) is used to evaluate unknown operating conditions as shown in Example 8.2.

Example 8.2 Cooling coil

A 150 m^3 bioreactor is operated at 35°C to produce fungal biomass from glucose. The rate of oxygen uptake by the culture is 1.5 kg m^{-3} h^{-1}; the agitator dissipates heat at a rate of 1 kW m^{-3}. 60 m^3 h^{-1} cooling water available from a nearby river at 10°C is passed through an internal coil in the fermentation tank. If the system operates at steady state, what is the exit temperature of the cooling water?

Solution:

Rate of heat generation by aerobic cultures is calculated directly from the oxygen demand. As outlined in Section 5.9.2, approximately 460 kJ heat is released for each gmol oxygen consumed. Therefore, the metabolic heat load is:

$$\Delta\hat{H}_{rxn} = \frac{-460\text{ kJ}}{\text{gmol}} \cdot \left|\frac{1\text{ gmol}}{32\text{ g}}\right| \cdot \left|\frac{1000\text{ g}}{1\text{ kg}}\right| \cdot (1.5\text{ kg m}^{-3}\text{h}^{-1}) \cdot \left|\frac{1\text{ h}}{3600\text{ s}}\right| \cdot 150\text{ m}^3$$

$$= -898\text{ kJ s}^{-1}$$

$$= -898\text{ kW}.$$

$\Delta\hat{H}_{rxn}$ is negative because fermentation is exothermic. The rate of heat dissipation by the agitator is:

$$(1\text{ kW m}^{-3})\ 150\text{ m}^3 = 150\text{ kW}.$$

We can now calculate $\hat{Q}$ from Eq. (8.33):

$$\hat{Q} = (898 + 150)\text{ kW} = 1048\text{ kW}.$$

The density of the cooling water is 1000 kg m^{-3}; therefore:

$$\hat{M}_c = 60\text{ m}^3\text{h}^{-1} \cdot \left|\frac{1\text{ h}}{3600\text{ s}}\right| \cdot \left|\frac{1000\text{ kg}}{1\text{ m}^3}\right| = 16.7\text{ kg s}^{-1}.$$

The heat capacity of water is 75.4 J $gmol^{-1}$ $°C^{-1}$ (Table B.3). Therefore:

$$C_{pc} = 75.4\text{ J gmol}^{-1}\,°\text{C}^{-1} \cdot \left|\frac{1\text{ gmol}}{18\text{ g}}\right| \cdot \left|\frac{1000\text{ g}}{1\text{ kg}}\right| = 4.19 \times 10^3\text{ J kg}^{-1}\,°\text{C}^{-1}.$$

We can now apply Eq. (8.32) by rearranging and solving for T_{co}:

$$T_{co} = T_{ci} + \frac{\hat{Q}}{\hat{M}_c C_{pc}}$$

$$T_{co} = 10°\text{C} + \frac{1048 \times 10^3\text{ J s}^{-1}}{(16.7\text{ kg s}^{-1})\,(4.19 \times 10^3\text{ J kg}^{-1}\,°\text{C}^{-1})} = 25.0°\text{C}.$$

The water outlet temperature is 25°C.

8.5.2 Logarithmic- and Arithmetic-Mean Temperature Differences

Application of the heat-exchanger design equation, Eq. (8.19), requires knowledge of the temperature-difference driving force for heat transfer, ΔT. ΔT is equal to the difference in temperature between hot and cold fluids. However, as we have seen in Figures 8.2, 8.4, 8.5 and 8.8, fluid temperatures vary with position in heat exchangers; for example, the temperature difference between hot and cold fluids at one end of the exchanger may be more or less than at the other end. The driving force for heat transfer therefore varies from point to point in the system. For application of Eq. (8.19), this difficulty is overcome by use of an average ΔT.

If the temperature varies in both fluids in either countercurrent or cocurrent flow, the *logarithmic-mean temperature difference* ΔT_L is used:

$$\Delta T_L = \frac{\Delta T_2 - \Delta T_1}{\ln(\Delta T_2/\Delta T_1)} = \frac{\Delta T_2 - \Delta T_1}{2.303 \log(\Delta T_2/\Delta T_1)} \tag{8.34}$$

where ΔT_1 and ΔT_2 are the temperature differences between hot and cold fluids at the ends of the equipment. ΔT_1 and ΔT_2 are calculated using the values for T_{hi}, T_{ho}, T_{ci} and T_{co} obtained from the energy balance. For convenience and to eliminate negative numbers and their logarithms, subscripts 1 and 2 can refer to either end of the exchanger. Eq. (8.34) has been derived using the following assumptions:

(i) the overall heat-transfer coefficient U is constant;
(ii) the specific heats of the hot and cold fluids are constant;
(iii) heat losses from the system are negligible; and
(iv) the system is at steady-state in either countercurrent or cocurrent flow.

The most questionable of these assumptions is that of constant U, since this coefficient varies with temperature of the fluids. However, because the change with temperature is gradual, when temperature differences in the system are moderate the assumption is not seriously in error. Other details of the derivation of Eq. (8.34) can be found elsewhere [1, 2].

Example 8.3 Log-mean temperature difference

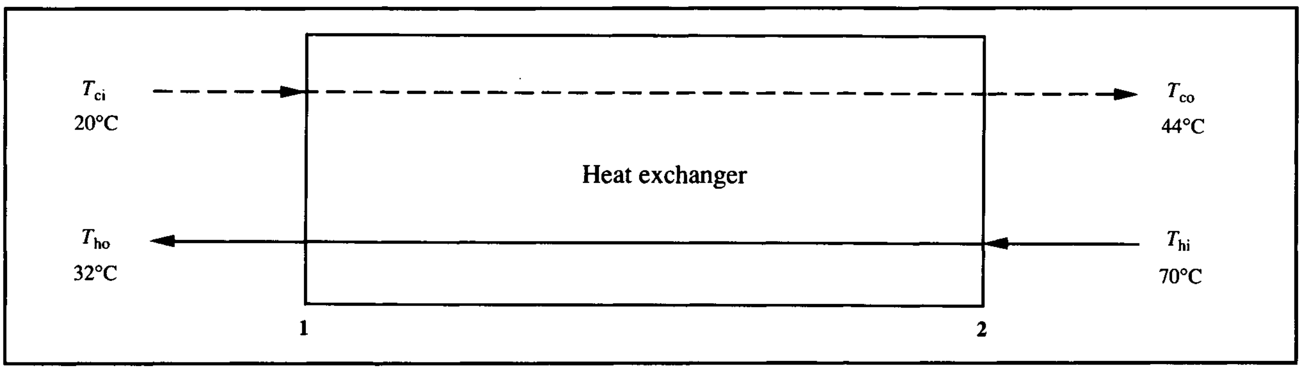
A liquid stream is cooled from 70°C to 32°C in a double-pipe heat exchanger. Fluid flowing countercurrently with this stream is heated from 20°C to 44°C. Calculate the log-mean temperature difference.

Solution:

The heat-exchanger configuration is shown in Figure 8E3.1. At the left-hand end of the equipment, $\Delta T_1 = (32 - 20)$°C = 12°C. At the other end, $\Delta T_2 = (70 - 44)$°C = 26°C. From Eq. (8.34):

$$\Delta T_L = \frac{(26 - 12)°C}{\ln(26/12)} = 18.1°C.$$

Figure 8E3.1 Flow configuration for heat exchanger.



As noted above, the log-mean temperature difference is applicable to systems with cocurrent or countercurrent flow. In multiple-pass shell-and-tube heat exchangers, flow is neither countercurrent nor cocurrent. In these units the flow pattern is complex, with cocurrent, countercurrent and cross-flow all present. For shell-and-tube heat exchangers with more than a single tube or shell pass, the log-mean temperature difference must be used with a suitable correction factor to account for the geometry of the exchanger. Correction factors for cross-flow are available in other references [1–3].

When one fluid in the heat-exchange system remains at a constant temperature such as in a fermenter, the ***arithmetic-mean temperature difference*** ΔT_A is the appropriate ΔT to use in heat-exchanger design:

$$\Delta T_A = \frac{2T_F - (T_1 + T_2)}{2} \tag{8.35}$$

where T_F is the temperature of fluid in the fermenter and T_1 and T_2 are the inlet and exit temperatures of the other fluid.

8.5.3 Calculation of Heat-Transfer Coefficients

As described in Sections 8.4.3 and 8.4.4, U in Eq. (8.19) can be determined as a combination of individual heat-transfer coefficients, properties of the separating wall, and, if applicable, fouling factors. Values of the individual heat-transfer coefficients h_h and h_c depend on the thickness of the fluid boundary layers, which is in turn strongly dependent on flow velocity and fluid properties such as viscosity and thermal conductivity. Increasing the level of turbulence and decreasing the viscosity will reduce the thickness of the liquid film, and hence increase the heat-transfer coefficient. Individual heat-transfer coefficients for flow in pipes or stirred vessels are usually evaluated using empirical correlations expressed in terms of dimensionless numbers.

The general form of correlations for heat-transfer coefficients is:

$$Nu = f\left(Re \text{ or } Re_i, Pr, Gr, \frac{D}{L}, \frac{\mu_b}{\mu_w}\right) \tag{8.36}$$

where f means 'some function of', and:

$$Nu = \text{Nusselt number} = \frac{hD}{k_{fb}} \tag{8.37}$$

$$Re = \text{Reynolds number for pipe flow} = \frac{Du\rho}{\mu_b} \tag{8.38}$$

$$Re_i = \text{impeller Reynolds number} = \frac{N_i D_i^2 \rho}{\mu_b} \tag{8.39}$$

$$Pr = \text{Prandtl number} = \frac{C_p \mu_b}{k_{fb}} \tag{8.40}$$

and

$$Gr = \text{Grashof number for heat transfer} = \frac{D^3 g \rho^2 \beta \Delta T}{\mu_b^2}. \tag{8.41}$$

Parameters in the above equations are as follows: h is the individual heat-transfer coefficient, D is the pipe or tank diameter, k_{fb} is the thermal conductivity of the bulk fluid, u is the linear velocity of fluid in the pipe, ρ is the average density of the fluid, μ_b is the viscosity of the bulk fluid, N_i is the rotational speed of the impeller, D_i is the impeller diameter, C_p is the average heat capacity of the fluid, g is gravitational acceleration, β is the coefficient of thermal expansion of the fluid, ΔT is the variation of fluid temperature in the system, L is pipe length, and μ_w is the viscosity of fluid at the wall.

The Nusselt number contains the heat-transfer coefficient h, and represents the ratio of rates of convective and conductive heat transfer. The Prandtl number represents the ratio of momentum and heat transfer; Pr contains physical constants which, for Newtonian fluids, are independent of flow conditions. The Grashof number represents the ratio of gravitational to viscous forces, and appears in correlations only when the fluid is not well mixed. Under these conditions the fluid density is no longer uniform and natural convection becomes an important heat-transfer mechanism. In most industrial applications, heat transfer occurs between turbulent fluids in pipes or in stirred vessels; forced convection in these systems is therefore more important than natural convection and the Grashof number is not of concern. The form of the correlation used to evaluate Nu and therefore h depends on the configuration of the heat-transfer equipment, the flow conditions and other factors.

A wide variety of heat-transfer situations is met in practice and there are many correlations available to biochemical engineers designing heat-exchange equipment. The most common heat-transfer applications are as follows:

(i) heat flow to or from fluids inside tubes, without phase change;
(ii) heat flow to or from fluids outside tubes, without phase change;
(iii) heat flow from condensing fluids; and
(iv) heat flow to boiling liquids.

Different equations are generally required to evaluate h_h and h_c depending on the flow geometry of the hot and cold fluids. Examples of correlations for heat-transfer coefficients in bioprocessing are given in the next section. Others can be found in the references listed at the end of this chapter.

8.5.3.1 *Flow in tubes without phase change*

There are several widely-accepted correlations for forced convection in tubes. The heat-transfer coefficient for fluid flowing inside a tube can be calculated from the following equation [2]:

$$Nu = 0.023\, Re^{0.8}\, Pr^{0.4}. \tag{8.42}$$

Eq. (8.42) is valid for either heating or cooling of liquids with viscosity close to water, and applies under the following conditions: $10^4 \leq Re \leq 1.2\times10^5$ (turbulent flow), $0.7 \leq Pr \leq 120$, and $L/D \geq 60$. Application of Eq. (8.42) to evaluate the tube-side heat-transfer coefficient in a heat exchanger is illustrated in Example 8.4.

Example 8.4 Tube-side heat-transfer coefficient

A single-pass shell-and-tube heat exchanger is used to heat a dilute salt solution used in large-scale protein chromatography. 25.5 $m^3\,h^{-1}$ solution passes through 42 parallel tubes inside the heat exchanger; the internal diameter of the tubes is 1.5 cm and the tube length is 4 m. The viscosity of the bulk salt solution is 10^{-3} kg $m^{-1}\,s^{-1}$, the density is 1010 kg m^{-3}, the average heat-capacity is 4 kJ $kg^{-1}\,°C^{-1}$ and the thermal conductivity is 0.64 W $m^{-1}\,°C^{-1}$. Calculate the heat-transfer coefficient.

Solution:
First we must evaluate Re and Pr. All parameter values for calculation of these dimensionless groups are known except u, the linear fluid velocity. u is obtained by dividing the volumetric flow rate of the fluid by the total flow cross-sectional area.

Total flow cross-sectional area = (cross-sectional area of each tube) (number of tubes)

$$= 42\,(\pi R^2)$$

$$= 42\,\pi \left(\frac{1.5\times10^{-2}\,\text{m}}{2}\right)^2$$

$$= 7.42\times10^{-3}\,\text{m}^2.$$

Therefore:

$$u = \frac{25.5\,\text{m}^3\,\text{h}^{-1}}{7.42\times10^{-3}\,\text{m}^2}\cdot\left|\frac{1\,\text{h}}{3600\,\text{s}}\right| = 0.95\,\text{m}\,\text{s}^{-1}.$$

From Eqs (8.38) and (8.40):

$$Re = \frac{(1.5\times10^{-2}\,\text{m})\,(0.95\,\text{m}\,\text{s}^{-1})\,(1010\,\text{kg}\,\text{m}^{-3})}{10^{-3}\,\text{kg}\,\text{m}^{-1}\,\text{s}^{-1}} = 1.44\times10^4$$

and

$$Pr = \frac{(4\times10^{-3}\,\text{J}\,\text{kg}^{-1}\,°\text{C}^{-1})\,(10^{-3}\,\text{kg}\,\text{m}^{-1}\,\text{s}^{-1})}{0.64\,\text{J}\,\text{s}^{-1}\,\text{m}^{-1}\,°\text{C}^{-1}} = 6.25.$$

Also, $L/D = 267$. As $10^4 \leq Re \leq 1.2 \times 10^5$, $0.7 \leq Pr \leq 120$ and $L/D \geq 60$, Eq. (8.42) is valid.

$$Nu = 0.023\,(1.44 \times 10^4)^{0.8}\,(6.25)^{0.4} = 101.6.$$

Calculating h from this value of Nu:

$$h = \frac{101.6\,(0.64\ \mathrm{W\,m^{-1}\,°C^{-1}})}{1.5 \times 10^{-2}\ \mathrm{m}} = 4335\ \mathrm{W\,m^{-2}\,°C^{-1}}.$$

The heat-transfer coefficient is $4.3\ \mathrm{kW\,m^{-2}\,°C^{-1}}$.

For very viscous liquids, because of the temperature variation across the thermal boundary layer, there may be a marked difference between the viscosity of fluid in bulk flow and the viscosity of fluid adjacent to the wall. A modified form of Eq. (8.42) includes a viscosity correction term [3, 4]:

$$Nu = 0.027\,Re^{0.8}\,Pr^{0.33}\left(\frac{\mu_b}{\mu_w}\right)^{0.14}. \tag{8.43}$$

When flow in pipes is laminar rather than turbulent, Eqs (8.42) and (8.43) do not apply. In liquids with $Re < 2100$, fluid buoyancy and natural convection play an important role in heat transfer. Heat-transfer correlations for laminar flow in tubes can be found in other texts, e.g. [2].

8.5.3.2 *Flow outside tubes without phase change*

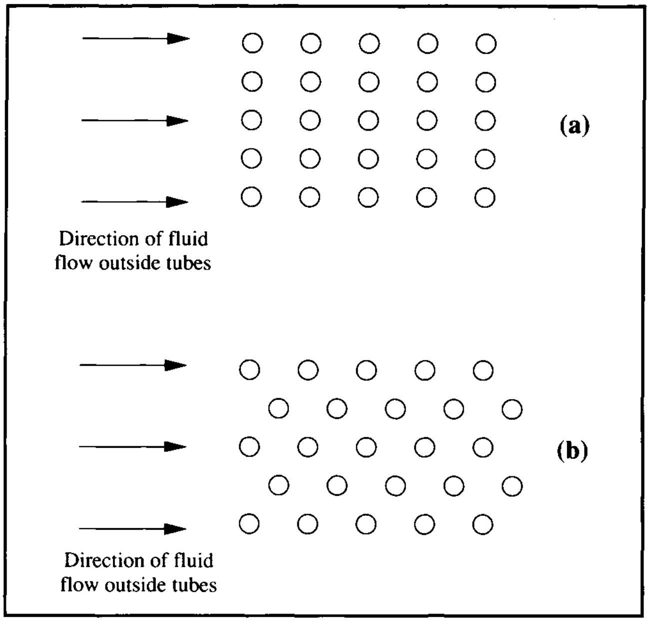
In the shell section of shell-and-tube heat exchangers, fluid flows around the outside of the tubes. The degree of turbulence, and therefore the external heat-transfer coefficient for the shell fluid, depend in part on the geometric arrangement of the tubes and baffles. Tubes in the exchanger may be arranged 'in line' as shown in Figure 8.14(a), or 'staggered' as shown in Figure 8.14(b). The degree of turbulence is considerably less for tubes in line than for staggered tubes. Because the area for flow through a bank of staggered tubes is continually changing, this is a difficult system to analyse.

The following correlation has been proposed for flow of fluid at right-angles to a bank of tubes more than 10 rows deep [5, 6]:

$$Nu = C\,Re_{max}^{0.6}\,Pr^{0.33}. \tag{8.44}$$

In Eq. (8.44) $C = 0.33$ for staggered tubes and $C = 0.26$ for in-line tubes. Re_{max} is the Reynolds number evaluated using Eq. (8.38) with D equal to the outside tube diameter and u equal to the maximum fluid velocity based on the minimum free area available for fluid flow. For flow outside tubes, D in Nu is usually taken as the outside tube diameter. Eq. (8.44) is valid for $Re_{max} > 6 \times 10^3$ [6].

Figure 8.14 Configuration of tubes in shell-and-tube heat exchanger: (a) tubes 'in line'; (b) 'staggered' tubes.



8.5.3.3 *Stirred liquids*

The heat-transfer coefficient in stirred vessels depends on the degree of agitation and the fluid properties. When heat is transferred to or from a helical coil in the vessel, h can be determined using the following equation [7]:

$$Nu = 0.87\, Re_i^{0.62}\, Pr^{0.33} \left(\frac{\mu_b}{\mu_w}\right)^{0.14} \tag{8.45}$$

where Re_i is given by Eq. (8.39), and D in Nu refers to the inside diameter of the tank. For low-viscosity fluids such as water, the viscosity at the wall μ_w can usually be assumed equal to the bulk viscosity μ_b. When heat is transferred to or from a jacket rather than a coil, the correlation is slightly modified [7]:

$$Nu = 0.36\, Re_i^{0.67}\, Pr^{0.33} \left(\frac{\mu_b}{\mu_w}\right)^{0.14}. \tag{8.46}$$

Example 8.5 Heat-transfer coefficient for stirred vessel

A stirred fermenter of diameter 5 m contains an internal helical coil for heat transfer. The fermenter is mixed using a turbine impeller 1.8 m in diameter operated at 60 rpm. The fermentation broth has the following properties: $\mu_b = 5 \times 10^{-3}$ Pa s; $\rho = 1000$ kg m^{-3}; $C_p = 4.2$ kJ kg^{-1} °C^{-1}; $k_{fb} = 0.70$ W m^{-1} °C^{-1}. Neglecting viscosity changes at the wall of the coil, calculate the heat-transfer coefficient.

Solution:

From Eqs (8.39) and (8.40):

$$Re_i = \frac{60\ \text{min}^{-1} \cdot \left|\frac{1\ \text{min}}{60\ \text{s}}\right| \cdot (1.8\ \text{m})^2\, (1000\ \text{kg m}^{-3})}{5 \times 10^{-3}\ \text{kg m}^{-1}\ \text{s}^{-1}} = 6.48 \times 10^5$$

$$Pr = \frac{(4.2 \times 10^3\ \text{J kg}^{-1}\ °\text{C}^{-1})\, (5 \times 10^{-3}\ \text{kg m}^{-1}\ \text{s}^{-1})}{0.70\ \text{J s}^{-1}\ \text{m}^{-1}\ °\text{C}^{-1}} = 30.$$

These values can be substituted into Eq. (8.45) to evaluate Nu:

$$Nu = 0.87\, (6.48 \times 10^5)^{0.62}\, (30)^{0.33} = 1.07 \times 10^4.$$

Calculating h from this value of Nu:

$$h = \frac{(1.07 \times 10^4)\, (0.70\ \text{W m}^{-1}\ °\text{C}^{-1})}{5\ \text{m}} = 1501\ \text{W m}^{-2}\ °\text{C}^{-1}.$$

The heat-transfer coefficient is 1.5 kW m^{-2} °C^{-1}.

8.6 Application of the Design Equations

The equations of Sections 8.4 and 8.5 provide the essential elements for design of heat-transfer systems. Eq. (8.19) is used as the design equation, with $\hat{Q}$ and the inlet and outlet temperatures available from energy-balance calculations as described in Section 8.5.1. The overall heat-transfer coefficient is evaluated from correlations such as those given in Section 8.5.3; additional terms are included if the heat-transfer surfaces are fouled. With these parameters at hand, the area required and the size and capital cost of the equipment can be determined. Because metabolic rates, and therefore heat-production rates, vary during fermentation, the rate at which heat is removed from the bioreactor must also be varied to maintain constant temperature. Heat-transfer design is based on the maximum heat load for the system. When the rate of heat generation drops, operating conditions can be changed to reduce the rate of heat removal; the simplest way of achieving this is to decrease the cooling-water flow rate.

The procedures outlined in this chapter represent the simplest and most direct approach to heat-transfer design. If several independent variables remain unfixed prior to design calculations, many different design outcomes are possible. When variables such as type of fluid, mass flow rates and terminal temperatures are unspecified, they can be manipulated to produce an *optimum design*, e.g. the design yielding the lowest total cost per year of operation. Computer packages which optimise heat-exchanger design are available commercially.

Calculation of equipment requirements for heating or cooling of fermenters can sometimes be simplified by considering the relative importance of each heat-transfer resistance. For large fermentation vessels containing cooling coils, the fluid velocity in the vessel is generally much slower than in the coils; accordingly, the tube-side thermal boundary layer is relatively thin and most of the heat-transfer resistance is located in the fermentation medium. Especially when there is no fouling present in the tubes, the heat-transfer coefficient for the cooling water can often be omitted when calculating U. Likewise, the wall resistance can sometimes be ignored as conduction of heat through metal is generally very rapid. An exception is stainless steel which is used widely in the fermentation industry; the low thermal conductivity of this material means that wall resistance should be taken into account if the wall thickness is greater than about 5 mm.

The influence of sparging on heat transfer in stirred bioreactors is not clear; however, as the effect is minor compared with other parameters, correlations developed for ungassed systems are applied to aerobic fermentations. For non-Newtonian broths, apparent viscosity can be substituted for μ_b in the dimensionless groups Re, Re_i and Pr; this substitution is not straightforward when rheological parameters such as n, K and τ_0 change during the culture period. Apparent viscosity also depends on the shear rate in the fermenter which varies greatly throughout the vessel. Correlations for heat-transfer coefficients such as those presented in this chapter were not developed for fermentation systems and must not be considered to give exact values. The actual flow behaviour in many bioprocesses is poorly characterised due to the effects of non-Newtonian rheology and the presence of cells and air bubbles.

Application of heat-exchanger design equations to specify a fermenter cooling-system is illustrated in Example 8.6.

Example 8.6 Cooling-coil length in fermenter design

A fermenter used for antibiotic production must be kept at 27°C. After considering the oxygen demand of the organism and the heat dissipation from the stirrer, the maximum heat-transfer rate required is estimated as 550 kW. Cooling water is available at 10°C; the exit temperature of the cooling water is calculated using an energy balance as 25°C. The heat-transfer coefficient for the fermentation broth is estimated from Eq. (8.45) as 2150 W m^{-2} °C^{-1}. The heat-transfer coefficient for the cooling water is calculated as 14 000 W m^{-2} °C^{-1}. It is proposed to install a helical cooling coil inside the fermenter; the outer diameter of the coil pipe is 8 cm, the pipe thickness is 5 mm and the thermal conductivity of the steel is 60 W m^{-1} °C^{-1}. An average internal fouling-factor of 8500 W m^{-2} °C^{-1} is expected; the fermenter side surface of the coil is kept relatively clean. What length of cooling coil is required?

Solution:

$\hat{Q} = 550 \times 10^3$ W. ΔT is calculated as the arithmetic-average temperature difference from Eq. (8.35):

$$\Delta T = \frac{2(27°\mathrm{C}) - (10+25)°\mathrm{C}}{2} = 9.5°\mathrm{C}.$$

U is calculated using Eq. (8.24) after omitting h_{fh} as there is no fouling layer on the hot side of the system:

$$\begin{aligned}
\frac{1}{U} &= \left(\frac{1}{2150} + \frac{5\times10^{-3}}{60} + \frac{1}{14\,000} + \frac{1}{8500}\right)\mathrm{m^2\,°C\,W^{-1}} \\
&= (4.65\times10^{-4} + 8.33\times10^{-5} + 7.14\times10^{-5} + 1.18\times10^{-4})\,\mathrm{m^2\,°C\,W^{-1}} \\
&= 7.38\times10^{-4}\,\mathrm{m^2\,°C\,W^{-1}} \\
U &= 1355\,\mathrm{W\,m^{-2}\,°C^{-1}}.
\end{aligned}$$

Note the relative magnitudes of the four contributions to U: the cooling-water coefficient and the wall resistance make a comparatively minor contribution and can often be neglected in design calculations.

We can now apply Eq. (8.19) to evaluate the required surface area A.

$$A = \frac{550 \times 10^3 \text{ W}}{(9.5°\text{C})\,(1355 \text{ W m}^{-2}\,°\text{C}^{-1})} = 42.7 \text{ m}^2.$$

The heat-transfer area is equal to the surface area of the pipe:

$$A = 2\pi RL$$

where R is the pipe radius and L is the pipe length. Therefore:

$$L = \frac{42.7 \text{ m}^2}{2\pi \left(\dfrac{8 \times 10^{-2}}{2}\right) \text{m}} = 169.9 \text{ m}.$$

The length of coil required is 170 m. The cost of such a length of pipe is a significant factor in the overall cost of the fermenter.

8.6.1 Relationship Between Heat Transfer, Cell Concentration and Stirring Conditions

The design equation (8.19) and the energy-balance equation (8.33) allow us to derive some important relationships for fermenter operation. Because cell metabolism is usually the largest source of heat in fermenters, the capacity of the system for heat removal can be linked directly to the maximum cell concentration in the reactor. Assuming that heat dissipated from the stirrer and the cooling effects of evaporation are negligible compared with the heat of reaction, Eq. (8.33) becomes:

$$\hat{Q} = -\Delta\hat{H}_{rxn}. \tag{8.47}$$

In aerobic fermentation, heat of reaction is related to the rate of oxygen consumption by the cells. As outlined in Section 5.9.2, approximately 460 kJ heat is released for each gmol oxygen consumed. If Q_O is the rate of oxygen uptake per unit volume in the fermenter:

$$\Delta\hat{H}_{rxn} = (-460 \text{ kJ gmol}^{-1})\, Q_O V \tag{8.48}$$

where V is the reactor volume. Typical units for Q_O are gmol $\text{m}^{-3}\,\text{s}^{-1}$. $\Delta\hat{H}_{rxn}$ in Eq. (8.48) is negative because the reaction is exothermic. Substituting this equation in Eq. (8.47):

$$\hat{Q} = (460 \text{ kJ gmol}^{-1})\, Q_O V. \tag{8.49}$$

If q_O is the ***specific oxygen-uptake rate***, or rate of oxygen consumption per unit cell, $Q_O = q_O\, x$ where x is cell concentration. Typical units for q_O are gmol $\text{g}^{-1}\,\text{s}^{-1}$. Therefore:

$$\hat{Q} = (460 \text{ kJ gmol}^{-1})\, q_O x V. \tag{8.50}$$

Substituting this into Eq. (8.19) gives:

$$(460 \text{ kJ gmol}^{-1})\, q_O x V = UA\Delta T. \tag{8.51}$$

The fastest rate of heat transfer occurs when the temperature difference between fermenter contents and cooling water is maximum. Hypothetically, this occurs when the cooling water remains at its inlet temperature, i.e. $\Delta T = (T_F - T_{ci})$ where T_F is the fermentation temperature and T_{ci} is the water inlet temperature. Therefore, we can determine from Eq. (8.51) the maximum cell concentration supported by the heat-transfer system:

$$x_{max} = \frac{UA\,(T_F - T_{ci})}{(460 \text{ kJ gmol}^{-1})\, q_O V}. \tag{8.52}$$

It is undesirable for biomass concentration in fermenters to be limited by heat-transfer capacity. Therefore, if the maximum cell concentration estimated using Eq. (8.52) is lower than that desired in the process, the heat-transfer facilities must be improved. For example, area A could be increased by installing a longer cooling coil, or the overall heat-transfer coefficient could be improved by increasing the stirrer speed. Eq. (8.52) was derived for fermenters in which shaft work could be ignored; if stirrer power adds significantly to the total heat load, x_{max} will be somewhat smaller than that calculated.

It should be evident from Chapters 7 and 8 that mixing and heat transfer are not independent functions in bioreactors. Agitation rate and stirrer size affect the value of the heat-transfer coefficient in the fermentation fluid; turbulence in the reactor decreases the thickness of the thermal boundary layer and facilitates rapid heat transfer. However, stirring also generates heat and contributes to the total heat load that must be removed from the reactor to maintain constant temperature. Heat removal can be a severe problem in bioreactors if the fluid is viscous; turbulence and high heat-transfer coefficients are difficult to achieve in highly viscous liquids without enormous power input which itself generates an extra heat load. These relationships are explored further in Problem 8.8.

8.7 Summary of Chapter 8

After studying Chapter 8, you should:

(i) understand the mechanisms of *conduction* and *convection* in heat transfer;
(ii) know *Fourier's law* of conduction in terms of the *thermal conductivity* of materials;
(iii) be able to describe equipment used for heat exchange in industrial bioprocesses;
(iv) understand the importance of the *thermal boundary layer* in heat transfer;
(v) know the *heat-transfer design equation* and the meaning of the *overall heat-transfer coefficient U*;
(vi) understand how the overall heat-transfer coefficient can be expressed in terms of individual resistances to heat transfer;
(vii) know how individual heat-transfer coefficients are estimated;
(viii) know how to incorporate fouling factors into heat-transfer analysis;
(ix) for heat exchange to or from fermentation vessels, know what are the major resistances to heat transfer; and
(x) be able to carry out simple calculations for design of heat-transfer systems.

Problems

8.1 Rate of conduction

(a) A furnace wall is constructed of firebrick 15-cm thick. The temperature inside the wall is 700°C; the temperature outside is 80°C. If the thermal conductivity of the brick under these conditions is 0.3 W m^{-1} K^{-1}, what is the rate of heat loss through 1.5 m^2 of wall surface?
(b) The 1.5 m^2 area in part (a) is insulated with 4-cm thick asbestos with thermal conductivity 0.1 W m^{-1} K^{-1}. What is the rate of heat loss now?

8.2 Overall heat-transfer coefficient

Heat is transferred from one fluid to a second fluid across a metal wall. The film coefficients are 1.2 and 1.7 kW m^{-2} K^{-1}. The metal is 6-mm thick and has a thermal conductivity of 19 W m^{-1} K^{-1}. On one side of the wall there is a scale deposit with a fouling factor estimated at 830 W m^{-2} K^{-1}. What is the overall heat-transfer coefficient?

8.3 Effect of cooling-coil length on coolant requirements

A fermenter is maintained at 35°C by water circulating at a rate of 0.5 kg s^{-1} in a cooling coil inside the vessel. The inlet and outlet temperatures of the water are 8°C and 15°C, respectively.

The length of the cooling coil is increased by 50%. In order to maintain the same fermentation temperature, the rate of heat removal must be kept the same. Determine the new cooling-water flow rate and outlet temperature by carrying out the following calculations. The heat capacity of the cooling water can be taken as 4.18 kJ kg^{-1} °C^{-1}.

(a) From a steady-state energy balance on the cooling water, calculate the rate of cooling with the original coil.
(b) Determine the mean temperature difference with the original coil.
(c) Evaluate UA for the original coil.
(d) If the length of the coil is increased by 50%, the area available for heat transfer, A', also increases by 50% so that $A' = 1.5A$. The value of the overall heat-transfer coefficient is not expected to change very much. For the new coil, what is the value of UA'?
(e) Evaluate the new cooling-water outlet temperature.
(f) By how much are the cooling-water requirements reduced after the new coil is installed?

8.4 Calculation of heat-transfer area in fermenter design

A 100 m^3 fermenter of diameter 5 m is stirred using a turbine impeller 1.7 m in diameter at a speed of 80 rpm. The culture fluid inside the fermenter has the following properties:

$$C_p = 4.2 \text{ kJ kg}^{-1}\,°\text{C}^{-1}$$
$$k_{fb} = 0.6 \text{ W m}^{-1}\,°\text{C}^{-1}$$
$$\rho = 10^3 \text{ kg m}^{-3}$$
$$\mu_b = 10^{-3} \text{ N s m}^{-2}.$$

Assume that the viscosity at the wall is equal to the bulk-fluid viscosity.

Heat is generated by the fermentation at a rate of 2500 kW. This heat is removed to cooling water flowing in a helical stainless-steel coil inside the vessel. The coil wall thickness is 6 mm and the thermal conductivity of the metal is 20 W m^{-1} °C^{-1}. There are no fouling layers present, and the heat-transfer coefficient for the cooling water can be neglected. The fermentation temperature is 30°C; cooling water enters the coils at 10°C.

(a) Calculate the fermenter-side heat-transfer coefficient.
(b) Calculate the overall heat-transfer coefficient U. What proportion of the total resistance to heat transfer is due to the pipe wall?
(c) The surface area needed for cooling depends on the cooling-water flow rate. Prepare a graph showing the outlet cooling-water temperature and the area required for heat transfer as functions of coolant flow-rate for $1.2 \times 10^5 \text{ kg h}^{-1} \leq \hat{M}_c \leq 2 \times 10^6 \text{ kg h}^{-1}$.
(d) For a cooling-water flow rate of 5×10^5 kg h^{-1}, estimate the length of cooling-coil needed if the diameter is 10 cm.

8.5 Effect of fouling on heat-transfer resistance

In current service, 20 kg s^{-1} cooling water at 12°C must be circulated through a coil inside a fermenter to maintain the temperature at 37°C. The coil is 150 m long with pipe diameter 12 cm; the exit water temperature is 28°C. After the inner and outer surfaces of the coil are cleaned it is found that only 13 kg s^{-1} cooling water is required to control the fermentation temperature.

(a) Calculate the overall heat-transfer coefficient before cleaning.
(b) What is the outlet water temperature after cleaning?
(c) What fraction of the total resistance to heat transfer before cleaning was due to fouling deposits?

8.6 Pre-heating of nutrient medium

Nutrient medium is to be heated from 10°C to 28°C is a single-pass countercurrent shell-and-tube heat exchanger before being pumped into a fed-batch fermenter. Medium passes through the tubes of the exchanger; the shell-side fluid is water which enters with flow rate 3×10^4 kg h^{-1} and temperature 60°C. Pre-heated medium is required at a rate of 50 m^3 h^{-1}. The density, viscosity and heat capacity of the medium are the same as water; the thermal conductivity of the medium is 0.54 W m^{-1} °C^{-1}.

It is proposed to use 30 steel tubes with inner diameter 5 cm; the tubes will be arranged in line. The pipe wall is 5-mm thick; the thermal conductivity of the metal is 50 W m^{-1} °C^{-1}. The maximum linear shell-side fluid velocity is estimated as 0.15 m s^{-1}. Estimate the tube length required by carrying out the following calculations.

(a) What is the rate of heat transfer?
(b) Calculate individual heat-transfer coefficients for the tube- and shell-side fluids.
(c) Calculate the overall heat-transfer coefficient.
(d) Calculate the log-mean temperature difference.
(e) Determine the heat-transfer area.
(f) What tube length is required?

8.7 Suitability of an existing cooling-coil

An enzyme manufacturer in the same industrial park as your antibiotic factory has a re-conditioned 20 m^3 fermenter for sale. You are in the market for a cheap 20 m^3 fermenter; however the vessel on offer is fitted with a 45-m steel helical cooling-coil with inner pipe-diameter 7.5 cm. You propose to use the fermenter for your newest production organism which is known to have a maximum oxygen demand of 90 mol m^{-3} h^{-1} at its optimum culture temperature of 28°C. You consider that the 3-m diameter vessel should be stirred with a 1-m diameter turbine-impeller operated at an average speed of 50 rpm. The fermentation fluid can be assumed to have the properties of water. If 20 m^3 h^{-1} cooling water at 12°C is available, should you make an offer for the second-hand fermenter and cooling coil?

8.8 Optimum stirring speed for removal of heat from viscous broth

The viscosity of a fermentation broth containing exopolysaccharide is about 10 000 cP. The broth is stirred in an aerated 10 m^3 fermenter of diameter 2.3 m using a single 0.78-m diameter turbine impeller. Other properties of the broth are as follows:

$C_p = 2\ \mathrm{kJ\,kg^{-1}\,°C^{-1}}$
$k_{fb} = 2\ \mathrm{W\,m^{-1}\,°C^{-1}}$
$\rho = 10^3\ \mathrm{kg\,m^{-3}}$.

The fermenter is equipped with an internal cooling-coil which provides a heat-transfer area of 14 m^2. Cooling water is provided; the average temperature difference for heat transfer is 20°C. Neglect any variation of viscosity at the wall of the coil. Assume that the power dissipated in aerated broth is 40% lower than in ungassed liquid.

(a) Using logarithmic coordinates, plot $\hat{Q}$ as a function of stirrer speed between 0.5 and 10 s^{-1}.
(b) From equations presented in Section 7.10, calculate $\hat{W}_s$, the power dissipated from the stirrer, as a function of stirrer speed. Plot these values on the same graph as $\hat{Q}$.
(c) If evaporation, heat losses and other factors have only a negligible effect on heat load, the difference between $\hat{Q}$ and $\hat{W}_s$ is equal to the rate of metabolic heat removal from the fermenter, $\Delta\hat{H}_{rxn}$. Plot $\Delta\hat{H}_{rxn}$ as a function of stirrer speed.
(d) At what stirrer speed is removal of metabolic heat most rapid?
(e) The specific rate of oxygen consumption is 6 mmol g^{-1} h^{-1}. If the fermenter is operated at the stirrer speed identified in (d), what is the maximum cell concentration?
(f) How do you interpret the intersection of the curves for $\hat{Q}$ and $\hat{W}_s$ at high stirrer speed in terms of the capacity of the system to handle exothermic reactions?

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Suggestions for Further Reading

Heat-Transfer Theory (see also refs 1, 2, 4 and 6)

Kern, D.Q. (1950) *Process Heat Transfer*, McGraw-Hill, Tokyo.

Heat-Transfer Equipment

Perry, R.H., D.W. Green and J.O. Maloney (Eds) (1984) *Chemical Engineers' Handbook*, 6th edn, Section 11, McGraw-Hill, New York.

Heat Transfer in Bioprocessing

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9

Mass Transfer

Mass transfer occurs in mixtures containing local concentration variations. For example, when dye is dropped into a pail of water, mass-transfer processes are responsible for movement of dye molecules through the water until equilibrium is established and the concentration is uniform. Mass is transferred from one location to another under the influence of a concentration difference or concentration gradient in the system.

There are many situations in bioprocessing where concentrations of compounds are not uniform; we rely on mechanisms of mass transfer to transport material from regions of high concentration to regions where the concentration is low. An example is the supply of oxygen in fermenters for aerobic culture. Concentration of oxygen at the surface of air bubbles is high compared with the rest of the fluid; this concentration gradient promotes oxygen transfer from the bubbles into the medium. Another example of mass transfer is extraction of penicillin from fermentation liquor using organic solvents such as butyl acetate. When solvent is added to the broth, the relatively low concentration of penicillin in the organic phase causes mass transfer of penicillin into the solvent. Solvent extraction is an efficient downstream-processing technique as it selectively removes the desired product from the rest of the fermentation fluid.

Mass transfer plays a vital role in many reaction systems. As distance between the reactants and site of reaction becomes greater, rate of mass transfer is more likely to influence or control the conversion rate. Taking again the example of oxygen in aerobic culture, if mass transfer of oxygen from the bubbles is slow, the rate of cell metabolism will become dependent on the rate of oxygen supply from the gas phase. Because oxygen is a critical component of aerobic fermentations and is so sparingly soluble in aqueous solutions, much of our interest in mass transfer lies with the transfer of oxygen across gas–liquid interfaces. However, liquid–solid mass transfer can also be important in systems containing clumps, pellets, flocs or films of cells or enzymes; in these cases, nutrients in the liquid phase must be transported into the solid before they can be utilised in reaction. Unless mass transfer is rapid, supply of nutrients will limit the rate of biological conversion.

In a solid or quiescent fluid, mass transfer occurs as a result of molecular diffusion. However, most mass-transfer systems contain moving fluid; in turbulent flow, mass transfer by molecular motion is supplemented by convective transfer. There is an enormous variety of circumstances in which convective mass transfer takes place. In this chapter, we will consider the theory of mass transfer with applications relevant to the bioprocessing industry.

9.1 Molecular Diffusion

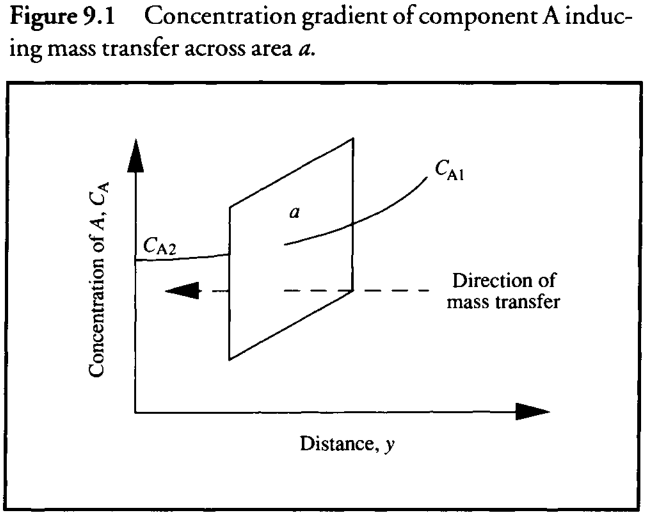
Molecular diffusion is the movement of component molecules in a mixture under the influence of a concentration difference in the system. Diffusion of molecules occurs in the direction required to destroy the concentration gradient. If the gradient is maintained by constantly supplying material to the region of high concentration and removing it from the region of low concentration, diffusion will be continuous. This situation is often exploited in mass-transfer operations and reaction systems.

9.1.1 Diffusion Theory

In this text, we confine our discussion of diffusion to *binary mixtures*, i.e. mixtures or solutions containing only two components. Consider a system containing molecular components A and B. Initially, the concentration of A in the system is not uniform; as indicated in Figure 9.1, concentration C_A varies from C_{A1} to C_{A2} is a function of distance y. In response to this concentration gradient, molecules of A will diffuse away from the region of high concentration until eventually the whole system acquires uniform composition. If there is no large-scale fluid motion in the system, e.g. due to stirring, mixing occurs solely by random molecular movement.

Assume that mass transfer of A occurs across area a perpendicular to the direction of diffusion. In single-phase systems, the rate of mass transfer due to molecular diffusion is given by *Fick's law of diffusion*, which states that mass flux is proportional to the concentration gradient:

Figure 9.1 Concentration gradient of component A inducing mass transfer across area a.



$$J_A = \frac{N_A}{a} = -\mathscr{D}_{AB} \frac{dC_A}{dy} . \tag{9.1}$$

In Eq. (9.1), J_A is the *mass flux* of component A, N_A is the *rate of mass transfer* of component A, a is the area across which mass transfer occurs, $\mathscr{D}_{AB}$ is the ***binary diffusion coefficient*** or ***diffusivity*** of component A in a mixture of A and B, C_A is the concentration of component A, and y is distance. dC_A/dy is the concentration gradient, or change in concentration of A with distance. As indicated in Eq. (9.1), mass flux is defined as the rate of mass transfer per unit area perpendicular to the direction of movement; J_A has units of, e.g. gmol s^{-1} m^{-2}. Corresponding units for N_A are gmol s^{-1}, for C_A gmol m^{-3}, and for $\mathscr{D}_{AB}$ m^2 s^{-1}. Mass rather than mole units may be used for J_A, N_A and C_A; Eq. (9.1) holds in either case. Eq. (9.1) indicates that rate of diffusion can be enhanced by increasing the area available for mass transfer, the concentration gradient in the system, and the magnitude of the diffusion coefficient. The negative sign in Eq. (9.1) indicates that the direction of mass transfer is always from high concentration to low concentration, opposite to that of the concentration gradient. In other words, if the slope of C_A versus y is positive as in Figure 9.1, the direction of mass transfer is in the negative y-direction, and vice versa.

The diffusion coefficient $\mathscr{D}_{AB}$ is a property of materials; values can be found in handbooks. $\mathscr{D}_{AB}$ reflects the ease with which diffusion takes place. Its value depends on both components of the mixture; for example, the diffusivity of carbon dioxide in water will be different from the diffusivity of carbon dioxide in another solvent such as ethanol. The value of $\mathscr{D}_{AB}$ is also dependent on temperature. Diffusivity of gases varies with pressure; for liquids there is an approximate linear dependence on concentration. Diffusivity values in liquids are several orders of magnitude smaller than in gases. As examples, $\mathscr{D}_{AB}$ for oxygen in air at 0°C and 1 atm is 1.78×10^{-5} m^2 s^{-1}; $\mathscr{D}_{AB}$ is 2.5×10^{-9} m^2 s^{-1} for oxygen in water at 25°C and 1 atm, and 6.9×10^{-10} m^2 s^{-1} for glucose in dilute solution in water at 25°C.

When diffusivity values are not available for the exact temperature and pressure of interest, $\mathscr{D}_{AB}$ can be estimated using equations. Relationships for calculating diffusivities are available from other references [1–3]. The theory of diffusion in liquids is not as well advanced as with gases; there are also fewer experimental data available for liquid systems.

9.1.2 Analogy Between Mass, Heat and Momentum Transfer

There is a close similarity between the processes of mass, heat and momentum transfer occurring as a result of molecular motion. This is suggested by the form of the equations for mass, heat and momentum fluxes:

$$J_A = -\mathscr{D}_{AB} \frac{dC_A}{dy} \tag{9.2}$$

$$\hat{q} = -k \frac{dT}{dy} \tag{8.2}$$

and

$$\tau = -\mu \frac{dv}{dy} . \tag{7.6}$$

The three processes represented above are quite different on the molecular level, but the basic equations have the same form. In each case, flux in the y-direction is directly proportional to the driving force (either dC_A/dy, dT/dy or dv/dy), with the proportionality constant ($\mathscr{D}_{AB}$, k or μ) a physical property of the material. The negative signs in Eqs (9.2), (8.2) and (7.6) indicate that transfer of mass, heat or momentum is always in the direction opposite to that of increasing concentration, temperature or velocity. The similarity in the form of the three rate equations makes it possible in some situations to apply analysis of one process to either of the other two.

The analogy of Eqs (9.2), (8.2) and (7.6) is valid for transport of mass, heat and momentum resulting from motion or vibration of molecules. Extension of the analogy to turbulent flow is generally valid for heat and mass transfer; however the analogy with momentum transfer presents a number of difficulties. Several analogy theories have been proposed in the chemical engineering literature to describe simultaneous transport phenomena in turbulent systems. Details are presented elsewhere [2, 4, 5].

9.2 Role of Diffusion in Bioprocessing

Fluid mixing is carried out in most industrial processes where mass transfer takes place. Bulk fluid motion causes more rapid large-scale mixing than molecular diffusion; why then is diffusive transport still important? Areas of bioprocessing in which diffusion plays a major role are described below.

(i) *Scale of mixing.* As discussed in Section 7.9.3, turbulence in fluids produces bulk mixing on a scale equal to the smallest eddy size. Within the smallest eddies, flow is largely streamline so that further mixing must occur by diffusion of fluid components. Mixing on a molecular scale therefore relies on diffusion as the final step in the mixing process.

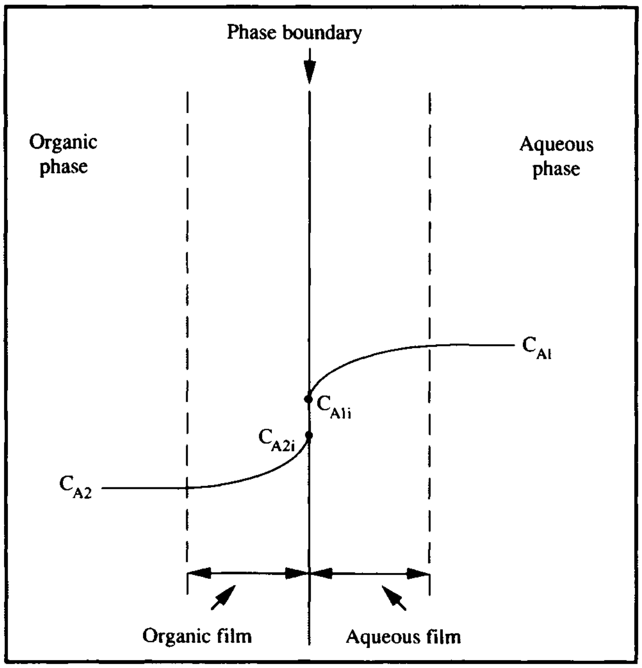
(ii) *Solid-phase reaction.* In biological systems, reactions are sometimes mediated by catalysts in solid form, e.g. clumps, flocs and films of cells and immobilised-enzyme and -cell particles. When cells or enzyme molecules are clumped together into a solid particle, substrates must be transported into the solid before reaction can take place. Mass transfer within solid particles is usually unassisted by bulk fluid convection; the only mechanism for intraparticle mass transfer is molecular diffusion. As the reaction proceeds, diffusion is also responsible for removal of product molecules away from the site of reaction. As discussed more fully in Chapter 12, when reaction is coupled with diffusion, the overall reaction rate can be significantly reduced if diffusion is slow.

(iii) *Mass transfer across a phase boundary.* Mass transfer between phases occurs often in bioprocessing. Oxygen transfer from gas bubbles to fermentation broth, penicillin recovery from aqueous to organic liquid, and glucose transfer from liquid medium into mould pellets are typical examples. When different phases come into contact, fluid velocity near the phase interface is significantly decreased and diffusion becomes crucial for mass transfer across the phase interface. This is discussed further in the next section.

9.3 Film Theory

The *two-film theory* is a useful model for mass transfer between phases. Mass transfer of solute from one phase to another involves transport from the bulk of one phase to the *phase boundary* or *interface*, and then from the interface to the bulk of the second phase. The film theory is based on the idea that a fluid film or *mass-transfer boundary layer* forms wherever there is contact between two phases.

Figure 9.2 Film resistance to mass transfer between two immiscible liquids.



Let us consider mass transfer of component A across the phase interface represented in Figure 9.2. Assume that the phases are two immiscible liquids such as water and chloroform, and that A is initially at higher concentration in the aqueous phase than in the organic phase. Each phase is well mixed and in turbulent flow. The concentration of A in the bulk aqueous phase is C_{A1}; the concentration of A in the bulk organic phase is C_{A2}.

According to the film theory, turbulence in each fluid dies out at the phase boundary. A thin film of relatively stagnant fluid exists on either side of the interface; mass transfer through this film is effected solely by molecular diffusion. The concentration of A changes near the interface as indicated in Figure 9.2; C_{A1i} is the interfacial concentration of A in the aqueous phase; C_{A2i} is the interfacial concentration of A in the organic phase. Most of the resistance to mass transfer resides in the liquid films rather than in the bulk liquid. For practical purposes it is generally assumed that there is negligible resistance to transport at the interface itself; this is equivalent to assuming that the phases are in equilibrium at the plane of contact. The difference between C_{A1i} and C_{A2i} at the interface accounts for the possibility that, at equilibrium, A may be more soluble in one phase than in the other. For example, if A were acetic acid in contact at the interface with both water and chloroform, the equilibrium concentration in water would be greater than that in chloroform by a factor of between 5 and 10. C_{A1i} would then be significantly higher than C_{A2i}.

Even though the bulk liquids in Figure 9.2 may be well mixed, diffusion of component A is crucial in effecting mass transfer because the local fluid velocities approach zero at the interface. The film theory as described above is applied extensively in analysis of mass transfer, although it is a greatly simplified representation. There are other models of mass transfer in fluids which lead to more realistic mathematical outcomes than the film theory [1, 4]. Nevertheless, irrespective of how mass transfer is visualised, diffusion is always an important mechanism of mass transfer close to the interface between fluids.

9.4 Convective Mass Transfer

The term *convective mass transfer* refers to mass transfer occurring in the presence of bulk fluid motion. Molecular diffusion will occur whenever there is a concentration gradient; however if the bulk fluid is also moving, the overall rate of mass transfer will be higher due to the contribution of convective currents. Analysis of mass transfer is most important in multi-phase systems where interfacial boundary layers provide significant mass-transfer resistance. Let us develop an expression for rate of mass transfer which is applicable to mass-transfer boundary layers.

Rate of mass transfer is directly proportional to the driving force for transfer, and the area available for the transfer process to take place. This can be expressed as:

Transfer rate ∝ (transfer area) × (driving force).

The proportionality coefficient in this equation is called the *mass-transfer coefficient*, so that:

Transfer rate = (mass-transfer coefficient) × (transfer area) × (driving force).

For each fluid on either side of a phase boundary, the driving force for mass transfer can be expressed in terms of a concentration difference. Therefore, rate of mass transfer to a phase boundary is given by the equation:

$$N_A = k\,a\,\Delta C_A = k\,a\,(C_{Ao} - C_{Ai}) \tag{9.3}$$

where N_A is the rate of mass transfer of component A, k is the mass-transfer coefficient, a is the area available for mass transfer, C_{Ao} is the bulk concentration of component A away from the phase boundary, and C_{Ai} is the concentration of A at the interface. Eq. (9.3) is usually used to represent the *volumetric rate of mass transfer*, so units of N_A are, for example, gmol m^{-3} s^{-1}. Consistent with this representation, a is the interfacial area per unit volume with dimensions L^{-1} and units of, for example, m^2 m^{-3} or m^{-1}. The dimensions of the mass-transfer coefficient are LT^{-1}; SI units are m s^{-1}. Eq. (9.3) indicates that the rate of convective mass transfer can be enhanced by increasing the area available for mass transfer, the concentration difference between the bulk fluid and the interface, and the magnitude of the mass-transfer coefficient. By analogy with Eq. (8.11) for heat transfer, Eq. (9.3) can also be written in the form:

$$N_A = \frac{\Delta C_A}{R_m} \tag{9.4}$$

where R_m is the resistance to mass transfer:

$$R_m = \frac{1}{k\,a}. \tag{9.5}$$

Mass transfer coupled with fluid flow is a more complicated process than diffusive mass transfer. The value of the mass-transfer coefficient reflects the contribution to mass transfer from all the processes in the system that affect the boundary layer. Like the heat-transfer coefficient in Chapter 8, k depends on the combined effects of flow velocity, geometry of the mass-transfer system, and fluid properties such as viscosity and diffusivity. Because the hydrodynamics of most practical systems are not easily characterised, k cannot be calculated reliably from first principles. Instead, it is measured experimentally or estimated using correlations available from the literature. In general, reducing the thickness of the boundary layer or improving the diffusion coefficient in the film will result in enhancement of k and improvement in the rate of mass transfer.

Three mass-transfer situations which occur in bioprocessing are *liquid–solid mass transfer*, *liquid–liquid mass transfer* between immiscible solvents and *gas–liquid mass transfer*. Use of Eq. (9.3) to determine the rate of mass transfer in these systems is discussed in the following sections.

9.4.1 Liquid–Solid Mass Transfer

Mass transfer between a moving liquid and a solid is important in biological processing in a variety of applications. Transport of substrates to solid-phase cell or enzyme catalysts has already been mentioned. Adsorption of molecules onto surfaces, such as in chromatography, requires transport from liquid phase to solid; liquid–solid mass transfer is also important in crystallisation as molecules move from the liquid to the face of the growing crystal. Conversely, the process of dissolving a solid in liquid requires liquid–solid mass transfer directed away from the solid surface.

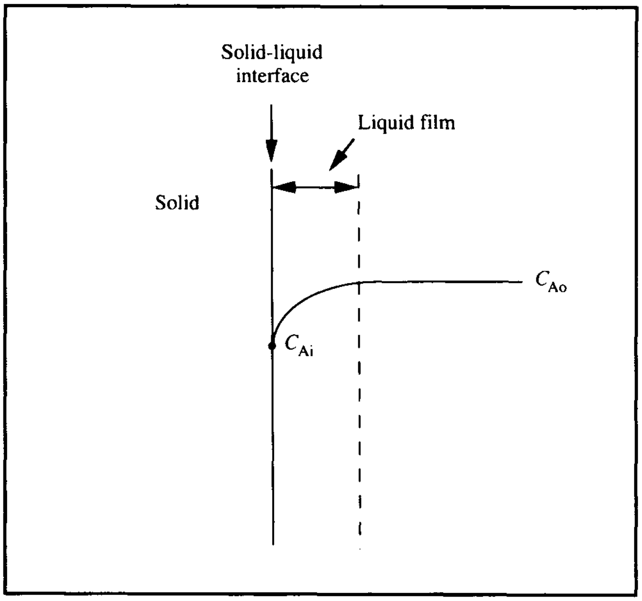
Let us assume that component A is required for reaction at the surface of a solid. The situation at the interface between flowing liquid containing A and the solid is illustrated in Figure 9.3. Near the interface, the fluid velocity is reduced and a boundary layer develops. As A is consumed by reaction, the local concentration of A at the surface decreases and a concentration gradient is established through the film. The concentration difference between the bulk liquid and the phase interface drives mass transfer of A from the liquid to the solid, allowing the reaction to continue. If the solid is non-porous, A does not penetrate further than the surface. The concentration of A at the phase boundary is C_{Ai}; the concentration of A in the bulk liquid outside the film is C_{Ao}. If a is the liquid–solid interfacial area per unit volume, the volumetric rate of mass transfer can be determined from Eq. (9.3) as:

$$N_A = k_L a (C_{Ao} - C_{Ai}) \tag{9.6}$$

where k_L is the ***liquid-phase mass-transfer coefficient.***

Application of Eq. (9.6) requires knowledge of the mass-transfer coefficient, the interfacial area between the phases, the bulk concentration of A, and the concentration of A at the interface. Bulk compositions are generally easy to measure; in simple cases, area a can be determined from the shape and size of the solid. The value of the mass-transfer coefficient is either measured or calculated using published correlations. Estimation of the interfacial concentration C_{Ai} is more difficult; measuring compositions at phase boundaries is not easy experimentally. To overcome this problem, we must consider the processes in the system which are linked to mass transfer of A. In the example of Figure 9.3, transport of A is linked to reaction at the surface of the solid, so that the value of C_{Ai} will depend on the rate of consumption of A at the interface. In practical terms, we can therefore calculate the rate of mass transfer of A only if we have information about the rate of reaction at the solid surface. Simultaneous reaction and mass transfer occurs in many bioprocesses; Chapter 12 treats solid-phase reaction coupled with mass transfer in more detail.

Figure 9.3 Concentration gradient for liquid–solid mass transfer.



9.4.2 Liquid–Liquid Mass Transfer

Liquid–liquid mass transfer between immiscible solvents is most often encountered in the product-recovery stages of bioprocessing. Organic solvents are used to isolate antibiotics, steroids and alkaloids from fermentation broths; two-phase aqueous systems are used in protein purification. Liquid–liquid mass transfer is also important when hydrocarbons are used as substrates in fermentation, e.g. in production of microbial biomass for single-cell protein.

The situation at the interface between two immiscible liquids is shown in Figure 9.2. Component A is present at bulk concentration C_{A1} in one liquid phase; this concentration falls to C_{A1i} at the interface. In the other liquid, the concentration of A falls from C_{A2i} at the interface to C_{A2} in the bulk. The rate of mass transfer N_A in each liquid phase can be obtained from Eq. (9.3):

$$N_{A1} = k_{L1} a (C_{A1} - C_{A1i}) \tag{9.7}$$

and

$$N_{A2} = k_{L2} a (C_{A2i} - C_{A2}) \tag{9.8}$$

where k_L is the liquid-phase mass-transfer coefficient, and subscripts 1 and 2 refer to the two liquid phases. As mentioned already in Section 9.4.1, application of Eqs (9.7) and (9.8) is difficult because we cannot easily measure interfacial concentrations. However, in this case these terms can be eliminated by considering the physical situation at the interface and manipulating the equations.

First, let us recognise that at steady state, because there can be no accumulation of A at the interface or anywhere else in the system, any A transported through liquid 1 must also be transported through liquid 2. This means that N_{A1} in Eq. (9.7) must be equal to N_{A2} in Eq. (9.8); we will call $N_{A1} = N_{A2} = N_A$. We can then rearrange Eqs (9.7) and (9.8):

$$\frac{N_A}{k_{L1}a} = C_{A1} - C_{A1i} \tag{9.9}$$

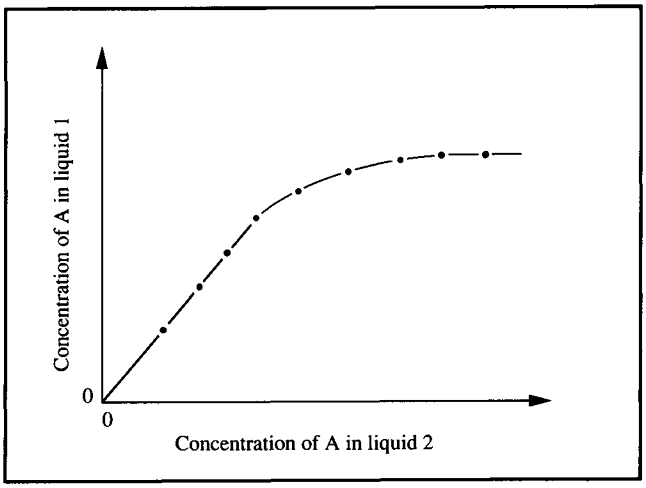
and

$$\frac{N_A}{k_{L2}a} = C_{A2i} - C_{A2}. \tag{9.10}$$

Normally, it can be assumed that there is negligible resistance to mass transfer at the actual interface, i.e. within distances corresponding to molecular free paths on either side of the phase boundary. This is equivalent to assuming that the phases are in equilibrium at the interface; therefore, C_{A1i} and C_{A2i} are equilibrium concentrations. The assumption of phase-boundary equilibrium has been subjected to many tests. As a result it is known that there are special situations, such as when there is adsorption of material at the interface, where the assumption is invalid. However, in ordinary situations, the evidence is that equilibrium does exist at the interface between phases. Note that we are not proposing to relate bulk concentrations C_{A1} and C_{A2} using equilibrium relationships, only C_{A1i} and C_{A2i}. If the bulk liquids were in equilibrium, no nett mass transfer would take place.

A typical equilibrium curve relating concentrations of solute A in two immiscible liquid phases is shown in Figure 9.4. The points making up the curve are obtained readily from experiments; alternatively equilibrium data can be found in handbooks. Equilibrium distribution of one solute between two phases is conveniently described in terms of the *distribution law*. At equilibrium, the ratio of solute concentrations in the two phases is given by the *distribution coefficient* or *partition coefficient*, *m*. As shown in Figure 9.4, when the concentration of A is low, the equilibrium curve is approximately a straight line, so that *m* is constant. The distribution law is accurate only if both solvents are immiscible and there is no chemical reaction.

Figure 9.4 Equilibrium curve for solute A in two immiscible solvents 1 and 2.



Therefore if C_{A1i} and C_{A2i} are equilibrium concentrations, they can be related using the distribution coefficient *m*.

$$m = \frac{C_{A1i}}{C_{A2i}} \tag{9.11}$$

i.e.

$$C_{A1i} = mC_{A2i} \tag{9.12}$$

or, alternatively:

$$C_{A2i} = \frac{C_{A1i}}{m}. \tag{9.13}$$

Eqs (9.12) and (9.13) can now be used to eliminate the interfacial concentrations from Eqs (9.9) and (9.10). First, we make a direct substitution:

$$\frac{N_A}{k_{L1}a} = C_{A1} - mC_{A2i} \tag{9.14}$$

and

$$\frac{N_A}{k_{L2}a} = \frac{C_{A1i}}{m} - C_{A2}. \tag{9.15}$$

If we now multiply Eq. (9.10) by m:

$$\frac{mN_A}{k_{L2}a} = mC_{A2i} - mC_{A2} \tag{9.16}$$

and divide Eq. (9.9) by m:

$$\frac{N_A}{m\,k_{L1}a} = \frac{C_{A1}}{m} - \frac{C_{A1i}}{m} \tag{9.17}$$

and add Eq. (9.14) to Eq. (9.16), and Eq. (9.15) to Eq. (9.17), we eliminate the interfacial-concentration terms completely:

$$N_A\left(\frac{1}{k_{L1}a}+\frac{m}{k_{L2}a}\right) = C_{A1} - mC_{A2} \tag{9.18}$$

$$N_A\left(\frac{1}{m\,k_{L1}a}+\frac{1}{k_{L2}a}\right) = \frac{C_{A1}}{m} - C_{A2}. \tag{9.19}$$

Eqs (9.18) and (9.19) combine mass-transfer resistances in the two liquid films, and relate the rate of mass transfer N_A to the bulk fluid concentrations C_{A1} and C_{A2}. The bracketed terms for the combined mass-transfer coefficients are used to define the *overall liquid-phase mass-transfer coefficient*, K_L. Depending on the form used for the concentration difference, we can define two overall mass-transfer coefficients:

$$\frac{1}{K_{L1}a} = \left(\frac{1}{k_{L1}a}+\frac{m}{k_{L2}a}\right) \tag{9.20}$$

and

$$\frac{1}{K_{L2}a} = \left(\frac{1}{m\,k_{L1}a}+\frac{1}{k_{L2}a}\right) \tag{9.21}$$

where K_{L1} is the overall mass-transfer coefficient based on the bulk concentration in liquid 1, and K_{L2} is the overall mass-transfer coefficient based on the bulk concentration in liquid 2.

We can now summarise the results to obtain two equations for the mass-transfer rate in liquid–liquid systems:

$$N_A = K_{L1}a(C_{A1} - mC_{A2}) \tag{9.22}$$

and

$$N_A = K_{L2}a\left(\frac{C_{A1}}{m} - C_{A2}\right) \tag{9.23}$$

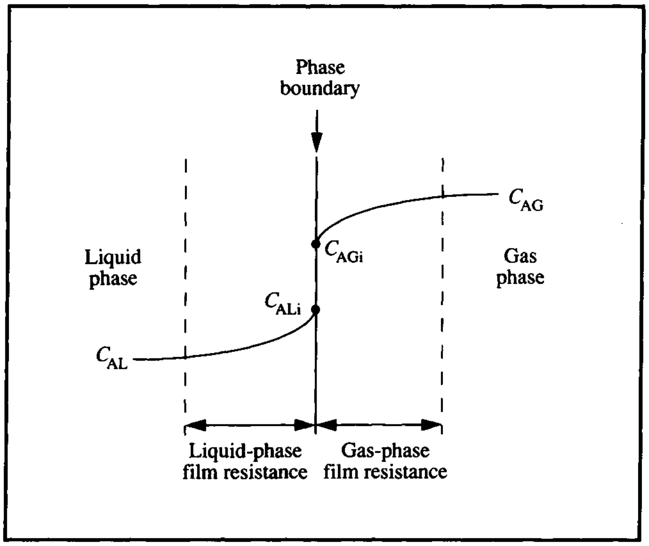
where K_{L1} and K_{L2} are given by Eqs (9.20) and (9.21). Use of either of these two equations requires knowledge of the concentrations of A in the bulk fluids, the partition coefficient m, the interfacial area a between the two liquid phases, and the value of either K_{L1} or K_{L2}. C_{A1} and C_{A2} are generally easy to measure. m can also be measured, or is found in handbooks of physical properties. The overall mass-transfer coefficients can be measured experimentally, or are estimated from published correlations for k_{L1} and k_{L2} in the literature. The only remaining parameter is the interfacial area, a. In many applications of liquid–liquid mass transfer it is difficult to know how much interfacial area is available between phases. For example, liquid–liquid extraction is often carried out in stirred tanks where an impeller is used to disperse and mix droplets of one phase through the other. The interfacial area under these circumstances will depend on the size, shape and number of the droplets, which will in turn depend on the intensity of agitation and properties of the fluid. Because these factors also affect the value of k_L, correlations for mass-transfer coefficients are often given in terms of k_La as a combined parameter.

Eqs (9.22) and (9.23) indicate that the rate of mass transfer between two phases is not dependent simply on the concentration difference; the equilibrium relationship is also an important factor. According to Eq. (9.22), the driving force for transfer of A out of liquid 1 is the difference between the bulk concentration C_{A1} *and the concentration of A in liquid 1 which would be in equilibrium with concentration* C_{A2} *in liquid 2.* Similarly, the driving force for mass transfer according to Eq. (9.23) is the difference between C_{A2} and the concentration of A in liquid 2 which would be in equilibrium with C_{A1} in liquid 1.

9.4.3 Gas–Liquid Mass Transfer

Gas–liquid mass transfer is of paramount importance in bioprocessing because of the requirement for oxygen in aerobic fermentations. Transfer of a solute such as oxygen from gas to

Figure 9.5 Concentration gradients for gas–liquid mass transfer.



liquid is analysed in a similar way to liquid–liquid and liquid–solid mass transfer.

Figure 9.5 shows the situation at an interface between gas and liquid phases containing component A. Let us assume that A is transferred from the gas phase into the liquid. The concentration of A in the liquid is C_{AL} in the bulk and C_{ALi} at the interface. In the gas, the concentration is C_{AG} in the bulk and C_{AGi} at the interface.

From Eq. (9.3), the rate of mass transfer of A through the gas boundary layer is:

$$N_{AG} = k_G\, a\,(C_{AG} - C_{AGi}) \tag{9.24}$$

and the rate of mass transfer of A through the liquid boundary layer is:

$$N_{AL} = k_L\, a\,(C_{ALi} - C_{AL}) \tag{9.25}$$

where k_G is the gas-phase mass-transfer coefficient and k_L is the liquid-phase mass-transfer coefficient. To eliminate C_{AGi} and C_{ALi}, we must manipulate the equations as in Section 9.4.2.

If we assume that equilibrium exists at the interface, C_{AGi} and C_{ALi} can be related. For dilute concentrations of most gases and for a wide range of concentration for some gases, equilibrium concentration in the gas phase is a linear function of liquid concentration. Therefore, we can write:

$$C_{AGi} = m C_{ALi} \tag{9.26}$$

or, alternatively:

$$C_{ALi} = \frac{C_{AGi}}{m} \tag{9.27}$$

where m is the distribution factor. These equilibrium relationships can be incorporated into Eqs (9.24) and (9.25) at steady state using procedures which parallel those already used for liquid–liquid mass transfer. The results are also similar:

$$N_A \left(\frac{1}{k_G a} + \frac{m}{k_L a} \right) = C_{AG} - m C_{AL} \tag{9.28}$$

$$N_A \left(\frac{1}{m\, k_G a} + \frac{1}{k_L a} \right) = \frac{C_{AG}}{m} - C_{AL}. \tag{9.29}$$

The combined mass-transfer coefficients in Eqs (9.28) and (9.29) can be used to define overall mass-transfer coefficients. The *overall gas-phase mass-transfer coefficient* K_G is defined by the equation:

$$\frac{1}{K_G a} = \frac{1}{k_G a} + \frac{m}{k_L a} \tag{9.30}$$

and the *overall liquid-phase mass-transfer coefficient* K_L is defined as

$$\frac{1}{K_L a} = \frac{1}{m\, k_G a} + \frac{1}{k_L a}. \tag{9.31}$$

The rate of mass transfer in gas–liquid systems can therefore be expressed using either of two equations:

$$N_A = K_G\, a\,(C_{AG} - m C_{AL}) \tag{9.32}$$

or

$$N_A = K_L\, a \left(\frac{C_{AG}}{m} - C_{AL} \right). \tag{9.33}$$

Eqs (9.32) and (9.33) are usually expressed using equilibrium concentrations. mC_{AL} is equal to C^*_{AG}, the gas-phase concentration of A in equilibrium with C_{AL}, and $(C_{AG}/_m)$ is equal to C^*_{AL}, the liquid-phase concentration of A in equilibrium with C_{AG}. Eqs (9.32) and (9.33) become:

$$N_A = K_G a (C_{AG} - C^*_{AG}) \tag{9.34}$$

and

$$N_A = K_L a (C^*_{AL} - C_{AL}). \tag{9.35}$$

In real mass-transfer systems, obtaining the values of C_{AG}, C_{AL} and m in Eqs (9.32) and (9.33) is reasonably straightforward. However, as in liquid–liquid mass-transfer systems, it is generally difficult to evaluate the interfacial area a. When gas is sparged through a liquid the interfacial area will depend on the size and number of bubbles present, which in turn depend on many other factors such as medium composition, stirrer speed and gas flow rate. Mass-transfer coefficients are measured experimentally or estimated using empirical correlations from the literature.

Eqs (9.34) and (9.35) can be simplified for systems in which most of the resistance to mass transfer lies in either the gas-phase interfacial film or the liquid-phase film. When solute A is very soluble in the liquid, for example in transfer of ammonia to water, the liquid-side resistance is small compared with that posed by the gas interfacial film. Therefore, from Eq. (9.5), if the liquid-side resistance is small $k_L a$ must be relatively large; from Eq. (9.30), $K_G a$ is then approximately equal to $k_G a$. Using this result in Eq. (9.34) gives:

$$N_A = k_G a (C_{AG} - C^*_{AG}). \tag{9.36}$$

Conversely, if A is poorly soluble in the liquid, e.g. oxygen in aqueous solution, the liquid-phase mass-transfer resistance dominates and $k_G a$ is much larger than $k_L a$. From Eq. (9.31), this means that $K_L a$ is approximately equal to $k_L a$, and Eq. (9.35) can be simplified to:

$$N_A = k_L a (C^*_{AL} - C_{AL}). \tag{9.37}$$

9.5 Oxygen Uptake in Cell Cultures

Cells in aerobic culture take up oxygen from the liquid. The rate of oxygen transfer from gas to liquid is therefore of prime importance, especially at high cell densities when cell growth is likely to be limited by availability of oxygen in the medium. An expression for rate of oxygen transfer from gas to liquid is given by Eq. (9.37); N_A is the rate of oxygen transfer per unit volume of fluid (gmol m^{-3} s^{-1}), k_L is the liquid-phase mass-transfer coefficient (m s^{-1}), a is the gas–liquid interfacial area per unit volume of fluid (m^2 m^{-3}), C_{AL} is the oxygen concentration in the broth (gmol m^{-3}), and C^*_{AL} is the oxygen concentration in the broth in equilibrium with the gas phase (gmol m^{-3}). The equilibrium concentration C^*_{AL} is also known as the *solubility* of oxygen in the broth. The difference $(C^*_{AL} - C_{AL})$ between the maximum possible and actual oxygen concentrations in the liquid represents the *concentration-difference driving force* for mass transfer.

The solubility of oxygen in aqueous solutions at ambient temperature and pressure is only about 10 ppm. This amount of oxygen is quickly consumed in aerobic cultures and must be constantly replaced by sparging. For an actively respiring yeast population with a cell density of 10^9 cells per ml, it can be calculated that the oxygen content of the broth must be replaced about 12 times per minute to keep up with cellular oxygen demand [6]. This is not an easy task because the low solubility of oxygen guarantees that the concentration difference $(C^*_{AL} - C_{AL})$ is always very small. Design of fermenters for aerobic operation must take these factors into account and provide optimum mass-transfer conditions.

9.5.1 Factors Affecting Cellular Oxygen Demand

The rate at which oxygen is consumed by cells in fermenters determines the rate at which it must be transferred from gas to liquid. Many factors influence oxygen demand; the most important of these are cell species, culture growth phase, and nature of the carbon source in the medium. In batch culture, rate of oxygen uptake varies with time. The reasons for this are twofold. First, the concentration of cells increases during the course of batch culture and the total rate of oxygen uptake is proportional to the number of cells present. In addition, the rate of oxygen consumption per cell, known as the *specific oxygen uptake rate*, also varies. Typically, specific oxygen demand passes through a maximum in early exponential phase as illustrated in Figure 9.6, even though the cell concentration is relatively small at that time. If Q_O is the oxygen uptake rate per volume of broth and q_O is the specific oxygen uptake rate:

$$Q_O = q_O x \tag{9.38}$$

where x is cell concentration. Typical units for q_O are g g^{-1} s^{-1}, and for Q_O, g l^{-1} s^{-1}.

Figure 9.6 Variation in specific rate of oxygen consumption and biomass concentration during batch culture. (From R.T. Darby and D.R. Goddard, 1950, Studies of the respiration of the mycelium of the fungus *Myrothecium verrucaria. Am. J. Bot.* 37, 379–387.)

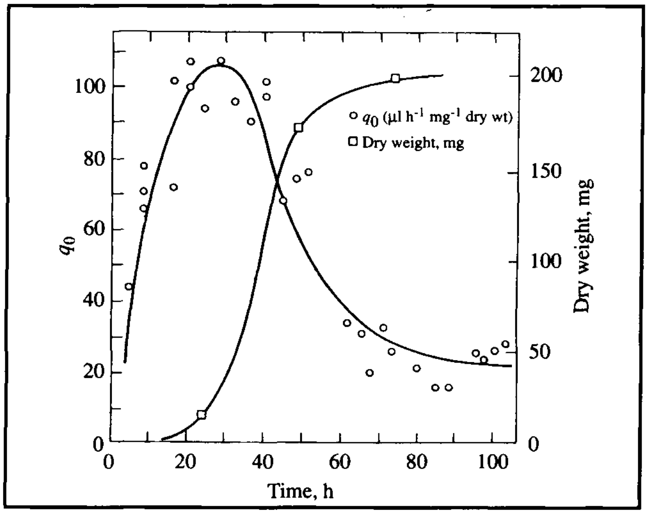
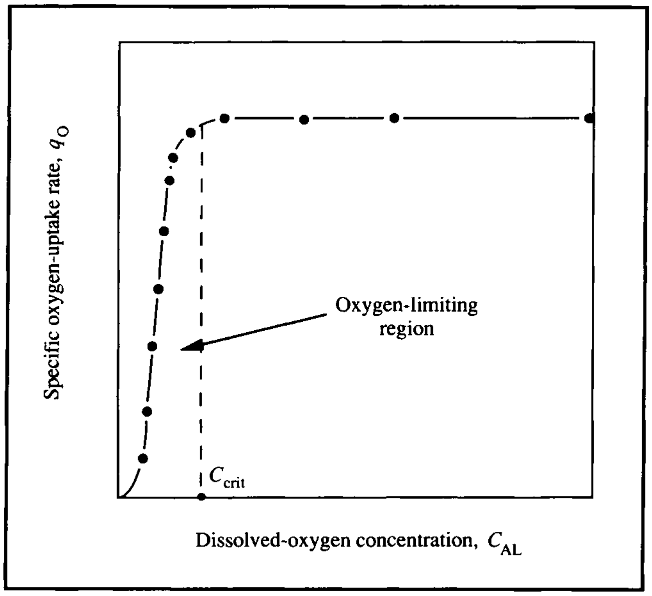


Figure 9.7 Relationship between specific rate of oxygen consumption by cells and dissolved-oxygen concentration.



The inherent demand of an organism for oxygen (q_O) depends primarily on the biochemical nature of the cell and its nutritional environment. However, when the level of dissolved oxygen in the medium falls below a certain point, the specific rate of oxygen uptake is also dependent on the oxygen concentration in the liquid. The dependence of q_O on C_{AL} is shown in Figure 9.7. If C_{AL} is above the *critical oxygen concentration* C_{crit}, q_O is a constant maximum and independent of C_{AL}. If C_{AL} is below C_{crit}, q_O is approximately linearly dependent on oxygen concentration.

To eliminate oxygen limitations and allow cell metabolism to function at its fastest, the dissolved-oxygen concentration at every point in the fermenter must be above C_{crit}. The exact value of C_{crit} depends on the organism, but under average operating conditions usually falls between 5% and 10% of air saturation. For cells with relatively high C_{crit} levels, the task of transferring sufficient oxygen to maintain $C_{AL} > C_{crit}$ is always more challenging than for cultures with low C_{crit} values.

Choice of substrate for the fermentation can also significantly affect oxygen demand. Because glucose is generally consumed more rapidly than other sugars or carbon-containing substrates, rates of oxygen demand are higher when glucose is used. For example, maximum oxygen-consumption rates of 5.5, 6.1 and 12 mmol l^{-1} h^{-1} have been observed for *Penicillium* mould growing on lactose, sucrose and glucose, respectively [7].

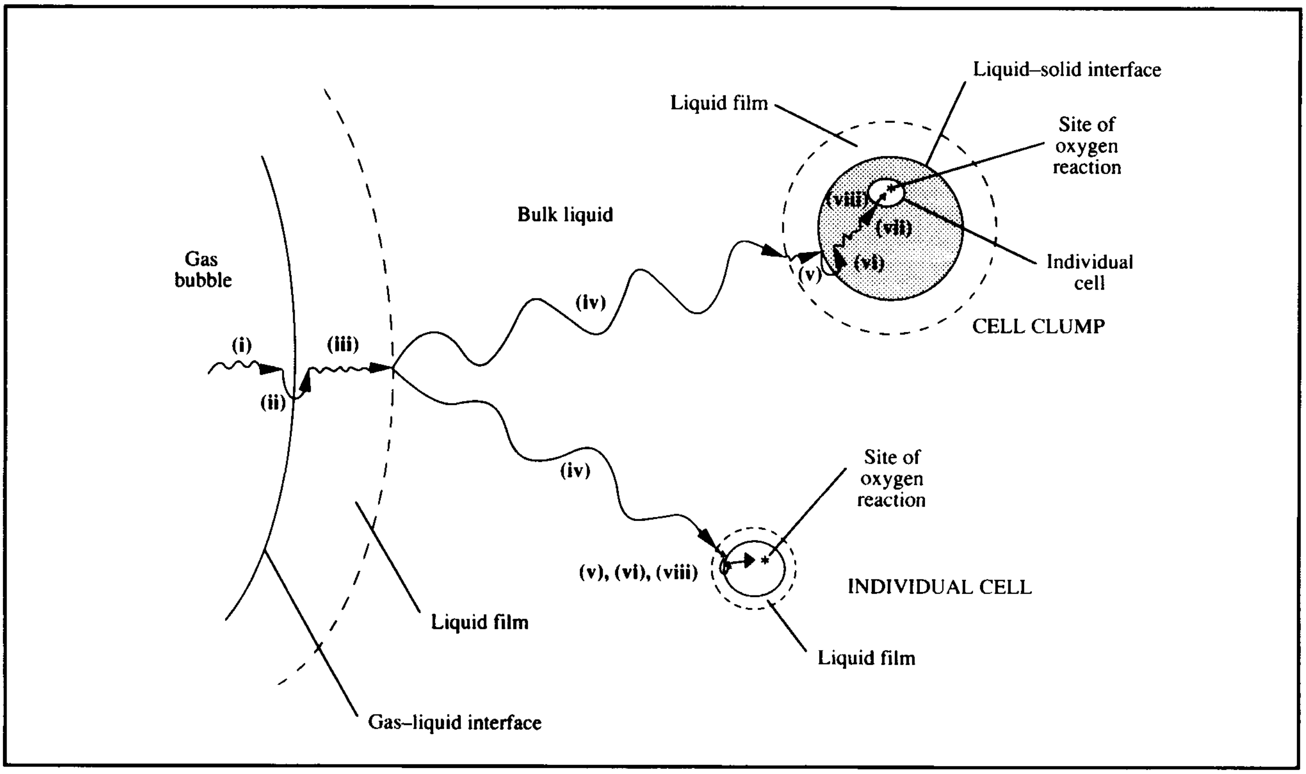
9.5.2 Oxygen Transfer From Gas Bubble to Cell

In aerobic fermentation, oxygen molecules must overcome a series of transport resistances before being utilised by the cells. Eight mass-transfer steps involved in transport of oxygen from the interior of gas bubbles to the site of intracellular reaction are represented diagrammatically in Figure 9.8. They are:

(i) transfer from the interior of the bubble to the gas–liquid interface;
(ii) movement across the gas–liquid interface;
(iii) diffusion through the relatively stagnant liquid film surrounding the bubble;
(iv) transport through the bulk liquid;
(v) diffusion through the relatively stagnant liquid film surrounding the cells;
(vi) movement across the liquid–cell interface;
(vii) if the cells are in a floc, clump or solid particle, diffusion through the solid to the individual cell; and
(viii) transport through the cytoplasm to the site of reaction.

Note that resistance due to the gas boundary layer on the inside of the bubble has been neglected; because of the low solubility of oxygen in aqueous solutions, we can assume that the liquid-film resistance dominates gas–liquid mass transfer (see

Figure 9.8 Steps for transfer of oxygen from gas bubble to cell.



Section 9.4.3). If the cells are individually suspended in liquid rather than in a clump, step (vii) disappears.

The relative magnitudes of the various mass-transfer resistances depend on the composition and rheological properties of the liquid, mixing intensity, bubble size, cell-clump size, interfacial adsorption characteristics and other factors. For most bioreactors the following analysis is valid.

(i) Transfer through the bulk gas phase in the bubble is relatively fast.

(ii) The gas–liquid interface itself contributes negligible resistance.

(iii) The liquid film around the bubbles is a major resistance to oxygen transfer.

(iv) In a well-mixed fermenter, concentration gradients in the bulk liquid are minimised and mass-transfer resistance in this region is small. However, rapid mixing can be difficult to achieve in viscous fermentation broths; if this is the case, oxygen-transfer resistance in the bulk liquid may be important.

(v) Because single cells are much smaller than gas bubbles, the liquid film surrounding each cell is much thinner than that around the bubbles and its effect on mass transfer can generally be neglected. On the other hand, if the cells form large clumps, liquid-film resistance can be significant.

(vi) Resistance at the cell–liquid interface is generally neglected.

(vii) When the cells are in clumps, intraparticle resistance is likely to be significant as oxygen has to diffuse through the solid pellet to reach the interior cells. The magnitude of this resistance depends on the size of the clumps.

(viii) Intracellular oxygen-transfer resistance is negligible because of the small distances involved.

When cells are dispersed in the liquid and the bulk fermentation broth is well mixed, *the major resistance to oxygen transfer is the liquid film surrounding the gas bubbles.* Transport through this film becomes the rate-limiting step in the complete process, and controls the overall mass-transfer rate. Consequently, the rate of oxygen transfer from the bubble all the way to the cell is dominated by the rate of step (iii). The mass-transfer rate for this step can be calculated using Eq. (9.37).

At steady state there can be no accumulation of oxygen at any location in the fermenter; therefore, the rate of oxygen transfer from the bubbles must be equal to the rate of oxygen consumption by the cells. If we make N_A in Eq. (9.37) equal to Q_O in Eq. (9.38) we obtain the following equation:

$$k_L a\,(C^*_{AL} - C_{AL}) = q_O x. \tag{9.39}$$

$k_L a$ is used to characterise the oxygen mass-transfer capability of fermenters. If $k_L a$ for a particular system is small, the ability of the reactor to deliver oxygen to the cells is limited. We can predict the response of the fermenter to changes in mass-transfer operating conditions using Eq. (9.39). For example if the rate of cell metabolism remains unchanged but $k_L a$ is increased, e.g. by raising the stirrer speed to reduce the thickness of the boundary layer around the bubbles, the dissolved-oxygen concentration C_{AL} must rise in order for the left-hand side of Eq. (9.39) to remain equal to the right-hand side. Similarly, if the rate of oxygen consumption by the cells accelerates while $k_L a$ is unaffected, C_{AL} must decrease.

We can use Eq. (9.39) to deduce some important relationships for fermenters. First, let us estimate the maximum cell concentration that can be supported by the fermenter's oxygen-transfer system. For a given set of operating conditions, the maximum rate of oxygen transfer occurs when the concentration-difference driving force $(C^*_{AL} - C_{AL})$ is highest, i.e. when the concentration of dissolved oxygen C_{AL} is zero. Therefore from Eq. (9.39), the maximum cell concentration that can be supported by the mass-transfer functions of the reactor is:

$$x_{max} = \frac{k_L a\, C^*_{AL}}{q_O}. \tag{9.40}$$

If x_{max} estimated using Eq. (9.40) is lower than the cell concentration required in the fermentation process, $k_L a$ must be improved. It is generally undesirable for cell density to be limited by rate of mass transfer. Comparison of x_{max} values evaluated using Eqs (8.52) and (9.40) can be used to gauge the relative effectiveness of heat and mass transfer in aerobic fermentation. For example, if x_{max} from Eq. (9.40) were small while x_{max} calculated from heat-transfer considerations were large, we would know that mass-transfer operations are more likely to limit biomass growth. If both x_{max} values are greater than that desired for the process, heat and mass transfer are adequate.

Another important parameter is the minimum $k_L a$ required to maintain $C_{AL} > C_{crit}$ in the fermenter. This can be determined from Eq. (9.39) as:

$$(k_L a)_{crit} = \frac{q_O x}{(C^*_{AL} - C_{crit})}. \tag{9.41}$$

Example 9.1 Cell concentration in aerobic culture

A strain of *Azotobacter vinelandii* is cultured in a 15 m^3 stirred fermenter for alginate production. Under current operating conditions $k_L a$ is 0.17 s^{-1}. Oxygen solubility in the broth is approximately 8×10^{-3} kg m^{-3}.

(a) The specific rate of oxygen uptake is 12.5 mmol g^{-1} h^{-1}. What is the maximum possible cell concentration?
(b) The bacteria suffer growth inhibition after copper sulphate is accidently added to the fermentation broth. This causes a reduction in oxygen uptake rate to 3 mmol g^{-1} h^{-1}. What maximum cell concentration can now be supported by the fermenter?

Solution:

(a) From Eq. (9.40):

$$x_{max} = \frac{(0.17\,\mathrm{s}^{-1})\,(8\times 10^{-3}\,\mathrm{kg\,m^{-3}})}{\dfrac{12.5\,\mathrm{mmol}}{\mathrm{g\,h}}\cdot\left|\dfrac{1\,\mathrm{h}}{3600\,\mathrm{s}}\right|\cdot\left|\dfrac{1\,\mathrm{gmol}}{1000\,\mathrm{mmol}}\right|\cdot\left|\dfrac{32\,\mathrm{g}}{1\,\mathrm{gmol}}\right|\cdot\left|\dfrac{1\,\mathrm{kg}}{1000\,\mathrm{g}}\right|}$$
$$= 1.2\times 10^4\,\mathrm{g\,m^{-3}} = 12\,\mathrm{g\,l^{-1}}.$$

(b) Assume that addition of copper sulphate does not affect C^*_{AL} or $k_L a$. If q_O is reduced by a factor of $12.5/3 = 4.167$, x_{max} is increased to:

$$x_{max} = 4.167\,(12\,\mathrm{g\,l^{-1}}) = 50\,\mathrm{g\,l^{-1}}.$$

To achieve the calculated cell densities all other conditions must be favourable, e.g. sufficient substrate and time must be provided.

9.6 Oxygen Transfer in Fermenters

The rate of oxygen transfer in fermentation broths is influenced by several physical and chemical factors that change either the value of k_L or the value of a, or the driving force for mass transfer, $(C^*_{AL} - C_{AL})$. As a general rule of thumb, k_L in fermentation liquids is about $3\text{–}4 \times 10^{-4}$ m s^{-1} for bubbles greater than 2–3 mm diameter; this can be reduced to 1×10^{-4} m s^{-1} for smaller bubbles depending on bubble rigidity. Once the bubbles are above 2–3 mm in size, k_L is relatively constant and insensitive to conditions. If substantial improvement in mass-transfer rates is required, it is usually more productive to focus on increasing the interfacial area a. Operating values in bioreactors for the combined coefficient k_La span a wide range over about three orders of magnitude; this is due mainly to the large variation in a. In production-scale fermenters, the value of k_La is typically in the range 0.02 s^{-1} to 0.25 s^{-1}.

In this section, several aspects of fermenter design and operation are discussed in terms of their effect on oxygen mass transfer.

9.6.1 Bubbles

The efficiency of gas–liquid mass transfer depends to a large extent on the characteristics of bubbles in the liquid medium. Bubble behaviour strongly affects the value of k_La; some properties of bubbles affect mainly the magnitude of k_L, whereas others change the interfacial area a. The important aspects of bubble behaviour in fermenters are described below.

Stirred fermenters are used most commonly for aerobic culture. In these vessels, oxygen is supplied to the medium by sparging swarms of air bubbles underneath the impeller. The action of the impeller then creates a dispersion of gas throughout the vessel. In small laboratory-scale fermenters all of the liquid is close to the impeller; bubbles in these systems are frequently subjected to severe distortions as they interact with turbulent liquid currents in the vessel. In contrast, bubbles in most industrial stirred tanks spend a large proportion of their time floating free and unimpeded through the liquid after initial dispersion at the impeller. Liquid in large fermenters away from the impeller does not possess sufficient energy for continuous break-up of bubbles. This is a consequence of scale; most laboratory fermenters operate with stirrer power between 10 and 20 kW m^{-3}, whereas large agitated vessels operate at 0.5–5 kW m^{-3}. The result is that virtually all large commercial-size stirred-tank reactors operate mostly in the free bubble-rise regime [8].

The most important property of air bubbles in fermenters is their size. For a given volume of gas, more interfacial area a is provided if the gas is dispersed into many small bubbles rather than a few large ones; therefore a major goal in bioreactor design is a high level of gas dispersion. However, there are other important benefits associated with small bubbles. Small bubbles have correspondingly slow bubble-rise velocities; consequently they stay in the liquid longer, allowing more time for the oxygen to dissolve. Small bubbles therefore create high *gas hold-up*, defined as the fraction of the fluid volume in the reactor occupied by gas:

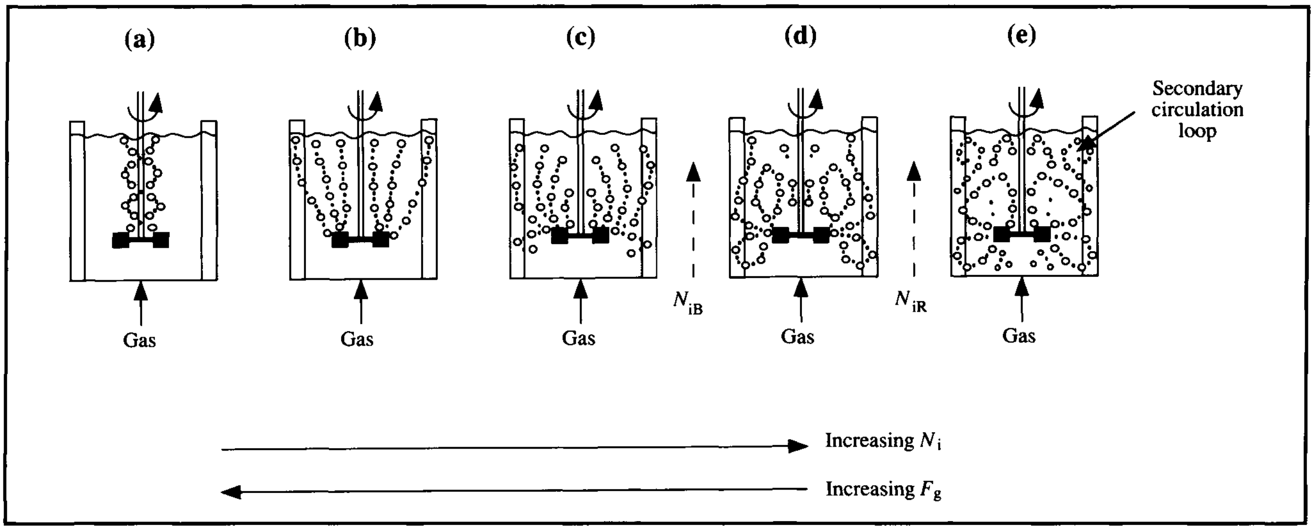
$$\varepsilon = \frac{V_G}{V_L + V_G} \tag{9.42}$$

where ε is the gas hold-up, V_G is the volume of gas bubbles in the reactor, and V_L is the volume of liquid. Because the total interfacial area for oxygen transfer depends on the total volume of gas in the system as well as on the average bubble size, high mass-transfer rates are achieved at high gas hold-ups. Gas hold-up values are very difficult to predict and may be anything from very low (0.01) up to a maximum in commercial-scale stirred fermenters of about 0.2. Under normal operating conditions, a significant fraction of the oxygen in fermentation vessels in contained in the gas hold-up. For example if the culture is sparged with air and the broth saturated with dissolved oxygen, for an air hold-up of only 0.03, about half the total oxygen in the system is in the gas phase.

While it is desirable to have small bubbles, there are practical limits. Bubbles << 1 mm diameter can become a nuisance in bioreactors. Oxygen concentration in these bubbles equilibrates with that in the medium within seconds, so that the gas hold-up no longer reflects the capacity of the system for mass transfer [9]. Problems with very small bubbles are exacerbated in viscous non-Newtonian broths; tiny bubbles remain lodged in these fluids for long periods of time because their velocity of rise is reduced. As a rule of thumb, relatively large bubbles must be employed in viscous cultures.

Bubble size also affects the value of k_L. In most fermentation broths, if the bubbles have diameters less than 2–3 mm, surface tension effects dominate the behaviour of the bubble surface. As a result, the bubbles behave as rigid spheres with immobile surfaces and no internal gas circulation. A rigid bubble surface gives lower k_L values; k_L decreases with decreasing bubble diameter below 2–3 mm. On the other hand, bubbles in fermentation media with sizes greater than about 3 mm develop internal circulation and relatively mobile surfaces, depending on liquid properties. Bubbles with mobile surfaces are able to wobble and move in spirals during free rise;

Figure 9.9 Flow pattern in stirred aerated bioreactors as a function of impeller speed N_i and gas flow rate F_g. (From A.W. Nienow, D.J. Wisdom and J.C. Middleton, 1978, The effect of scale and geometry on flooding, recirculation, and power in gassed stirred vessels. *Proc. 2nd Eur. Conf. on Mixing*, Cambridge, England, 1977, pp. F1-1–F1-16, BHRA Fluid Engineering, Cranfield.)



this behaviour has a marked beneficial effect on k_L and rate of mass transfer.

To summarise the influence of bubble size on oxygen mass transfer, small bubbles are generally beneficial because of the increased gas hold-up and larger interfacial surface-area. However, k_L for bubbles less than about 3 mm diameter is reduced due to surface effects. Very small bubbles << 1 mm should be avoided, especially in viscous broth.

9.6.2 Sparging, Stirring and Medium Properties

Because bubble size is such a critical parameter affecting oxygen transfer, it is useful to consider the physical processes in fermenters which determine bubble size. These processes include *bubble formation*, *gas dispersion* and *coalescence*.

Air bubbles are formed at the sparger. There exists a large variety of sparger designs, including simple open pipes, perforated tubes, porous diffusers, and complex two-phase injector devices. Bubbles leaving the sparger usually fall within a relatively narrow size range depending on the sparger type. This size range is a significant parameter in design of air-driven fermenters such as bubble and airlift columns because there is no other mechanism for bubble dispersion in these reactors. However in stirred vessels, design of the sparger and the mechanics of bubble formation are of secondary importance compared with the effects of the impeller. As a result of continual bubble break-up and dispersion by the impeller and coalescence from bubble collisions, bubble sizes in stirred reactors often bear little relationship to those formed at the sparger. Coalescence of small bubbles into bigger bubbles is generally undesirable because it reduces the total interfacial area and gas hold-up. Frequency of coalescence depends mainly on the liquid properties. In a coalescing liquid, a large fraction of bubble collisions results in the formation of bigger bubbles, while in non-coalescing liquids colliding bubbles do not coalesce readily. Salts act to suppress coalescence; therefore, fermentation media are usually non-coalescing to some extent depending on composition. This is an advantage for oxygen mass transfer.

Flow patterns set up in stirred vessels in the absence of aeration have been described in Chapter 7. As illustrated in Figure 9.9, when air is sparged different gas flow patterns develop depending on the relative rates of gas input and stirring. As shown in Figure 9.9(a), if the agitator speed N_i is low and the gas feed rate F_g is high, gas envelopes the impeller without dispersion and the flow pattern is dominated by air flow up the stirrer shaft. *Impeller flooding* is said to occur; this means that the gas-handling capacity of the stirrer is smaller than the amount introduced. Flooding should be avoided because an impeller surrounded by gas no longer contacts the liquid properly, resulting in poor mixing and gas dispersion. As the impeller speed increases, gas is captured behind the agitator blades and is dispersed into the liquid. N_{iB} is the minimum stirrer speed required to just completely disperse the gas,

Figure 9.10 Dependence of $k_L a$ on stirrer speed N_i in an agitated tank with 0.381 cm s^{-1} superficial gas velocity. Symbols represent various concentrations of sodium polyacrylate and sodium carboxyl methyl cellulose in aqueous solution. (From H. Yagi and F. Yoshida, 1975, Gas absorption by Newtonian and non-Newtonian fluids in sparged agitated vessels. *Ind. Eng. Chem. Process Des. Dev.* **14**, 488–493.)

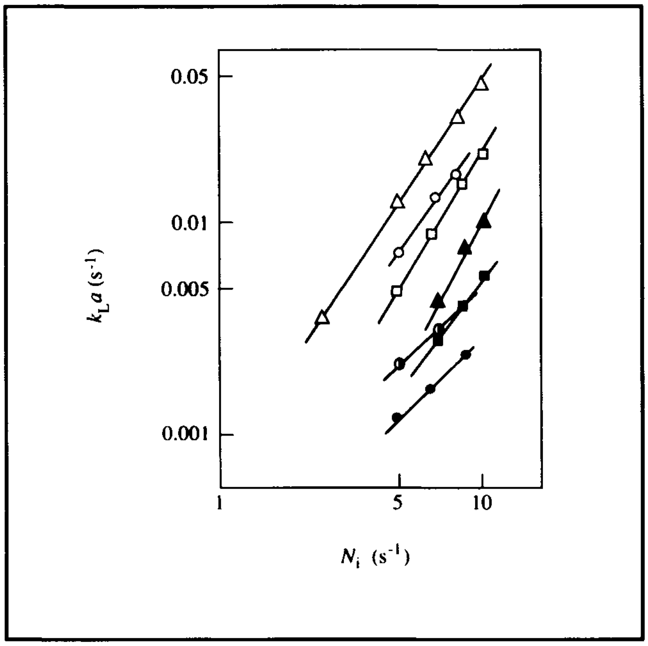
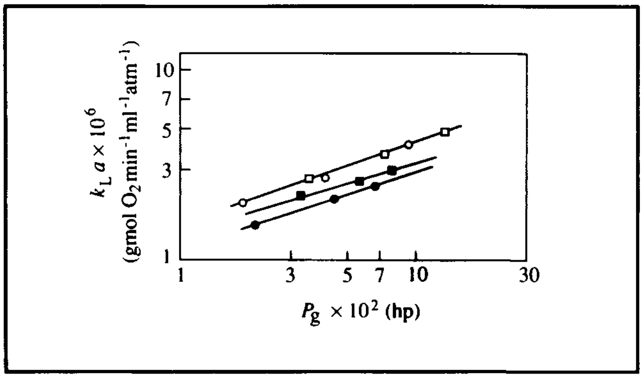


Figure 9.11 Effect on $k_L a$ of number of impellers used to mix a viscous mycelial suspension. The ratio of liquid depth to tank diameter is 2.41. P_g is power consumption with gas sparging. (○) Two impellers, apparent viscosity = 500 cP, impeller spacing/impeller diameter = 2.06; (□) three impellers, apparent viscosity = 500 cP, impeller spacing/impeller diameter = 1.37; (●) two impellers, apparent viscosity = 700 cP, impeller spacing/impeller diameter = 2.06; (■) three impellers, apparent viscosity = 700 cP, impeller spacing/impeller diameter = 1.37. (From H. Taguchi, 1971, The nature of fermentation fluids. *Adv. Biochem. Eng.* **1**, 1–30.)



including below the impeller. The minimum agitator tip speed for dispersion of air bubbles even at very low gas flow rates has been estimated by Westerterp *et al.* [10] as roughly 1.5–2.5 m s^{-1}; tip speed = $\pi N_i D_i$ where N_i is impeller speed and D_i is impeller diameter. With further increases in stirrer speed, small recirculation patterns start to emerge as indicated in Figure 9.9(e); N_{iR} is the speed at which gross recirculation of gas back to the agitator starts to occur.

Gas dispersion in stirred vessels takes place mainly in the immediate vicinity of the impeller. As shown in Figure 7.27 (see p. 154), when gas is present in stirred liquids it is drawn into low-pressure cavities behind the stirrer blades. Gas from the sparger together with a large fraction of the recirculating gas in the system is entrained in these cavities. Gas contacting the impeller blades leads to a decrease in the friction or drag coefficient associated with impeller rotation and a concomitant reduction in power consumption. As the impeller blades rotate at high speed, small gas bubbles are thrown out from the back of the cavities into the bulk liquid under the influence of dispersion processes that are not yet completely understood. In a non-coalescing liquid, the bubbles remain close to the size produced at the back of the cavities. Because bubbles formed at the sparger are immediately drawn into the impeller zone, dispersion of gas in stirred vessels is largely independent of sparger design; when the sparger is located under the stirrer, it has been shown that sparger type does not significantly affect mass transfer.

Under typical fermenter operating conditions, increasing the stirrer speed improves the value of $k_L a$, as shown in Figure 9.10. In contrast, except at very low sparging rates, increasing the gas flow is generally considered to exert only a minor influence on $k_L a$. As indicated in Figure 9.11, increasing the number of impellers on the stirrer shaft does not necessarily improve $k_L a$ even though the power consumption is increased. The quantity of gas passing through the upper impellers is small compared with the lower impeller, so that any additional gas dispersion is not significant.

9.6.3 Antifoam Agents

Most cell cultures produce a variety of foam-producing and foam-stabilising agents, such as proteins, polysaccharides and fatty acids. Foam build-up in fermenters is very common, particularly in aerobic systems. Foaming causes a range of reactor operating problems; foam control is therefore an important consideration in fermentation design. Excessive foam overflowing from the top of the fermenter provides a route for entry of contaminating organisms and causes blockage of outlet gas lines. Liquid and cells trapped in the foam represent a loss of bioreactor volume; conditions in the foam may not be favourable for metabolic activity. In addition, fragile cells can be damaged by collapsing foam.

Addition of special antifoam compounds to the medium is the most common method of reducing foam build-up in fermenters. However, antifoam compounds affect the surface chemistry of bubbles and their tendency to coalesce, and have a significant effect on $k_L a$. Most antifoam agents are strong surface tension-lowering substances. Decrease in surface tension reduces the average bubble diameter, thus producing higher values of a. However, this is countered by a reduction in mobility of the gas–liquid interface which lowers the value of k_L. With most silicon-based antifoams, the decrease in k_L is generally larger than the increase in a so that, overall, $k_L a$ is reduced [11, 12]. The resulting decrease in rate of oxygen transfer can be dramatic, by up to a factor of 10.

In order to maintain the non-coalescing character of the medium and high $k_L a$ values, mechanical rather than chemical methods of disrupting foam are preferred because the liquid properties are not changed. Mechanical foam breakers, such as high-speed discs rotating at the top of the vessel and centrifugal foam destroyers, are suitable when foam development is moderate. However, some of these devices need large quantities of power to operate in commercial-scale vessels; their limited foam-destroying capacity is also a problem with highly-foaming cultures. In many cases, use of chemical antifoam agents is unavoidable.

9.6.4 Temperature

The temperature of aerobic fermentations affects both the solubility of oxygen C^*_{AL} and the mass-transfer coefficient k_L. Increasing temperature causes C^*_{AL} to drop, so that the driving force for mass transfer $(C^*_{AL} - C_{AL})$ is reduced. At the same time, diffusivity of oxygen in the liquid film surrounding the bubbles is increased, resulting in an increase in k_L. The net effect of temperature on oxygen transfer depends on the range of temperature considered. For temperatures between 10°C and 40°C, increase in temperature is more likely to increase the rate of oxygen transfer. Above 40°C the solubility of oxygen drops significantly, adversely affecting the driving force and rate of mass transfer.

9.6.5 Gas Pressure and Oxygen Partial Pressure

Pressure and oxygen partial pressure of the gas used to aerate fermenters affect the value of C^*_{AL}. The equilibrium relationship between these parameters for dilute liquid solutions is given by *Henry's law*:

$$p_{AG} = p_T \, y_{AG} = H \, C^*_{AL} \tag{9.43}$$

where p_{AG} is the partial pressure of component A in the gas, p_T is total gas pressure, y_{AG} is the mole fraction of A in the gas, H is *Henry's constant* which is a function of temperature, and C^*_{AL} is the solubility of component A in the liquid. From Eq. (9.43), if the total gas pressure p_T or concentration of oxygen in the gas y_{AG} is increased at constant temperature, C^*_{AL} and therefore the mass-transfer driving force $(C^*_{AL} - C_{AL})$ also increase.

In some fermentations, oxygen-enriched air or pure oxygen is used to improve mass transfer. Alternatively, oxygen solubility is increased by sparging compressed air at high pressure. Both these strategies increase the operating cost of the fermenter; it is also possible in some cases that the culture will suffer inhibitory effects from exposure to very high oxygen partial pressures.

9.6.6 Presence of Cells

Oxygen transfer is influenced by the presence of cells in fermentation broths; the nature of the effect depends on the species of organism, its morphology and concentration. Cells with complex morphology generally lead to lower transfer rates. Cells interfere with bubble break-up and coalescence; cells, proteins and other molecules which adsorb at gas–liquid interfaces also cause *interfacial blanketing* which reduces the contact area between gas and liquid. The quantitative effect of interfacial blanketing is highly system-specific. Because concentrations of cells, substrates and products change throughout batch fermentation, the value of $k_L a$ can also vary. An example of change in $k_L a$ due to these factors is shown in Figure 9.12.

9.7 Measuring Dissolved-Oxygen Concentrations

The concentration of dissolved oxygen C_{AL} in fermenters is normally measured using a *dissolved-oxygen electrode.* There are two types in common use: *galvanic electrodes* and *polarographic*

Figure 9.12 Variation in $k_L a$ during a batch 300-l streptomycete fermentation. (From C.M. Tuffile and F. Pinho, 1970, Determination of oxygen-transfer coefficients in viscous streptomycete fermentations. *Biotechnol. Bioeng.* 12, 849–871.)

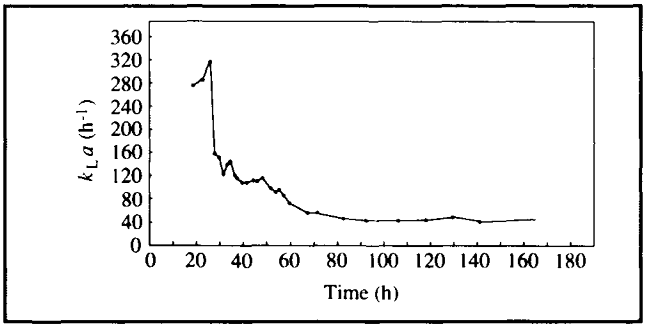
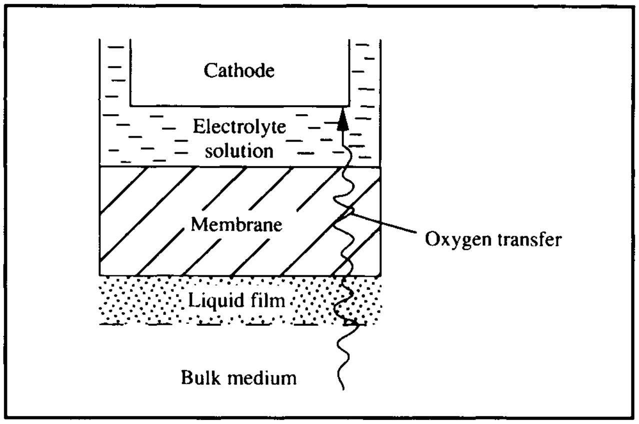


Figure 9.13 Diffusion of oxygen from bulk liquid to the cathode of an oxygen electrode.



electrodes. Details of the construction and operating principles of these probes can be found in other references [13, 14]. In both designs, a membrane which is permeable to oxygen separates the fermentation fluid from the electrode. As illustrated in Figure 9.13, oxygen diffuses through the membrane to the cathode, where it reacts to produce a current between anode and cathode proportional to the oxygen partial pressure in the fermentation broth. An electrolyte solution in the electrode supplies ions which take part in the reactions, and must be replenished at regular intervals. Depending on flow conditions in the bulk medium, liquid properties, and the rate of oxygen utilisation by the probe, a liquid boundary layer develops at the interface between the solid probe and the liquid.

As indicated in Figure 9.13, supply of oxygen molecules from the bulk medium to the cathode is in itself a mass-transfer process. Because there is no bulk fluid motion in the membrane or electrolyte solution and little motion in the liquid film at the membrane interface, operation of the probe relies on diffusion of oxygen across these thicknesses. This takes time, so the response of an electrode to sudden changes in dissolved-oxygen level is subject to delay. The *electrode response time* can be measured by quickly transferring the probe from a beaker containing medium saturated with nitrogen to one saturated with air. The response time is defined as the time taken for the probe to indicate 63% of the total change in dissolved-oxygen level. For commercially-available steam-sterilisable electrodes, response times are usually 10–100 s. Electrode response can usually be improved if the bulk liquid is stirred rapidly; this decreases the thickness of the liquid film at the membrane surface. Micro-probes for dissolved-oxygen measurement are also available. The smaller cathode size and lower rate of oxygen consumption by these instruments means that their response is quicker; so much so that micro-probes can be used to measure dissolved oxygen levels in unagitated systems. Steam-sterilisable probes are inserted directly into fermentation vessels for on-line monitoring of dissolved oxygen. Repeated calibration of dissolved-oxygen probes is usually necessary; fouling by cells attaching to the membrane surface, electronic noise due to air bubbles passing close to the membrane, and signal drift are the main operating problems.

Both galvanic and polarographic electrodes measure the partial pressure of dissolved oxygen or *oxygen tension* in the fermentation broth, not the dissolved-oxygen concentration. To convert this to dissolved-oxygen concentration, it is necessary to know the solubility of oxygen in the liquid at the temperature and pressure of measurement.

9.8 Estimating Oxygen Solubility

The concentration difference $(C^*_{AL} - C_{AL})$ is the driving force for oxygen mass transfer. Because this difference is usually very small, it is important that the solubility C^*_{AL} be known accurately; small errors in C^*_{AL} will result to large errors in $(C^*_{AL} - C_{AL})$. Experimentally-determined values for solubility of oxygen in water can be found in many literature sources, e.g. [15–17]; Table 9.1 shows the solubility of oxygen in water under 1 atm oxygen pressure at various temperatures. However, fermentations are not carried out using pure water and pure oxygen. Because oxygen partial pressure in the gas phase and presence of dissolved material in the liquid are major factors affecting oxygen solubility, the values given in Table 9.1 cannot be applied directly to bioprocessing systems.

Table 9.1 Solubility and Henry's constant for oxygen in pure water under 1 atm oxygen pressure

(*Calculated from data in* International Critical Tables, *1928, vol. III, p. 257. McGraw-Hill, New York*)

Temperature (°C)	*Oxygen solubility* ($kg\ m^{-3}$)	*Henry's constant* ($atm\ m^3\ kg^{-1}$)
0	7.03×10^{-2}	14.2
10	5.49×10^{-2}	18.2
15	4.95×10^{-2}	20.2
20	4.50×10^{-2}	22.2
25	4.14×10^{-2}	24.2
26	4.07×10^{-2}	24.6
27	4.01×10^{-2}	24.9
28	3.95×10^{-2}	25.3
29	3.89×10^{-2}	25.7
30	3.84×10^{-2}	26.1
35	3.58×10^{-2}	27.9
40	3.37×10^{-2}	29.7

Table 9.2 Solubility of oxygen in water under 1 atm air pressure

(*Calculated from data in Table 9.1 and Henry's law*)

Temperature (°C)	*Oxygen solubility* ($kg\ m^{-3}$)
0	1.48×10^{-2}
10	1.15×10^{-2}
15	1.04×10^{-2}
20	9.45×10^{-3}
25	8.69×10^{-3}
26	8.55×10^{-3}
27	8.42×10^{-3}
28	8.29×10^{-3}
29	8.17×10^{-3}
30	8.05×10^{-3}
35	7.52×10^{-3}
40	7.07×10^{-3}

9.8.1 Effect of Oxygen Partial Pressure

According to the *International Critical Tables* [18], the mole fraction of oxygen in air is 0.2099, so the partial pressure of oxygen at 1 atm air pressure is 0.2099 atm. At a given temperature, the effect of gas-phase oxygen partial pressure on solubility is given by Henry's law, Eq. (9.43). Therefore, the solubility of oxygen in water under 1 atm air pressure is 0.2099 times that under 1 atm pure oxygen. Values for solubility of oxygen in water sparged with air are given in Table 9.2.

9.8.2 Effect of Temperature

The variation of oxygen solubility with temperature is shown in Tables 9.1 and 9.2 for water in the range 0–40°C. Solubility falls with increasing temperature. Oxygen solubility in pure water between 0° and 36°C has been correlated by the following equation [15]:

$$C^*_{AL} = 14.161 - 0.3943T + 0.007714T^2 - 0.0000646T^3 \tag{9.44}$$

where C^*_{AL} is oxygen solubility in units of $mg\ l^{-1}$, and T is temperature in °C.

9.8.3 Effect of Solutes

Presence of solutes such as salts, acids and sugars has a significant effect on oxygen solubility in water, as indicated in Tables 9.3 and 9.4. These data indicate that oxygen solubility is decreased by the ions and sugars normally added to fermentation media. The effect on oxygen solubility of ionic and non-ionic solutes such as molasses, corn-steep liquor, protein and antifoam agents is reported in several publications [19–24]. Quicker *et al.* [23] have developed an empirical correlation to correct values of oxygen solubility in water for the effects of cations, anions and sugars:

Table 9.3 Solubility of oxygen in aqueous solutions at 25°C under 1 atm oxygen pressure

(Calculated from data in International Critical Tables, *1928, vol. III, p. 271. McGraw-Hill, New York)*

Concentration	*Oxygen solubility* (kg m^{-3})		
(M)	HCl	$\frac{1}{2}H_2SO_4$	NaCl
0	4.14×10^{-2}	4.14×10^{-2}	4.14×10^{-2}
0.5	3.87×10^{-2}	3.77×10^{-2}	3.43×10^{-2}
1.0	3.75×10^{-2}	3.60×10^{-2}	2.91×10^{-2}
2.0	3.50×10^{-2}	3.28×10^{-2}	2.07×10^{-2}

Table 9.4 Solubility of oxygen in aqueous solutions of sugars under 1 atm oxygen pressure

(Calculated from data in International Critical Tables, *1928, vol. III, p. 272. McGraw-Hill, New York)*

Sugar	*Concentration* (gmol per kg H_2O)	*Temperature* (°C)	*Oxygen solubility* (kg m^{-3})
Glucose	0	20	4.50×10^{-2}
	0.7	20	3.81×10^{-2}
	1.5	20	3.18×10^{-2}
	3.0	20	2.54×10^{-2}
Sucrose	0	15	4.95×10^{-2}
	0.4	15	4.25×10^{-2}
	0.9	15	3.47×10^{-2}
	1.2	15	3.08×10^{-2}

$$\log_{10}\left(\frac{C^*_{AL0}}{C^*_{AL}}\right) = 0.5 \sum_i H_i z_i^2 C_{iL} + \sum_j K_j C_{jL} \tag{9.45}$$

where:

C^*_{AL0} = oxygen solubility at zero solute concentration (mol m^{-3})
C^*_{AL} = oxygen solubility (mol m^{-3})
H_i = constant for ionic component i (m^3 mol^{-1})
z_i = valency of ionic component i
C_{iL} = concentration of ionic component i in the liquid (mol m^{-3})
K_j = constant for non-ionic component j (m^3 mol^{-1})
C_{jL} = concentration of non-ionic component j in the liquid (mol m^{-3})

Values of H_i and K_j for use in Eq. (9.45) are listed in Table 9.5. In a typical fermentation medium, oxygen solubility is between 5% and 25% lower than in water as a result of solute effects.

9.9 Mass-Transfer Correlations

In general, there are two approaches to evaluating k_L and a: calculation using empirical correlations, and experimental measurement. In both cases, separate determination of k_L and a is laborious and sometimes impossible. It is convenient therefore to directly evaluate the product k_La; the combined term k_La is often referred to as the mass-transfer coefficient rather than just k_L. In this section we consider methods for calculating k_La using published correlations.

In aerobic fermenters, k_L and a are dependent on the hydrodynamic conditions around the gas bubbles. Relationships between k_La and parameters such as bubble diameter, liquid velocity, density, viscosity and oxygen diffusivity have been investigated extensively, and empirical correlations between

Table 9.5 Values of H_i and K_j in Eq. (9.45) at 25°C

(From A. Schumpe, I. Adler and W.-D. Deckwer, 1978, Solubility of oxygen in electrolyte solutions, Biotechnol. Bioeng. *20, 145–150; and G. Quicker, A. Schumpe, B. König and W.-D. Deckwer, 1981, Comparison of measured and calculated oxygen solubilities in fermentation media,* Biotechnol. Bioeng. *23, 635–650)*

Cation	$H_i \times 10^3$ ($m^3\ mol^{-1}$)	*Anion*	$H_i \times 10^3$ ($m^3\ mol^{-1}$)	*Sugar*	$K_j \times 10^3$ ($m^3\ mol^{-1}$)
H^+	−0.774	OH^-	0.941	Glucose	0.119
K^+	−0.596	Cl^-	0.844	Lactose	0.197
Na^+	−0.550	CO_3^{2-}	0.485	Sucrose	0.149*
NH_4^+	−0.720	SO_4^{2-}	0.453		
NEt_4^+	−0.912	NO_3^-	0.802		
Mg^{2+}	−0.314	HCO_3^-	1.058		
Ca^{2+}	−0.303	$H_2PO_4^-$	1.037		
Mn^{2+}	−0.311	HPO_4^{2-}	0.485		
		PO_4^{3-}	0.320		

Approximately valid for sucrose concentrations up to about 200 g l^{-1}.

mass-transfer coefficients and important operating variables have been developed. Theoretically, these correlations allow prediction of mass-transfer coefficients based on information gathered from a large number of previous experiments. In practice, however, the accuracy of published correlations applied to biological systems is generally poor. The main reason is that mass transfer is strongly affected by the additives usually present in fermentation media. Because fermentation liquids contain varying levels of substrates, products, salts, surface-active agents and cells, the surface chemistry of bubbles and therefore the mass-transfer situation become very complex. Most available correlations for oxygen mass-transfer coefficients were determined using pure air in water, and it is very difficult to correct these correlations for different liquid compositions. The effective mass-transfer area a is also influenced by interfacial blanketing of the bubble–liquid surface by cells and other components of the broth. Prediction of $k_L a$ under these conditions is problematic.

When mass-transfer coefficients are required for large-scale equipment, another factor related to hydrodynamic conditions limits the applicability of published correlations. Most studies of oxygen mass-transfer have been carried out in laboratory-scale stirred reactors, which are characterised by high turbulence throughout most of the vessel. The gas phase in small-scale agitated tanks is well dispersed; break-up and coalescence of bubbles occur constantly due to the high level of turbulence and frequent bubble collisions. In contrast, bubbles in industrial-scale stirred tanks are mostly in free rise. The result is that, due to the different hydrodynamic regimes present in small- and large-scale vessels, mass-transfer correlations for stirred tanks developed in the laboratory tend to overestimate the oxygen-transfer capacity of commercial-scale systems. Better results can be achieved with a two-compartment model of large fermenters, applying different correlations for the mixed zone close to the impeller and the bubble zone away from the impeller. Application of this technique for calculation of oxygen transfer-rate is described by Oosterhuis and Kossen [25].

In this section we will consider one empirical correlation for oxygen mass transfer; its application in fermentation systems is subject to the problems mentioned above. A widely-used correlation for stirred vessels relates $k_L a$ directly to gas velocity and power input to the stirrer; all the effects of flow and turbulence on bubble dispersion and the mass-transfer boundary layer are represented by the power term. An expression for stirred fermenters containing non-coalescing non-viscous media is [26]:

$$k_L a = 2.0 \times 10^{-3} \left(\frac{P}{V} \right)^{0.7} u_G^{0.2} . \tag{9.46}$$

In Eq. (9.46), $k_L a$ is the combined mass-transfer coefficient in units of s^{-1}, P is the power dissipated by the stirrer in W, and

V is the fluid volume in m^3. u_G is the *superficial gas velocity* in $m\ s^{-1}$; superficial gas velocity is defined as the volumetric gas flow rate divided by the cross-sectional area of the fermenter. Eq. (9.46) was obtained with water containing ions in vessel volumes $2 \times 10^{-3} < V < 4.4\ m^3$, and $500 < P/V < 10\ 000\ W\ m^{-3}$.

Experimental results for $k_L a$ are reported to agree with Eq. (9.46) to within 20–40%; however application of this correlation to production-size vessels up to 25 m^3 in volume has been found to overestimate rate of oxygen transfer by about 100% [25]. Note that this correlation does not depend on the sparger or stirrer design; the power dissipated by the impeller determines $k_L a$ independent of stirrer type. Like most published correlations, Eq. (9.46) does not take into account the non-Newtonian behaviour of many culture fluids, the effect of added sugars and antifoam agents, and the presence of solids such as cells.

Eq. (9.46) suggests that $k_L a$ can be increased by raising the superficial gas velocity in the reactor. However, the exponent on u_G in Eq. (9.46) is much less than unity, so the effect of gas flow rate is relatively minor. There is usually limited scope for increasing u_G depending on vessel configuration; at high u_G the liquid contents can be blown out of the fermenter. The maximum operating value for u_G also depends on the stirrer speed; as discussed in Section 9.6.2, unless the impeller is able to disperse all the air impinging on it, impeller flooding will occur. Because the exponent on P in Eq. (9.46) is also less than one, increasing $k_L a$ by raising either the air flow rate or power input becomes progressively less efficient and more costly as the inputs increase.

9.10 Measurement of $k_L a$

Because of the difficulty in predicting $k_L a$ in bioreactors using correlations, mass-transfer coefficients for oxygen are usually determined experimentally. This is not without its own problems however, as discussed below. Whatever method is used to measure $k_L a$, the measurement conditions should match those in the fermenter during normal operation. Techniques for measuring $k_L a$ have been reviewed by van't Riet [26].

9.10.1 Oxygen-Balance Method

This technique is based on the equation for gas–liquid mass transfer, Eq. (9.37). In the experiment, the oxygen content of gas streams flowing to and from the fermenter are measured. From a mass balance at steady state, the difference in oxygen flow between inlet and outlet must be equal to the rate of oxygen transfer from gas to liquid:

$$N_A = \frac{1}{V_L}\left[(F_g\, C_{AG})_i - (F_g\, C_{AG})_o\right] \tag{9.47}$$

where V_L is the volume of liquid in the fermenter, F_g is the volumetric gas flow rate, C_{AG} is the gas-phase concentration of oxygen, and subscripts i and o refer to inlet and outlet gas streams, respectively. The first term on the right-hand side of Eq. (9.47) represents the rate at which oxygen enters the fermenter in the inlet-gas stream; the second term is the rate at which oxygen leaves. The difference between them is the rate at which oxygen is transferred out of the gas into the liquid, N_A. Because gas concentrations are generally measured as partial pressures, the ideal gas law Eq. (2.32) can be incorporated into Eq. (9.47) to obtain an alternative expression:

$$N_A = \frac{1}{R\,V_L}\left[\left(\frac{F_g p_{AG}}{T}\right)_i - \left(\frac{F_g p_{AG}}{T}\right)_o\right] \tag{9.48}$$

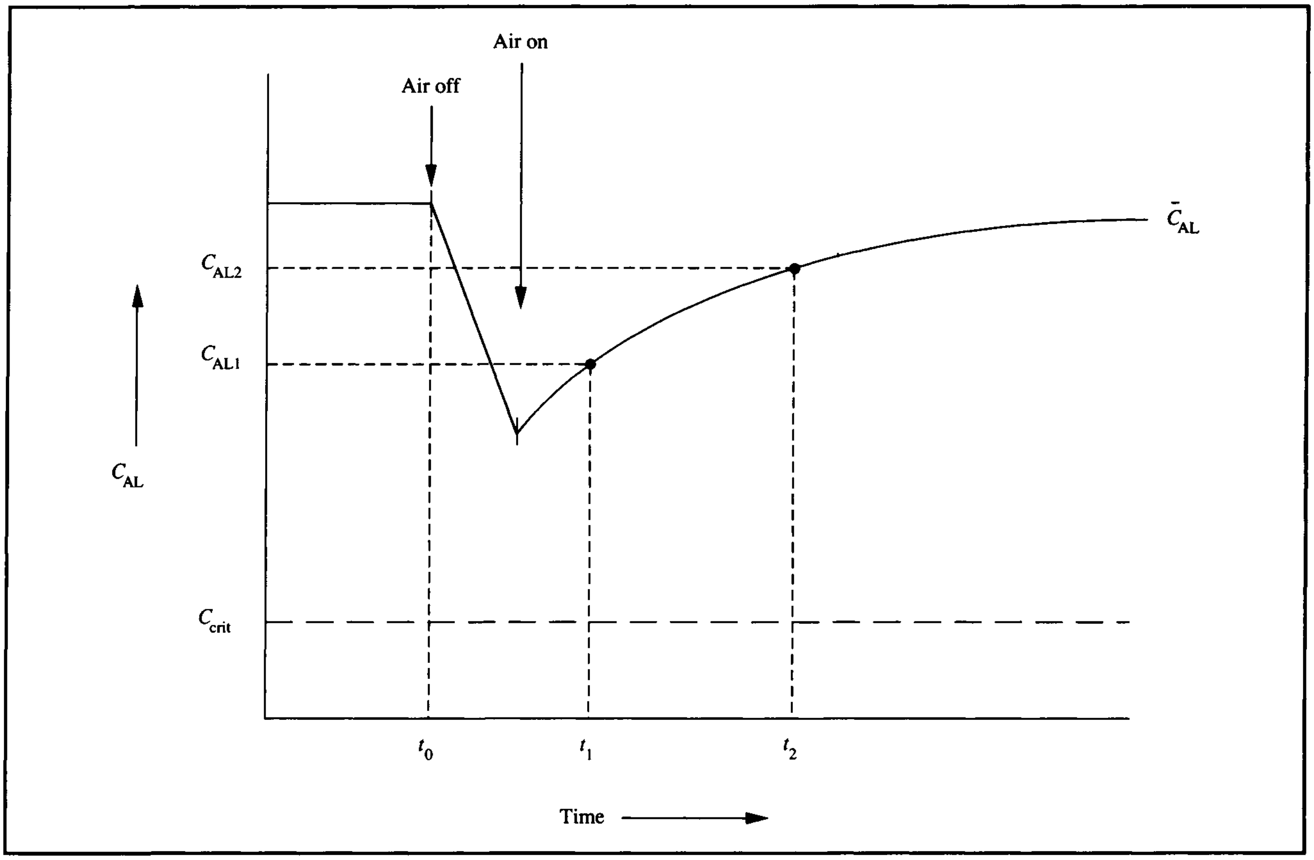
where R is the universal gas constant (see Table 2.5 on p. 20), p_{AG} is the oxygen partial pressure in the gas and T is absolute temperature. Because oxygen partial pressures in the inlet and exit gas streams are usually not very different during operation of fermenters, they must be measured very accurately, e.g. using mass spectrometry. The temperature and flow rate of the gases must also be measured carefully to ensure an accurate value of N_A is determined. Once N_A is known and C_{AL} and C^*_{AL} found using the methods described in Sections 9.7 and 9.8, $k_L a$ can be calculated from Eq. (9.37).

The steady-state oxygen-balance method is the most reliable procedure for measuring $k_L a$, and allows determination from a single-point measurement. An important advantage is that the method can be applied to fermenters during normal operation. It depends, however, on accurate measurement of gas composition, flow rate, pressure and temperature; large errors as high as ± 100% can be introduced if measurement techniques are inadequate. Considerations for design of laboratory equipment to ensure accurate oxygen uptake measurements are described by Brooks *et al.* [27].

9.10.2 Dynamic Method

This method for measuring $k_L a$ is based on an unsteady-state mass balance for oxygen. The main advantage of the dynamic

Figure 9.14 Variation of oxygen tension for dynamic measurement of $k_L a$.



method over the steady-state technique is the low cost of the equipment needed.

There are several different versions of the dynamic method; only one will be described here. Initially, the fermenter contains cells in batch culture. As shown in Figure 9.14, at some time t_0 the broth is de-oxygenated either by sparging nitrogen into the vessel or by stopping the air flow if the culture is oxygen-consuming. Dissolved-oxygen concentration C_{AL} drops during this period. Air is then pumped into the broth at a constant flow-rate and the increase in C_{AL} monitored as a function of time. It is important that the oxygen concentration remains above C_{crit} so that the rate of oxygen uptake by the cells is independent of oxygen level. Assuming re-oxygenation of the broth is fast relative to cell growth, the dissolved-oxygen level will soon reach a steady state value $\bar{C}_{AL}$ which reflects a balance between oxygen supply and oxygen consumption in the system. C_{AL1} and C_{AL2} are two oxygen concentrations measured during re-oxygenation at times t_1 and t_2, respectively. We can develop an equation for $k_L a$ in terms of these experimental data.

During the re-oxygenation step, the system is not at steady state. The rate of change in dissolved-oxygen concentration during this period is equal to the rate of oxygen transfer from gas to liquid, minus the rate of oxygen uptake by the cells:

$$\frac{dC_{AL}}{dt} = k_L a(C^*_{AL} - C_{AL}) - q_O x \tag{9.49}$$

where $q_O x$ is the rate of oxygen consumption. We can determine an expression for $q_O x$ by considering the final steady dissolved-oxygen concentration, $\bar{C}_{AL}$. When $C_{AL} = \bar{C}_{AL}$, $dC_{AL}/dt = 0$ because there is no change in C_{AL} with time. Therefore, from Eq. (9.49):

$$q_O x = k_L a(C^*_{AL} - \bar{C}_{AL}). \tag{9.50}$$

Substituting this result into Eq. (9.49) and cancelling the

$k_L a C^*_{AL}$ terms gives:

$$\frac{dC_{AL}}{dt} = k_L a(\bar{C}_{AL} - C_{AL}). \tag{9.51}$$

Assuming $k_L a$ is constant with time, we can integrate Eq. (9.51) between t_1 and t_2 using the integration rules described in Appendix D. The resulting equation for $k_L a$ is:

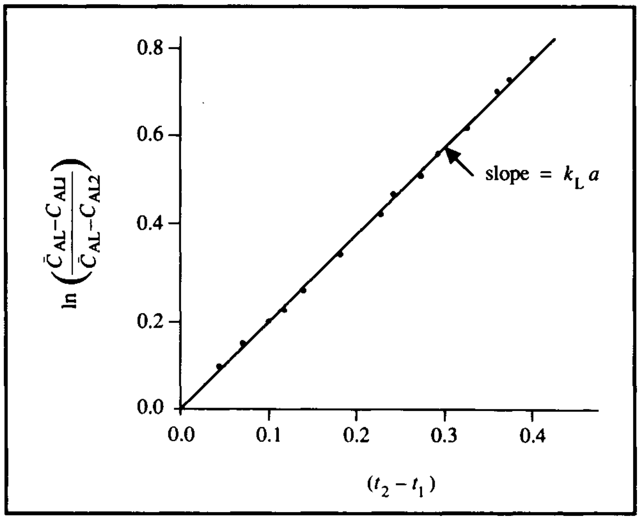
$$k_L a = \frac{\ln\left(\frac{\bar{C}_{AL} - C_{AL1}}{\bar{C}_{AL} - C_{AL2}}\right)}{t_2 - t_1}. \tag{9.52}$$

$k_L a$ can be estimated using two points from Figure 9.14 or, more accurately, from several values of (C_{AL1}, t_1) and (C_{AL2}, t_2). When

$$\ln\left(\frac{\bar{C}_{AL} - C_{AL1}}{\bar{C}_{AL} - C_{AL2}}\right)$$

is plotted against $(t_2 - t_1)$ as shown in Figure 9.15, the slope is $k_L a$. Eq. (9.52) can be applied to actively respiring cultures, or to systems without oxygen uptake. In the latter case, $\bar{C}_{AL} = C^*_{AL}$.

Figure 9.15 Evaluating $k_L a$ using the dynamic method.



Example 9.2 Estimating $k_L a$ using the dynamic method

A 20-l stirred fermenter containing a *Bacillus thuringiensis* culture at 30°C is used for production of microbial insecticide. $k_L a$ is determined using the dynamic method. Air flow is shut off for a few minutes and the dissolved-oxygen level drops; the air supply is then re-connected. When steady state is established, the dissolved-oxygen tension is 78% air saturation. The following results are obtained.

Time (s)	5	15
Oxygen tension (% air saturation)	50	66

(a) Estimate $k_L a$.
(b) An error is made determining the steady-state oxygen level which, instead of 78%, is taken as 70%. What is the percentage error in $k_L a$ resulting from this 10% error in $\bar{C}_{AL}$?

Solution:

(a) $\bar{C}_{AL} = 78\%$ air saturation. Let us define $t_1 = 5$ s, $C_{AL1} = 50\%$, $t_2 = 15$ s and $C_{AL2} = 66\%$. From Eq. (9.52):

$$k_L a = \frac{\ln\left(\frac{78 - 50}{78 - 66}\right)}{(15 - 5)\,\text{s}} = 0.085\,\text{s}^{-1}.$$

(b) If $\bar{C}_{AL}$ is taken to be 70% air saturation:

$$k_L a = \frac{\ln\left(\frac{70-50}{70-66}\right)}{(15-5)\,\mathrm{s}} = 0.16\,\mathrm{s}^{-1}.$$

The error in $k_L a$ is almost 100%. This example illustrates how important it is to obtain accurate data for $k_L a$ determination.

An oxygen probe with fast response time is required for measurement of C_{AL}, otherwise the dynamic method will not give accurate results. Frequently, however, the response time of the electrode is similar in magnitude to the time for mass transfer. When this is the case, values of C_{AL} measured during the experiment do not reflect the instantaneous oxygen concentration. Electrode response time should always be measured in conjunction with the dynamic method for $k_L a$ determination. In principle, the probe response time (Section 9.7) should be much smaller than the mass-transfer response time, $^1/_{k_L a}$. Van't Riet [26] has estimated that acceptable results are obtained using commercial electrodes with response times between 2 and 3 s for $k_L a$ values up to 0.1 s^{-1}. When the response time is closer to $^1/_{k_L a}$, the measurements should be corrected for electrode dynamics. Further discussion and analysis of this problem can be found in the literature [26, 28–34].

Another factor affecting accuracy of the dynamic method is the average residence time of gas in the system when deoxygenation with nitrogen is followed by a switch to aeration at the beginning of the measurement. Because there is a nitrogen gas hold-up in the vessel when air is re-introduced, measurement of C_{AL} does not reflect the kinetics of simple oxygen transfer until a hold-up of air is established. This takes longer in large vessels; dynamic methods for $k_L a$ measurement are therefore restricted to vessels with height less than 1 m [35]. Dynamic methods are inappropriate for viscous broths for similar reasons; the large residence times of bubbles in viscous broths affect the accuracy of measurement [9]. Corrections to account for gas-phase dynamics in small vessels are described by Dunn and Einsele [29] and Dang *et al.* [32]. When the gas velocity is low and the hold-up high, such large corrections are required that the original measurements become meaningless.

9.10.3 Sulphite Oxidation

This method is based on oxidation of sodium sulphite to sulphate in the presence of a catalyst such as Cu^{2+}. Although the sulphite method has been used extensively, the results appear to depend on operating conditions in an unknown way, and usually give higher $k_L a$ values than other techniques. Accordingly, its application is discouraged [26].

9.11 Oxygen Transfer in Large Vessels

Special difficulties are associated with measurement of oxygen transfer in large fermenters. Problems of residual gas hold-up with the dynamic method have already been mentioned. Implicit in application of the experimental techniques described in Section 9.10 is the assumption that $k_L a$ is constant throughout the entire reactor. This requires that the gas and liquid phases be perfectly mixed with uniform turbulence. In commercial-size reactors (> 10 m^3), perfect mixing is difficult to achieve; as a result the calculated value of $k_L a$ may depend on the position in the tank where the measurements of C_{AL} are made. Even when mixing is good, variation in composition of the gas phase is inevitable as changing static pressure and continuing dissolution of oxygen reduce the oxygen partial pressure in the bubbles as they rise.

Significant variation between inlet and outlet oxygen partial pressures affects the value for C^*_{AL} used in mass-transfer calculations. Allowance can be made for this by determining an average concentration driving force $(C^*_{AL} - C_{AL})$ across the system. A suitable average is the *logarithmic-mean concentration difference,* $(C^*_{AL} - C_{AL})_L$:

$$(C^*_{AL} - C_{AL})_L = \frac{(C^*_{AL} - C_{AL})_o - (C^*_{AL} - C_{AL})_i}{\ln\left[\frac{(C^*_{AL} - C_{AL})_o}{(C^*_{AL} - C_{AL})_i}\right]}. \tag{9.53}$$

In Eq. (9.53), subscripts i and o represent the gas inlet and outlet ends of the vessel, respectively.

9.12 Summary of Chapter 9

At the end of Chapter 9 you should:

(i) be able to describe the *two-film theory* of mass transfer between phases;

(ii) know *Fick's law* in terms of the *diffusion coefficient*, $\mathscr{D}_{AB}$;
(iii) be able to describe in simple terms the mathematical analogy between mass, heat and momentum transfer;
(iv) know the empirical equation for rate of mass transfer in terms of the *mass-transfer coefficient* and *concentration-difference driving force*;
(v) know the importance of the *critical oxygen concentration*;
(vi) be able to identify which steps are most likely to present major resistances to oxygen mass transfer from bubbles to cells;
(vii) understand how oxygen mass-transfer and $k_L a$ can limit the biomass density in fermenters;
(viii) understand the mechanisms of *gas dispersion* and *coalescence* in stirred fermenters and the importance of bubble size in determining *gas hold-up* and $k_L a$;
(ix) know how temperature, total pressure, oxygen partial pressure and presence of dissolved and suspended material in the medium affect oxygen solubility and rates of oxygen mass transfer in fermenters; and
(x) know the techniques and limitations of *steady-state* and *dynamic methods* for experimental determination of $k_L a$ for oxygen transfer.

Problems

9.1 Rate-controlling processes in fermentation

Serratia marcescens bacteria are used for production of threonine. The maximum specific oxygen uptake rate of *S. marcescens* in batch culture is 5 mmol O_2 g^{-1} h^{-1}. The bacteria are grown in a stirred fermenter to a cell density of 40 g l^{-1}; $k_L a$ under these circumstances is 0.15 s^{-1}. At the fermenter operating temperature and pressure, the solubility of oxygen in the culture liquid is 8×10^{-3} kg m^{-3}. Is the rate of cell metabolism limited by mass-transfer, or dependent solely on metabolic kinetics?

9.2 $k_L a$ required to maintain critical oxygen concentration

A genetically-engineered strain of yeast is cultured in a bioreactor at 30°C for production of heterologous protein. The oxygen requirement is 80 mmol l^{-1} h^{-1}; the critical oxygen concentration is 0.004 mM. The solubility of oxygen in the fermentation broth is estimated to be 10% lower than in water due to solute effects.

(a) What is the minimum mass-transfer coefficient necessary to sustain this culture if the reactor is sparged with air at approximately 1 atm pressure?
(b) What mass-transfer coefficient is required if pure oxygen is used instead of air?

9.3 Single-point $k_L a$ determination using the oxygen-balance method

A 200-litre stirred fermenter contains a batch culture of *Bacillus subtilis* bacteria at 28°C. Air at 20°C is pumped into the vessel at a rate of 1 vvm; (vvm stands for volume of gas per volume of liquid per minute). The average pressure in the fermenter is 1 atm. The volumetric flow rate of off-gas from the fermenter is measured as 189 l min^{-1}. The exit gas stream is analysed for oxygen and is found to contain 20.1% O_2. The dissolved-oxygen concentration in the broth is measured using an oxygen electrode as 52% air saturation. The solubility of oxygen in the fermentation broth at 28°C and 1 atm air pressure is 7.8×10^{-3} kg m^{-3}.

(a) Calculate the oxygen transfer rate.
(b) Determine the value of $k_L a$ for the system.
(c) The oxygen analyser used to measure the exit gas composition has been incorrectly calibrated. If the oxygen content has been overestimated by 10%, what error is associated with the result for $k_L a$?

9.4 $k_L a$ measurement

Escherichia coli bacteria are cultured at 35°C in the following medium:

Component	g l^{-1}
glucose	20
sucrose	8.5
$CaCO_3$	1.3
$(NH_4)_2SO_4$	1.3
Na_2HPO_4	0.09
KH_2PO_4	0.12

The stirred fermenter used for this culture has an operating volume of 20 m^3 and a liquid height of 3.5 m. Air at 25°C is sparged into the bottom of the vessel at a rate of 25 m^3 min^{-1}. Oxygen tension in the fermenter is measured using polarographic electrodes located at the top and bottom of the vessel. At the top the reading is 50% air saturation; the reading at the bottom is 65%. The gas flow rate leaving the fermenter is measured with a rotary gas meter and is found to be 407 l s^{-1}. The oxygen content of the off-gas is 20.15%.

(a) Calculate the pressure at the sparger. (Static pressure p_s due to the height of liquid is given by the equation: $p_s = \rho g h$, where ρ is liquid density, g is gravitational acceleration, and h is liquid height.)
(b) Estimate the solubility of oxygen in the fermentation broth at 35°C and 1 atm air pressure using the correlation of Eq. (9.45).
(c) Estimate the solubility of oxygen at the bottom of the tank.
(d) Calculate the logarithmic-mean concentration driving force, $(C^*_{AL} - C_{AL})_L$.
(e) What is the oxygen transfer rate?
(f) Determine the value of $k_L a$.
(g) What is the maximum cell concentration that can be supported in this fermenter if the oxygen demand of the organism is 7.4 mmol $g^{-1} h^{-1}$?

9.5 Dynamic $k_L a$ measurement

The dynamic method is used to measure $k_L a$ in a fermenter operated at 30°C. Data for dissolved-oxygen concentration as a function of time during the re-oxygenation step is as follows:

Time (s)	C_{AL} (% air saturation)
10	43.5
15	53.5
20	60.0
30	67.5
40	70.5
50	72.0
70	73.0
100	73.5
130	73.5

The equilibrium concentration of oxygen in the broth is 7.9×10^{-3} kg m^{-3}. Determine $k_L a$.

9.6 Measurement of $k_L a$ as a function of stirrer speed: the oxygen-balance method of Mukhopadhyay and Ghose

Combining Eqs (9.37) and (9.48) gives:

$$k_L a\,(C^*_{AL} - C_{AL}) = \frac{1}{R V_L}\left[\left(\frac{F_g p_{AG}}{T}\right)_i - \left(\frac{F_g p_{AG}}{T}\right)_o\right]. \tag{9P6.1}$$

Let us make the following assumptions.

(i) the volumetric flow rate of exit gas is equal to that of the inlet, i.e. $(F_g)_i = (F_g)_o = F_g$; and
(ii) inlet and outlet gases have the same density, i.e.

$$\left(\frac{p_T}{RT}\right)_i = \left(\frac{p_T}{RT}\right)_o = \frac{p_T}{RT}$$

where p_T is total gas pressure.

Applying these assumptions, Eq. (9P6.1) can be written as:

$$k_L a\,(C^*_{AL} - C_{AL}) = \frac{p_T F_g}{R T V_L}\,[(y_{AG})_i - (y_{AG})_o] \tag{9P6.2}$$

where y_{AG} is the mole fraction of oxygen in the gas. Rearranging gives an expression for C_{AL}:

$$C_{AL} = C^*_{AL} - \frac{p_T F_g}{k_L a R T V_L}\,[(y_{AG})_i - (y_{AG})_o]. \tag{9P6.3}$$

In Eq. (9P6.3) the only variables are C_{AL} and $(y_{AG})_o$; C^*_{AL}, p_T, F_g, $k_L a$, R, T, V_L and $(y_{AG})_i$ are assumed to be constant. Accordingly, a linear plot of C_{AL} versus $[(y_{AG})_i - (y_{AG})_o]$ should give a straight line with slope

$$\frac{-p_T F_g}{k_L a R T V_L}$$

and intercept C^*_{AL}.

Data collected during fermentation of *Pseudomonas ovalis* B1486 at varying stirrer speeds are given by Mukhopadhyay and Ghose [36]. The fermenter volume was 3 litres. Air flow into the vessel was maintained at 1 vvm (vvm means volume of gas per volume of liquid per minute). The air pressure was 3 atm and the temperature 29°C. The following data were measured.

Fermentation time (h)	*Agitator speed*					
	300 rpm		*500 rpm*		*700 rpm*	
	C_{AL} (ppm)	$(y_{AG})_o$	C_{AL} (ppm)	$(y_{AG})_o$	C_{AL} (ppm)	$(y_{AG})_o$
0	5.9	21.0	5.9	21.0	5.9	21.0
4	–	–	5.6	20.9	5.7	20.9
5	5.3	20.9	–	–	–	–
6	–	–	5.2	20.8	5.4	20.8
7	4.7	20.8	4.9	20.7	5.1	20.7
8	4.1	20.7	4.4	20.6	4.7	20.6
9	3.4	20.6	4.0	20.5	4.1	20.4
10	3.4	20.6	4.0	20.5	4.1	20.4
11	3.5	20.7	4.2	20.6	4.2	20.5

(a) Determine $k_L a$ for each stirrer speed.
(b) What is the solubility of oxygen in the fermentation broth?
(c) By considering Eq. (9.39) and Eq. (9P6.2) together, calculate the maximum cellular oxygen uptake rate at each stirrer speed.

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Suggestions for Further Reading

Mass-transfer theory (see also refs 1, 2 and 4)

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Oxygen transfer in fermenters

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10

Unit Operations

Bioprocesses treat raw materials and generate useful products. Individual operations or steps within the process that change or separate components are called unit operations. Although the specific objectives of bioprocesses vary from factory to factory, each processing scheme can be viewed as a series of component operations which appear again and again in different systems. For example, most bioprocesses involve one or more of the following unit operations: centrifugation, chromatography, cooling, crystallisation, dialysis, distillation, drying, evaporation, filtration, heating, humidification, membrane separation, milling, mixing, precipitation, solids handling, solvent extraction. A particular sequence of unit operations used for manufacture of enzymes is shown in the flow sheet of Figure 10.1. Although the same operations are involved in other processes, the order in which they are carried out, the conditions used and the actual materials handled account for the differences in final results. Engineering principles for design of unit operations are independent of specific industries or applications.

In a typical fermentation process, raw materials are altered most significantly by reactions occurring in the fermenter. However physical changes before and after fermentation are also important to prepare the substrates for reaction and to extract and purify the desired product from the culture broth. The term 'unit operation' usually refers to the physical steps in processes; chemical or biochemical transformations are the subject of reaction engineering which is considered in detail in Chapters 11–13.

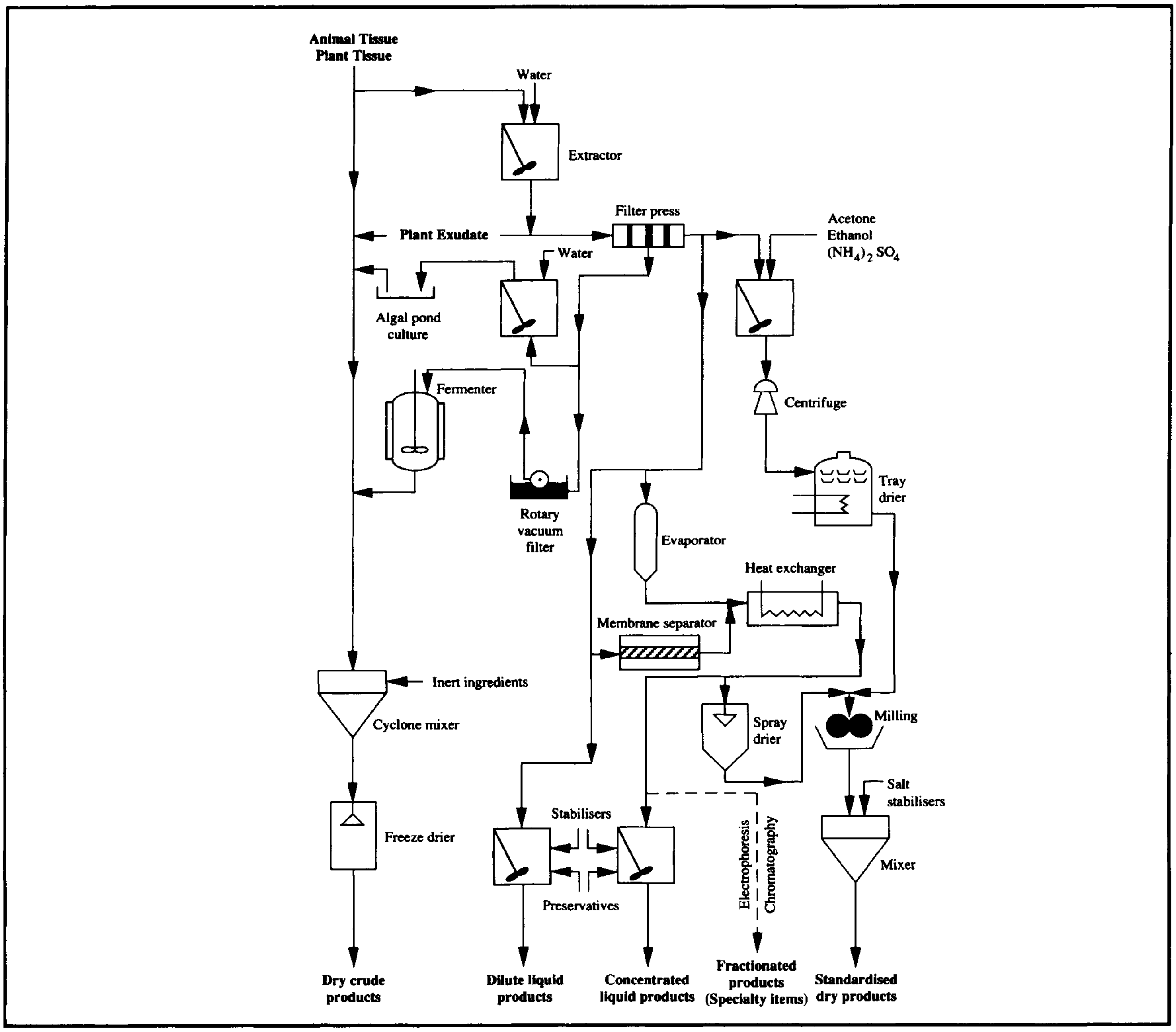
Fermentation broths are complex mixtures of components containing products in dilute solution. In bioprocessing, any treatment of the culture broth after fermentation is known as *downstream processing*. The purpose of downstream processing is to concentrate and purify the product for sale; in most cases this requires only physical modification. Although each recovery scheme will be different, downstream processing follows a general sequence of steps.

(i) *Cell removal.* A common first step in product recovery is removal of cells from the fermentation liquor. This is necessary if the biomass itself is the desired product, e.g. bakers' yeast, or if the product is contained within the cells. Removal of cells can also assist recovery of product from the liquid phase. Filtration and centrifugation are typical unit operations for cell removal.
(ii) *Primary isolation.* A wide variety of techniques is available for primary isolation of fermentation products from cells or cell-free broth. The method used depends on the physical and chemical properties of the product and surrounding material. The aim of primary isolation is to remove components with properties significantly different from those of the product. Typically, processes for primary isolation treat large volumes of material and are relatively non-selective; however significant increases in product quality and concentration can be accomplished. Unit operations such as adsorption, liquid extraction and precipitation are used for primary isolation.
(iii) *Purification.* Processes for purification are highly selective and separate the product from impurities with similar properties. Typical unit operations are chromatography, ultrafiltration and fractional precipitation.
(iv) *Final isolation.* The final purity required depends on the product application. Crystallisation, followed by centrifugation or filtration and drying, are typical operations used for high-quality products such as pharmaceuticals.

A typical profile of product quality through the various stages of downstream processing is given in Table 10.1.

Downstream processing can account for a substantial part of the total production cost of a fermentation product. For example, the ratio of fermentation cost to cost of product recovery is approximately 60:40 for antibiotics such as penicillin. For newer antibiotics this ratio is reversed; product recovery is more costly than fermentation. Many modern products of biotechnology such as recombinant proteins and monoclonal antibodies require expensive downstream processing which accounts for 80–90% of process costs [1]. Starting product levels before recovery have a strong influence on cost;

Figure 10.1 Unit operations used in manufacture of enzymes. (From B. Atkinson and F. Mavituna, 1991, *Biochemical Engineering and Biotechnology Handbook*, 2nd edn, Macmillan, Basingstoke; and W.T. Faith, C.E. Neubeck and E.T. Reese, 1971, Production and applications of enzymes, *Adv. Biochem. Eng.* 1, 77–111.)



purification is more expensive when the concentration of product in the biomass or fermentation broth is low. As illustrated in Figure 10.2, the higher the starting concentration the cheaper is the final product. Each downstream-processing step involves some product loss; these losses can be substantial for multi-step procedures. For example, if 80% of the product is retained at each purification step, after a five-step process the overall product recovery is only about one-third. If the starting concentration is very low, more recovery stages are required with higher attendant losses and costs. This situation can be improved by either enhancing product synthesis during fermentation or developing better downstream-processing techniques which minimise product loss.

There is an extensive literature on downstream processing, much of it dealing with recent advances. To thoroughly cover all unit operations used in bioprocessing is beyond the scope of this book; at least one entire separate volume would be required. Rather than attempt such a treatise, this chapter considers the

Table 10.1 Typical profile of product quality during downstream processing

(From P.A. Belter, E.L. Cussler and W.-S. Hu, 1988, Bioseparations: Downstream Processing For Biotechnology, *John Wiley, New York)*

Step	*Typical unit operation*	*Product concentration* ($g\,l^{-1}$)	*Product quality* (%)
Harvest broth	–	0.1–5	0.1–1
Cell removal	Filtration	1–5	0.2–2
Primary isolation	Extraction	5–50	1–10
Purification	Chromatography	50–200	50–80
Final isolation	Crystallisation	50–200	90–100

engineering principles of a small selection of unit operations commonly applied for recovery of fermentation products. Information about other equally-important unit operations can be found in references listed at the end of the chapter.

10.1 Filtration

In filtration, solid particles are separated from a fluid–solid mixture by forcing the fluid through a *filter medium* or *filter cloth* which retains the particles. Solids are deposited on the filter and, as the deposit or *filter cake* increases in depth, pose a resistance to further filtration. Filtration can be performed using either vacuum or positive-pressure equipment. The pressure difference exerted across the filter to separate fluid from the solids is called the filtration *pressure drop.*

Ease of filtration depends on the properties of the solid and fluid; filtration of crystalline, incompressible solids in low-viscosity liquids is relatively straightforward. In contrast, fermentation broths can be difficult to filter because of the small size and gelatinous nature of the cells and the viscous non-Newtonian behaviour of the broth. Most microbial filter cakes are *compressible,* i.e. the porosity of the cake declines as pressure drop across the filter increases. This can be a major problem causing reduced filtration rates and greater loss of product. Filtration of fermentation broths is usually carried out under non-aseptic conditions; the process must therefore be efficient and reliable to avoid undue contamination and degradation of labile products.

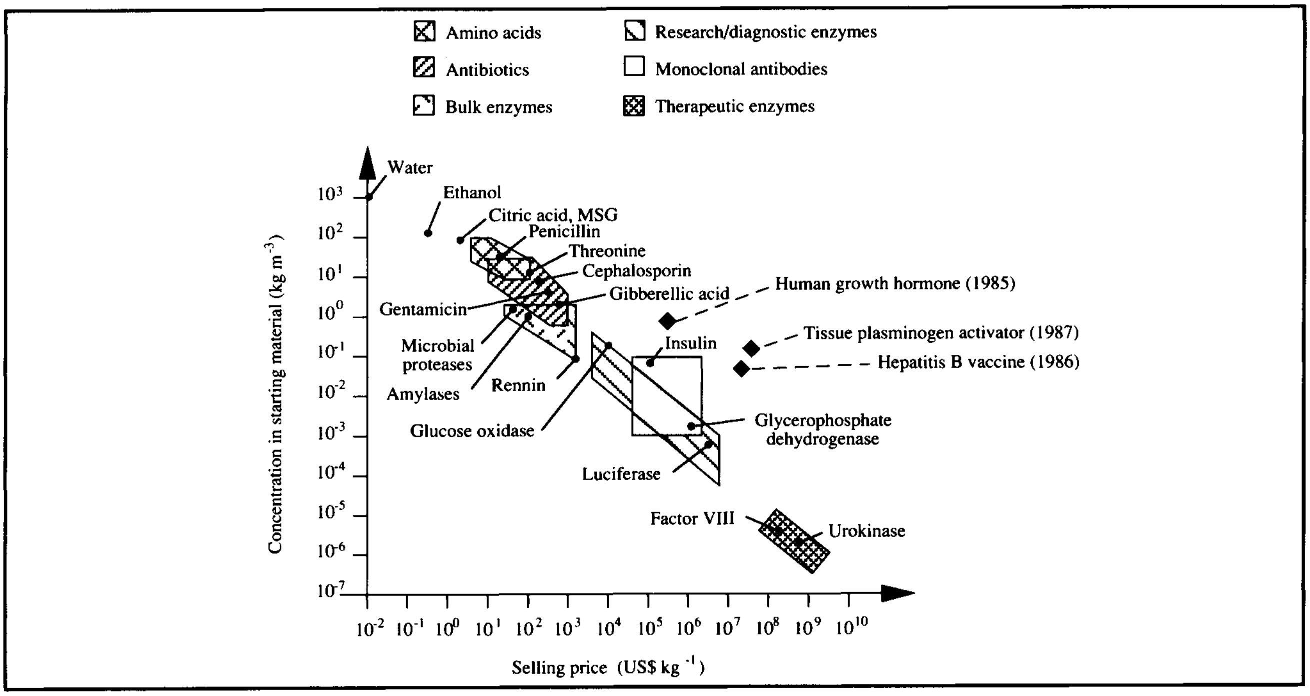
10.1.1 Filter Aids

Filter aids such as diatomaceous earth have found widespread use in the fermentation industry to improve the efficiency of filtration. Diatomaceous earth, also known as kieselguhr, is the fused skeletal remains of diatoms. Packed beds of granulated kieselguhr have very high porosity; only about 15% of the total volume of packed kieselguhr is solid, the rest is empty space. Such high porosity facilitates liquid flow around the particles and improves rate of filtration through the bed.

Filter aids are applied in two ways. As shown in Figure 10.3, filter aid can be used as a pre-coat on the filter medium to prevent blockage or 'blinding' of the filter by solids which would otherwise wedge themselves into the pores of the cloth. Filter aid can also be added to the fermentation broth to increase the porosity of the cake as it forms. This is only recommended when the fermentation product is extracellular; for intracellular products requiring further processing of the cells, severe handling problems can arise if the cake is contaminated with filter aid. Filter aid adds to the cost of filtration; the minimum quantity needed to achieve the desired result must be established experimentally. Kieselguhr absorbs liquid; therefore, if the fermentation product is in the liquid phase, some will be lost. Another disadvantage is that as filtration rate increases with the assistance of filter aid, filtrate clarity is reduced. Disposal of waste cell material is more difficult if it contains kieselguhr; for example, biomass cannot be used as animal feed unless the filter aid is removed.

Fermentation broths can be pre-treated to improve filtration characteristics. Heating to denature proteins enhances the filterability of mycelial broths such as in penicillin production. Alternatively, electrolytes may be added to promote coagulation of colloids into larger, denser particles which are easier to filter. Ease of filtration is also affected by the duration of the fermentation; this affects the composition and viscosity of the medium and properties of the cell cake.

Figure 10.2 Relationship between selling price and concentration before downstream processing for several fermentation products. (From J.L. Dwyer, 1984, Scaling up bio-product separation with high performance liquid chromatography, *Bio/Technology* 2, 957–964; and J. van Brunt, 1988, How big is big enough? *Bio/Technology* 6, 479–485.)



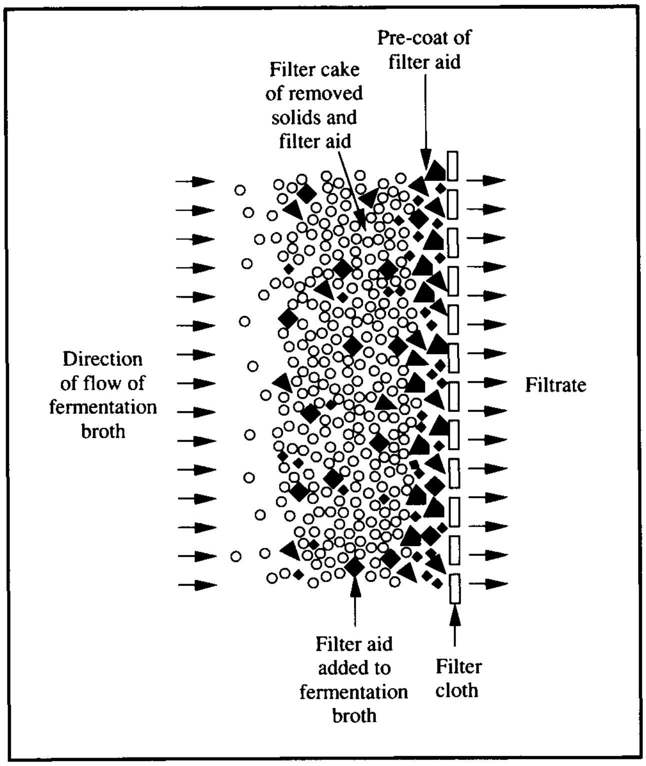
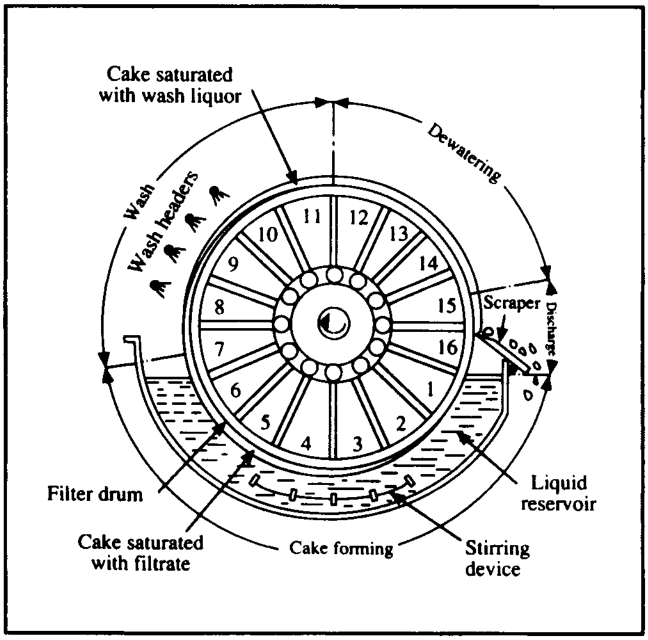


Figure 10.3 Use of filter aid in filtration of fermentation broth.

10.1.2 Filtration Equipment

Plate filters are suitable for filtration of small fermentation batches; this type of filter gradually accumulates biomass and must be periodically opened and cleared of filter cake. Larger processes require continuous filters. ***Rotary-drum vacuum filters***, such as that shown in Figure 10.4, are the most widely-used filtration devices in the fermentation industry. A horizontal drum 0.5–3 m in diameter is covered with filter cloth and rotated slowly at 0.1–2 rpm. The cloth is partially immersed in an agitated reservoir containing material to be filtered. As a section of drum enters the liquid, a vacuum is applied from the interior of the drum. A cake forms on the face of the cloth while liquid is drawn through internal pipes to a collection tank. As the drum rotates out of the reservoir, the surface of the filter is sprayed with wash liquid which is drawn through the cloth and collected in a separate holding tank. After washing, the cake is dewatered by continued application of the vacuum. The vacuum is turned off as the drum reaches the discharge zone where the cake is removed by means of a scraper, knife or strings. Air pressure may be applied at this

Figure 10.4 Continuous rotary-drum vacuum filter. (From G.G. Brown, A.S. Foust, D.L. Katz, R. Schneidewind, R.R. White, W.P. Wood, G.M. Brown, L.E. Brownell, J.J. Martin, G.B. Williams, J.T. Banchero and J.L. York, 1950, *Unit Operations*, John Wiley, New York.)



stage to help dislodge the filter cake from the cloth. After the cake is removed, the drum re-enters the reservoir for another filtration cycle.

10.1.3 Filtration Theory

Filtration theory is used to estimate rate of filtration. For a given pressure drop across the filter, rate of filtration is greatest just as filtering begins. This is because resistance to filtration is at a minimum when there are no deposited solids. Orientation of particles in the initial cake deposit is very important and can significantly influence the structure and permeability of the whole filter bed. Excessive pressure drop and high initial rates of filtration can result in plugging of the filter cloth and very high subsequent resistance to flow. Flow resistance due to the filter cloth can be considered constant if particles do not penetrate the material; resistance due to the cake increases with thickness.

Rate of filtration is usually measured as the rate at which liquid filtrate is collected. Filtration rate depends on the area of the filter cloth, the viscosity of the fluid, the pressure difference across the filter, and the resistance to filtration offered by the cloth and deposited filter cake. At any instant during filtration, rate of filtration is given by the equation:

$$\frac{1}{A}\frac{dV_f}{dt} = \frac{\Delta p}{\mu_f\left[\alpha\left(\frac{M_c}{A}\right)+r_m\right]} \tag{10.1}$$

where A is filter area, V_f is volume of filtrate, t is filtration time, Δp is pressure drop across the filter, μ_f is filtrate viscosity, M_c is the total mass of solids in the cake, α is the average *specific cake resistance* and r_m is the *filter medium resistance*. r_m includes the effect of the filter cloth and any particles wedged in it during the initial stages of filtration. α has dimensions LM^{-1}; r_m has dimensions L^{-1}. ${}^{dV_f}/_{dt}$ is the filtrate flow rate or *volumetric rate of filtration*. Area A represents the capital cost of the filter; the bigger the area required to achieve a given filtration rate, the larger is the equipment and related investment. α is a measure of the resistance of the filter cake to flow; its value depends on the shape and size of the particles, the size of the interstitial spaces between them, and the mechanical stability of the cake. Resistance due to the filter medium is often negligible compared with cake resistance.

If the filter cake is incompressible, the specific cake resistance α does not vary with pressure drop across the filter. However, cakes from fermentation broths are seldom incompressible; as these cakes compress, filtration rates decline. For a compressible cake, α can be related empirically to Δp as follows:

$$\alpha = \alpha'(\Delta p)^s \tag{10.2}$$

where s is cake *compressibility* and α' is a constant dependent largely on the size and morphology of particles in the cake. The value of s is zero for rigid incompressible solids; for highly compressible material s is close to unity. α is also related to the average properties of particles in the cake as follows:

$$\alpha = \frac{K_v a^2(1-\varepsilon)}{\varepsilon^3 \rho_p} \tag{10.3}$$

where K_v is a factor depending on the shape of the particles, a is the specific surface area of the particles, ε is *porosity* of the cake and ρ_p is density of the particles. a and ε are defined as follows:

$$a = \frac{\text{surface area of a single particle}}{\text{volume of a single particle}}\ ; \tag{10.4}$$

$$\varepsilon = \frac{\text{total volume of the cake} - \text{volume of solids in the cake}}{\text{total volume of the cake}} \tag{10.5}$$

For compressible cakes, both ε and K_v depend on filtration pressure drop.

It is useful to consider methods for improving rate of filtration for a given batch of material. Various strategies can be deduced from the relationship between variables in Eq. (10.1).

(i) *Increase the filter area A.* When all other parameters remain constant, rate of filtration is improved if A is increased. However, this requires installation of larger filtration equipment and greater capital cost.
(ii) *Increase the filtration pressure drop Δp.* The problem with this approach for compressible cakes is that α increases with Δp as indicated in Eq. (10.2); higher α results in lower filtration rate. In practice, pressure drops are usually kept below 0.5 atm to minimise cake resistance. Improving filtration rate by increasing the pressure drop can only be achieved by reducing s, the compressibility of the cake. Addition of filter aid in the broth can reduce s to some extent.
(iii) *Reduce the cake mass M_c.* This is achieved in continuous rotary filtration by reducing the thickness of cake deposited per revolution of the drum, and ensuring that the scraper leaves minimal cake residue on the filter cloth.
(iv) *Reduce the liquid viscosity μ_f.* Material to be filtered is sometimes diluted if the starting viscosity is very high.
(v) *Reduce the specific cake resistance α.* From Eq. (10.3), possible methods of reducing α for compressible cakes are as follows:
    (a) *Increase the porosity ε.* Cake porosity usually decreases as cells are filtered. Application of filter aid reduces this effect.
    (b) *Reduce the shape factor of the particles K_v.* In the case of mycelial broths, it may be possible to change the morphology of the cells by manipulating fermentation conditions.
    (c) *Reduce the specific surface area of the particles a.* Increasing the average size of the particles and minimising variation in particle size reduce the value of a. Changes in fermentation conditions and broth pretreatment are used to achieve these effects.

Integration of Eq. (10.1) allows us to calculate the time required to filter a given volume of material. Before carrying out the integration, let us substitute an expression for mass of solids in the cake as a function of filtrate volume:

$$M_c = c\,V_f \tag{10.6}$$

where c is the mass of solids deposited per volume of filtrate and is related to the concentration of solid in the material to be filtered. Substituting Eq. (10.6) into Eq. (10.1), the expression for rate of filtration becomes:

$$\frac{1}{A}\frac{dV_f}{dt} = \frac{\Delta p}{\mu_f\left[\alpha\left(\frac{cV_f}{A}\right)+r_m\right]}. \tag{10.7}$$

A filter can be operated in two different ways. If the pressure drop across the filter is kept constant, the filtration rate will become progressively smaller as resistance due to the cake increases. On the other hand, in constant-rate filtration the flow rate is maintained by gradually increasing the pressure drop. Filtrations are most commonly carried out at constant pressure. When this is the case, Eq. (10.7) can be integrated directly because V_f and t are the only variables; for a given filtration device and material to be filtered, each of the remaining parameters is constant.

It is convenient for integration to write Eq. (10.7) in its reciprocal form:

$$A\frac{dt}{dV_f} = \mu_f\alpha c\frac{V_f}{A\Delta p} + \frac{\mu_f r_m}{\Delta p}. \tag{10.8}$$

At the beginning of filtration $t=0$ and $V_f=0$; this is the initial condition for integration. Separating variables and placing constants outside the integral signs gives:

$$A\int dt = \left(\frac{\mu_f\alpha c}{A\,\Delta p}\right)\int V_f\,dV_f + \left(\frac{\mu_f r_m}{\Delta p}\right)\int dV_f. \tag{10.9}$$

Carrying out the integration:

$$At = \left(\frac{\mu_f\alpha c}{2A\,\Delta p}\right)V_f^2 + \left(\frac{\mu_f r_m}{\Delta p}\right)V_f. \tag{10.10}$$

Thus, for constant-pressure filtration, Eq. (10.10) can be used to calculate either filtrate volume V_f or filtration time t provided all the constants are known. α and r_m for a particular filtration must be evaluated beforehand. For experimental determination of α and r_m, Eq. (10.10) is rearranged by dividing both sides of the equation by AV_f:

$$\frac{t}{V_f} = \left(\frac{\mu_f\alpha c}{2A^2\Delta p}\right)V_f + \left(\frac{\mu_f r_m}{A\Delta p}\right). \tag{10.11}$$

Eq. (10.11) can be written more simply as:

$$\frac{t}{V_f} = K_1 V_f + K_2 \tag{10.12}$$

where:

$$K_1 = \frac{\mu_f \alpha c}{2A^2 \Delta p} \tag{10.13}$$

and

$$K_2 = \frac{\mu_f r_m}{A \, \Delta p}. \tag{10.14}$$

K_1 and K_2 are constant during constant-pressure filtration. Eq. (10.12) is therefore an equation of a straight line when t/V_f is plotted against V_f. The slope K_1 depends on the filtration pressure drop and properties of the cake; the intercept K_2 also depends on pressure drop but is independent of cake properties. α is calculated from the slope; r_m is determined from the intercept. Eq. (10.12) is valid for compressible cakes; however K_1 becomes a more complex function of Δp than is directly apparent from Eq. (10.13) because of the dependence of α on pressure.

Eq. (10.11) is the basic filtration equation for industrial-scale equipment such as the rotary-drum vacuum filter shown in Figure 10.4; rates of filtration during cake formation and washing determine the size of the filter required for the process. Simple modifications to Eq. (10.11) give equations which are directly applicable to rotary-drum filters [2].

Example 10.1 Filtration of mycelial broth

A 30-ml sample of broth from a penicillin fermentation is filtered in the laboratory on a 3 cm^2 filter at a pressure drop of 5 psi. The filtration time is 4.5 min. Previous studies have shown that filter cake of *Penicillium chrysogenum* is significantly compressible with $s = 0.5$. If 500 litres broth from a pilot-scale fermenter must be filtered in 1 hour, what size filter is required if the pressure drop is:

(a) 10 psi?
(b) 5 psi?

Resistance due to the filter medium is negligible.

Solution:
Properties of the filtrate and mycelial cake can be determined from results of the laboratory experiment. If r_m can be eliminated from Eq. (10.11):

$$\frac{t}{V_f} = \left(\frac{\mu_f \alpha c}{2A^2 \Delta p}\right) V_f.$$

Substituting α for a compressible cake from Eq. (10.2):

$$\frac{t}{V_f} = \left(\frac{\mu_f \alpha' (\Delta p)^{s-1} c}{2A^2}\right) V_f \tag{1}$$

and rearranging:

$$\mu_f \alpha' c = \frac{2A^2 t}{(\Delta p)^{s-1} V_f^2}.$$

Substituting values:

$$\mu_f \alpha' c = \frac{2(3\ \text{cm}^2)^2\,(4.5\ \text{min})}{(5\ \text{psi})^{0.5-1}\,(30\ \text{cm}^3)^2} = 0.201\ \text{cm}^{-2}\,(\text{psi})^{0.5}\ \text{min}.$$

This value for $\mu_f \alpha' c$ is used to evaluate the area required for pilot-scale filtration. From (1):

$$A^2 = \left(\frac{\mu_f \alpha' c(\Delta p)^{s-1}}{2t}\right) V_f^2$$

therefore:

$$A = \left(\frac{\mu_f \alpha' c(\Delta p)^{s-1}}{2t}\right)^{1/2} V_f.$$

(a) Substituting values when $\Delta p = 10$ psi:

$$A = \left(\frac{(0.201\ \mathrm{cm}^{-2}\ (\mathrm{psi})^{0.5}\ \mathrm{min})\ .\ (10\ \mathrm{psi})^{0.5-1}}{2\ (1\ \mathrm{h})\ .\ \left|\frac{60\ \mathrm{min}}{1\ \mathrm{h}}\right|}\right)^{1/2} (500\ \mathrm{l}).\ \left|\frac{1000\ \mathrm{cm}^3}{1\ \mathrm{l}}\right|$$

$$= 1.15 \times 10^4\ \mathrm{cm}^2 = 1.15\ \mathrm{m}^2.$$

(b) When $\Delta p = 5$ psi:

$$A = \left(\frac{(0.201\ \mathrm{cm}^{-2}\ (\mathrm{psi})^{0.5}\ \mathrm{min})\ .\ (5\ \mathrm{psi})^{0.5-1}}{2\ (1\ \mathrm{h})\ .\ \left|\frac{60\ \mathrm{min}}{1\ \mathrm{h}}\right|}\right)^{1/2} (500\ \mathrm{l}).\ \left|\frac{1000\ \mathrm{cm}^3}{1\ \mathrm{l}}\right|$$

$$= 1.37 \times 10^4\ \mathrm{cm}^2 = 1.37\ \mathrm{m}^2.$$

Halving the pressure drop increases the area required by only 20% because at 5 psi the cake is less compressed and more porous than at 10 psi.

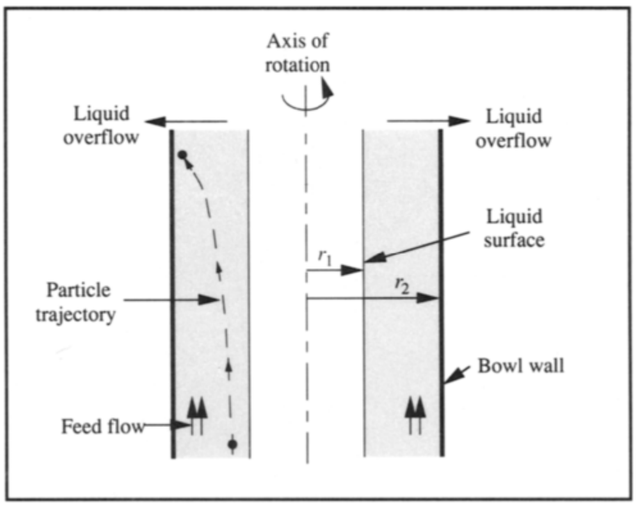
Example 10.1 underlines the importance of laboratory testing in design of filtration systems. Experiments are required to evaluate properties of the cake such as compressibility and specific resistance; these parameters cannot be calculated from theory. Experimental observations are also necessary to evaluate a wide range of other important filtration characteristics. These include filtrate clarity, ease of washing, dryness of the final cake, ease of cake removal, and effects of filter aids and broth pre-treatment. It is essential that any laboratory tests be conducted with the same materials as the large-scale process. Variables such as temperature, age of the broth, and presence of contaminants and cell debris have significant effects on filtration characteristics.

In general, fungal mycelia are filtered relatively easily because mycelial filter cake has sufficiently large porosity. Yeast and bacteria are much more difficult to handle because of their small size. Alternative filtration methods which eliminate the filter cake are becoming more accepted for bacterial and yeast separations. *Microfiltration* is achieved by developing large cross-flow fluid velocities across the filter surface while the velocity normal to the surface is relatively small. Build-up of filter cake and problems of high cake resistance are therefore prevented. Microfiltration is not covered in this text; further details are available elsewhere, e.g. [3, 4].

10.2 Centrifugation

Centrifugation is used to separate materials of different density when a force greater than gravity is desired. In

Figure 10.5 Separation of solids in a tubular-bowl centrifuge.



bioprocessing, centrifugation is used to remove cells from fermentation broth, to eliminate cell debris, to collect precipitates, and to prepare fermentation media such as in clarification of molasses or production of wort for brewing. Equipment for centrifugation is more expensive than for filtration; however centrifugation is often effective when the particles are very small and difficult to filter. Centrifugation of fermentation broth produces a thick, concentrated cell sludge or cream containing more liquid than filter cake.

Steam-sterilisable centrifuges are applied when either the cells or fermentation liquid is recycled to the fermenter or when product contamination must be prevented. Industrial centrifuges generate large amounts of heat due to friction; it is therefore necessary to have good ventilation and cooling. Aerosols created by fast-spinning centrifuges have been known to cause infections and allergic reactions in factory workers so that isolation cabinets are required for certain applications.

Centrifugation is most effective when the particles to be separated are large, the liquid viscosity is low, and the density difference between particles and fluid is great. It is also assisted by large centrifuge radius and high rotational speed. In centrifugation of biological solids such as cells, the particles are very small, the viscosity of the medium can be relatively high, and the particle density is very similar to the suspending fluid. These disadvantages are easily overcome in the laboratory with small centrifuges operated at high speed. However, problems arise in industrial centrifugation when large quantities of material must be treated. Centrifuge capacity cannot be increased by simply increasing the size of the equipment without limit; mechanical stress in centrifuges increases in proportion to (radius)^2 so that safe operating speeds are substantially lower in large equipment. The need for continuous throughput of material in industrial applications also restricts practical operating speeds. To overcome these difficulties, a

Figure 10.6 Disc-stack bowl centrifuge with continuous discharge of solids.

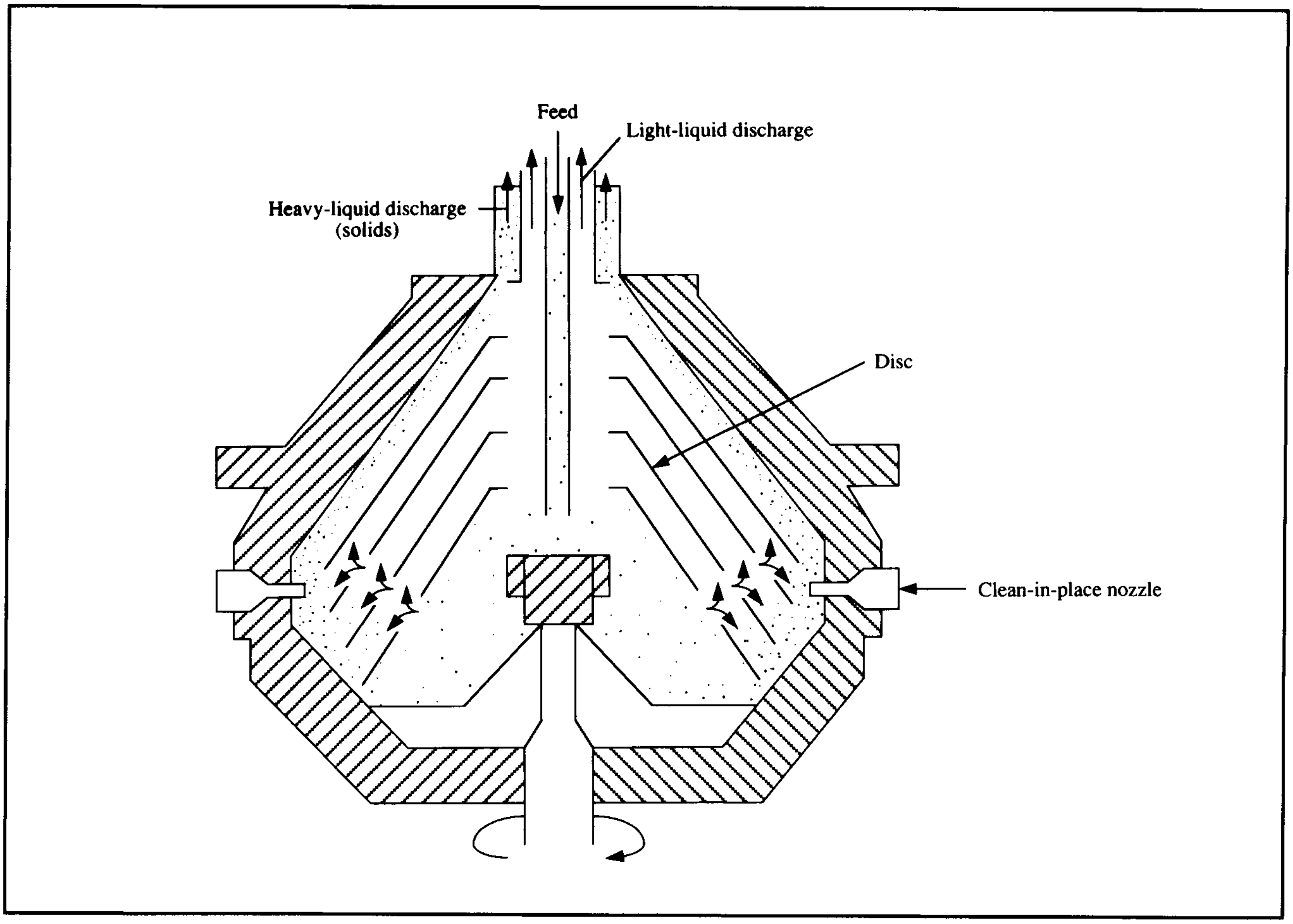
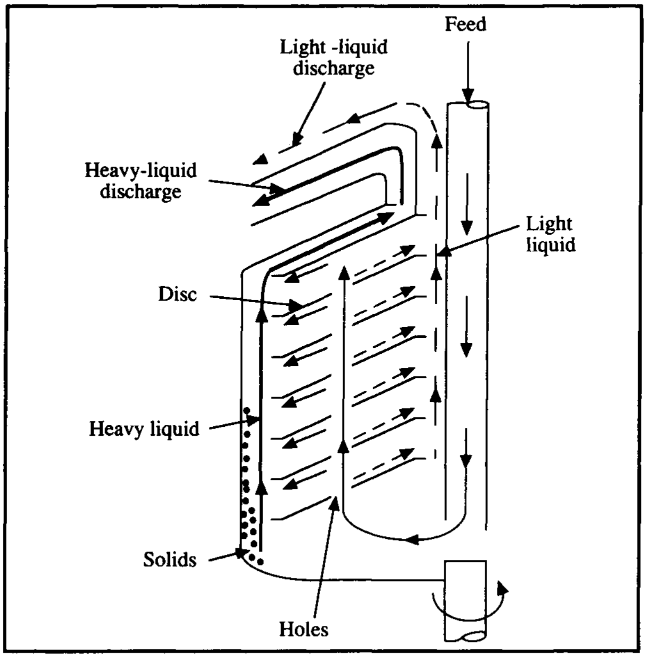


Figure 10.7 Mechanism of solids separation in a disc-stack bowl centrifuge. (From C.J. Geankoplis, 1983, *Transport Processes and Unit Operations*, 2nd edn, Allyn and Bacon, Boston.)



range of centrifuges has been developed for bioprocessing. Types of centrifuge commonly used in industrial operations are described below.

10.2.1 Centrifuge Equipment

Centrifuge equipment is classified according to internal structure. The *tubular-bowl centrifuge* has the simplest configuration and is widely employed in the food and pharmaceutical industries. Feed enters under pressure through a nozzle at the bottom, is accelerated to rotor speed, and moves upwards through the cylindrical bowl. As the bowl rotates, particles travelling upward are spun out and collide with the walls of the bowl as illustrated schematically in Figure 10.5. Solids are removed from the liquid if they move with sufficient velocity to reach the wall of the bowl within the residence time of liquid in the machine. As the feed rate is increased the liquid layer moving up the wall of the centrifuge becomes thicker; this reduces performance of the centrifuge by increasing the distance a particle must travel to reach the wall. Liquid from the feed spills over a weir at the top of the bowl; solids which have collided with the walls are collected separately. When the thickness of sediment collecting in the bowl reaches the position of the liquid-overflow weir, separation efficiency declines rapidly. This limits the capacity of the centrifuge. Tubular centrifuges are applied mainly for difficult separations requiring high centrifugal forces. Solids in tubular centrifuges are accelerated by forces between 13 000 and 16 000 times the force of gravity.

A type of narrow tubular-bowl centrifuge is the *ultracentrifuge*. This device is used for recovery of fine precipitates from high-density solutions, for breaking down emulsions, and for separation of colloidal particles such as ribosomes and mitochondria. It produces centrifugal forces 10^5–10^6 times the force of gravity. The bowl is usually air-driven and operated at low pressure or in an atmosphere of nitrogen to reduce generation of frictional heat. The main commercial application of ultracentrifuges has been in production of vaccines to separate viral particles from cell debris. Ultracentrifugation has also been tested for removal of very fine cell debris in isolation of enzymes, and for ribosome removal in purification of RNA polymerase. A typical ultracentrifuge operates discontinuously so its processing capacity is restricted by the need to empty the bowl manually. Continuous ultracentrifuges are available commercially; however safe operating speeds with these machines are not as high as in batch equipment.

An alternative to the tubular centrifuge is the *disc-stack bowl centrifuge*. Disc-stack centrifuges are common in bioprocessing. There are many types of disc centrifuge; the principal difference between them is the method used to discharge the accumulated solid. In simple disc centrifuges, solids must be removed periodically by hand. Continuous or intermittent discharge of solids is possible in a variety of disc centrifuges without reducing the bowl speed. Some centrifuges are equipped with peripheral nozzles for continuous solids removal; others have valves for intermittent discharge. Another method is to concentrate the solids in the periphery of the bowl and then discharge them at the top of the centrifuge using a paring device; the equipment configuration for this mode of operation is shown in Figure 10.6. A disadvantage of centrifuges with automatic discharge of solids is that the solids must remain sufficiently wet to flow through the machine. Extra nozzles may be provided for cleaning the bowl should blockages of the system occur.

Disc-stack centrifuges contain conical sheets of metal called discs which are stacked one on top of the other with clearances as small as 0.3 mm. The discs rotate with the bowl and their function is to split the liquid into thin layers. As shown in Figure 10.7, feed is released near the bottom of the centrifuge and travels upwards through matching holes in the discs. Between the discs, heavy components of the feed are thrown outward under the influence of centrifugal forces as lighter liquid is displaced towards the centre of the bowl. As they are flung out, the solids strike the undersides of the discs and slide down to the bottom edge of the bowl. At the same time, the lighter liquid flows in and over the upper surfaces of the discs to be discharged from the top of the bowl. Heavier liquid containing solids can be discharged either at the top of the centrifuge or through nozzles around the periphery of the bowl. Disc-stack centrifuges used in bioprocessing typically develop forces of 5000–15 000 times gravity. As a guide, the minimum density difference between solid and liquid for successful separation in a disc-stack centrifuge is approximately 0.01–0.03 kg m^{-3}; in practical operations at appropriate flow rates the minimum particle diameter separated is about 0.5 μm [5].

10.2.2 Centrifugation Theory

The particle velocity achieved in a particular centrifuge compared with the settling velocity which would occur under the influence of gravity characterises the effectiveness of centrifugation. The terminal velocity during gravity settling of a small spherical particle in dilute suspension is given by Stoke's law:

$$u_g = \frac{\rho_p - \rho_f}{18\,\mu}\, D_p^2\, g \tag{10.15}$$

where u_g is the sedimentation velocity under gravity, ρ_p is density of the particle, ρ_f is density of the liquid, μ is viscosity of the liquid, D_p is particle diameter and g is gravitational acceleration. In a centrifuge, the corresponding terminal velocity is:

$$u_c = \frac{\rho_p - \rho_f}{18\,\mu}\, D_p^2\, \omega^2 r \tag{10.16}$$

where u_c is particle velocity in the centrifuge, ω is angular velocity of the bowl in units of rad s^{-1} and r is radius of the centrifuge drum. The ratio of velocity in the centrifuge to velocity under gravity is called the *centrifuge effect* or *g-number*, and is usually denoted Z. Therefore:

$$Z = \frac{\omega^2 r}{g}\,. \tag{10.17}$$

The force developed in a centrifuge is Z times the force of gravity, and is often expressed as so many g-forces. Industrial centrifuges have Z factors from 300 to 16 000; for small laboratory centrifuges Z may be up to 500 000 [6].

Sedimentation occurs in a centrifuge as particles moving away from the centre of rotation collide with the walls of the centrifuge bowl. Increasing the velocity of motion will improve the rate of sedimentation. From Eq. (10.16), particle velocity in a given centrifuge can be increased by:

(i) increasing the centrifuge speed ω;
(ii) increasing the particle diameter D_p;
(iii) increasing the density difference between particle and liquid, $\rho_p - \rho_f$; and
(iv) decreasing the viscosity of the suspending fluid, μ.

However, whether the particles reach the walls of the bowl also depends on the time of exposure to the centrifugal force. In batch centrifuges such as those used in the laboratory, centrifuge time is increased by running the equipment longer. In continuous-flow devices such as the disc-stack centrifuge, the residence time is increased by decreasing the feed flow rate.

Performance of centrifuges of different size can be compared using a parameter called the *sigma factor* Σ. Physically, Σ represents the cross-sectional area of a gravity settler with the same sedimentation characteristics as the centrifuge. For continuous centrifuges, Σ is related to the feed rate of material as follows:

$$\Sigma = \frac{Q}{2\,u_g} \tag{10.18}$$

where Q is the volumetric feed rate and u_g is the terminal velocity of the particles in a gravitational field. If two centrifuges perform with equal effectiveness:

$$\frac{Q_1}{\Sigma_1} = \frac{Q_2}{\Sigma_2} \tag{10.19}$$

where subscripts 1 and 2 denote the two centrifuges. Eq. (10.19) can be used to scale-up centrifuge equipment. Equations for evaluating Σ depend on the centrifuge design. For a disc-stack bowl centrifuge [7]:

$$\Sigma = \frac{2\pi\,\omega^2 (N-1)}{3g\,\tan\theta}\,(r_2^3 - r_1^3) \tag{10.20}$$

where ω is angular velocity in rad s^{-1}, N is number of discs in the stack, r_2 is outer radius of the disc, r_1 is inner radius of the disc, g is gravitational acceleration, and θ is the half-cone angle of the disc. For a tubular-bowl centrifuge, the following equation is accurate to within 4% [8, 9]:

$$\Sigma = \frac{\pi\,\omega^2 b}{2g}\,(3r_2^2 + r_1^2) \tag{10.21}$$

where b is length of the bowl, r_1 is radius of the liquid surface and r_2 is radius of the inner wall of the bowl. Because r_1 and r_2 in a tubular-bowl centrifuge are about equal, Eq. (10.21) can be approximated as:

$$\Sigma = \frac{2\pi\,\omega^2 b r^2}{g} \tag{10.22}$$

where r is an average radius roughly equal to either r_1 or r_2.

The above equations for Σ are based on ideal operating conditions. Because different types of centrifuge deviate to varying degrees from ideal operation, Eq. (10.19) cannot generally be used to compare different centrifuge configurations. Performance of any centrifuge can deviate from theoretical prediction due to factors such as particle shape and size distribution, aggregation of particles, non-uniform flow distribution in the centrifuge and interaction between particles during sedimentation. Experimental tests must be performed to account for these factors.

Example 10.2 Cell recovery in a disc-stack centrifuge

A continuous disc-stack centrifuge is operated at 5000 rpm for separation of bakers' yeast. At a feed rate of 60 l min^{-1}, 50% of the cells are recovered. At constant centrifuge speed, solids recovery is inversely proportional to flow rate.

(a) What flow rate is required to achieve 90% cell recovery if the centrifuge speed is maintained at 5000 rpm?
(b) What operating speed is required to achieve 90% recovery at a feed rate of 60 l min^{-1}?

Solution:

(a) If solids recovery is inversely proportional to feed rate, the flow rate required is:

$$\frac{50\%}{90\%}\,(60\,\mathrm{l\,min^{-1}}) = 33.3\,\mathrm{l\,min^{-1}}.$$

(b) Eq. (10.19) relates operating characteristics of centrifuges achieving the same separation. From (a), 90% recovery is achieved at $Q_1 = 33.3$ l min^{-1} and $\omega_1 = 5000$ rpm. $Q_2 = 60$ l min^{-1}. From Eq. (10.19):

$$\frac{Q_1}{Q_2} = \frac{\Sigma_1}{\Sigma_2} = \frac{33.3\,\mathrm{l\,min^{-1}}}{60\,\mathrm{l\,min^{-1}}} = 0.56.$$

Because the same centrifuge is used and all the geometric parameters are the same, from Eq. (10.20):

$$\frac{\Sigma_1}{\Sigma_2} = \frac{\omega_1^2}{\omega_2^2} = 0.56.$$

Therefore:

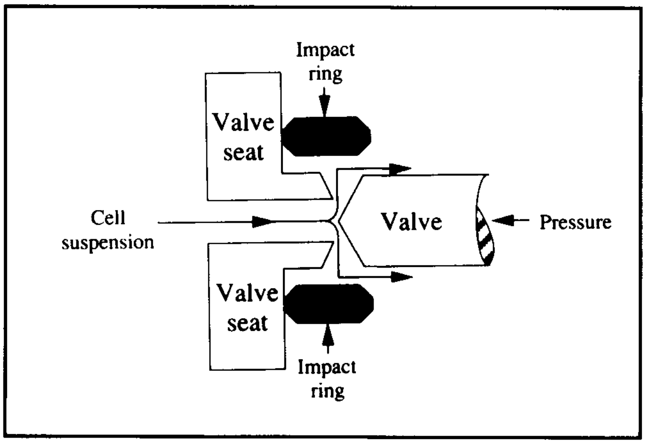
$$\omega_2^2 = \frac{\omega_1^2}{0.56} = \frac{(5000\,\mathrm{rpm})^2}{0.56} = 4.46\times 10^7\,\mathrm{rpm}^2.$$

Taking the square root, $\omega_2 = 6680$ rpm.

10.3 Cell Disruption

Downstream processing of fermentation broths usually begins with separation of cells by filtration or centrifugation. The next step depends on location of the desired product. For substances such as ethanol, citric acid and antibiotics which are excreted from cells, product is recovered from the cell-free broth using unit operations such as those described later in this

Figure 10.8 Cell disruption in a high-pressure homogeniser.



chapter. Biomass separated from the liquid is discarded or sold as a by-product.

For products such as enzymes and recombinant proteins which remain in the biomass, cell disruption must be carried out to release the desired material. A variety of methods is available to disrupt cells. Mechanical options include grinding with abrasives, high-speed agitation, high-pressure pumping and ultrasound. Non-mechanical methods such as osmotic shock, freezing and thawing, enzymic digestion of cell walls, and treatment with solvents and detergents can also be applied.

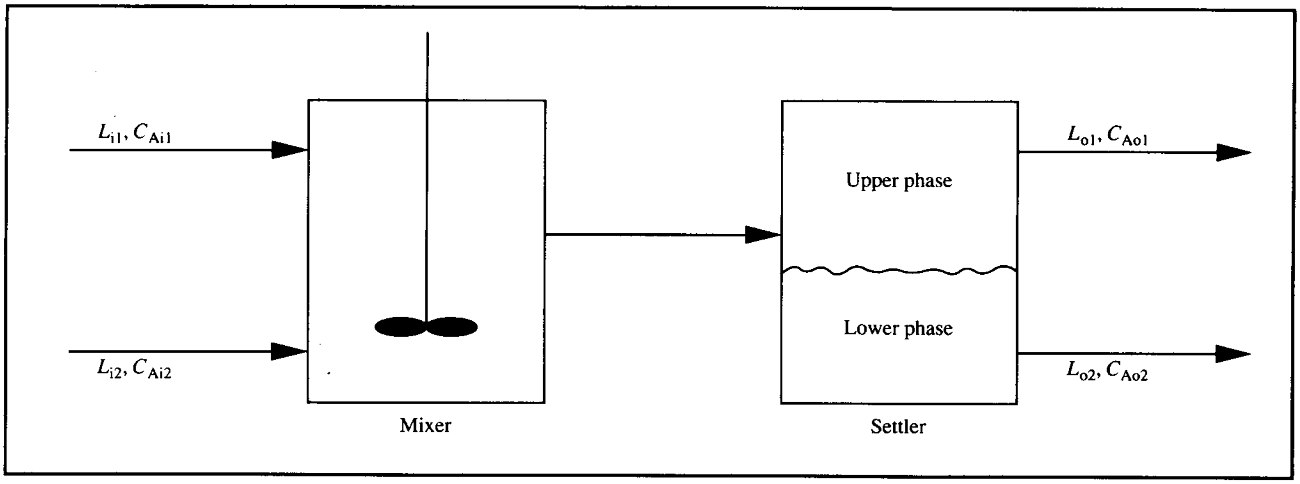
A widely-used technique for cell disruption is high-pressure homogenisation. Shear forces generated in this treatment are sufficient to completely disrupt many types of cell. A common apparatus for homogenisation of cells is the Manton–Gaulin homogeniser. As indicated in Figure 10.8, this high-pressure pump incorporates an adjustable valve with restricted orifice through which cells are forced at pressures up to 550 atm. The homogeniser is of general applicability for cell disruption, although the homogenising valve can become blocked when used with highly filamentous organisms.

The following equation relates disruption of cells to operating conditions in the Manton–Gaulin homogeniser [10]:

$$\ln\left(\frac{R_m}{R_m - R}\right) = kNp^{\alpha}. \tag{10.23}$$

In Eq. (10.23), R_m is the maximum amount of protein available for release, R is the amount of protein released after N passes through the homogeniser, k is a temperature-dependent rate constant, and p is the operating pressure. The exponent α is a measure of the resistance of the cells to disruption. For *Saccharomyces cerevisiae* yeast, α has been determined as 2.9 [10]; however the exponent for a particular organism depends to some extent on growth conditions [11]. The strong dependence of protein release on pressure suggests that high-pressure operation is beneficial; complete disruption in a single pass may be possible if the pressure is sufficiently high. Reduction in the number of passes through the homogeniser is generally preferable because multiple passes produce fine cell debris which can cause problems in subsequent clarification steps.

Figure 10.9 An ideal stage.



Release of protein is markedly dependent on temperature; protein recovery increases at elevated temperatures up to 50°C. However, cooling to 0–4°C is recommended during operation of homogenisers to minimise protein denaturation. Procedures for scale-up of homogenisers are not well developed. Methods which work well in the laboratory may give variable results when used at a larger scale.

10.4 The Ideal-Stage Concept

So far, we have considered only the initial steps of downstream processing: cell isolation and disruption. An important group of unit operations used for primary isolation of fermentation products relies on mass transfer to achieve separation of components between phases. Equipment for these separations sometimes consists of a series of stages in which the phases make intimate contact so that material can be transferred between them. Even if the equipment itself is not constructed in discrete stages, mass transfer can be considered to occur in stages. The effectiveness of each stage in accomplishing mass transfer depends on many factors, including equipment design, physical properties of the phases, and equilibrium relationships.

Consider the simple mixer-settler device of Figure 10.9 for extraction of component A from one liquid phase to another. The mass-transfer principles of liquid–liquid extraction have already been described in Chapter 9; unit operations for extraction are also discussed in the next section. Here, we will use the mixer-settler to explain the general concept of ideal stages. Two immiscible liquids enter the mixing vessel with volumetric flow rates L_{i1} and L_{i2}; these streams contain A at concentrations C_{Ai1} and C_{Ai2}, respectively. The two liquid phases are vigorously mixed together and A is transferred from one phase to the other. The mixture then passes to a settler where phase separation is allowed to occur under the influence of gravity. The flow rate of light liquid out of the settler is L_{o1}; the flow rate of heavy liquid is L_{o2}. Concentrations of A in these streams are C_{Ao1} and C_{Ao2}, respectively.

Operation of the mixer-settler relies on the liquids entering not being in equilibrium as far as concentration of A is concerned, i.e. C_{Ai1} and C_{Ai2} are not equilibrium concentrations. This means that there exists a driving force in the mixer for change of concentration as A is transferred from one liquid to the other. Depending on the ability of the mixer to promote mass transfer between the liquids and the effectiveness of the settler in allowing phase separation, when the liquids leave the system they will have been brought closer to equilibrium. If the mixer-settler were an *ideal stage* the two streams leaving it would be in equilibrium. No further mass transfer of A would result from additional contact between the phases; C_{Ao1} and C_{Ao2} would be equilibrium concentrations and the device would be operating at maximum efficiency. In reality, stages are not ideal; the change in concentration achieved in a real stage is always less than in an ideal stage so that extra stages are necessary to achieve the desired separation. The relative performance of an actual stage compared with that of an ideal stage is expressed as the *stage efficiency*.

The concept of ideal stages is used in design calculations for several unit operations. The important elements in analysis of these operations are material balances, energy balances if applicable, and the equilibrium relationships between phases. Application of these principles is illustrated in the following sections.

10.5 Aqueous Two-Phase Liquid Extraction

Liquid extraction is used to isolate many pharmaceutical products from animal and plant sources. In liquid extraction of fermentation products, components dissolved in liquid are recovered by transfer into an appropriate solvent. Extraction of penicillin from aqueous broth using solvents such as butyl acetate, amyl acetate or methyl isobutyl ketone, and isolation of erythromycin using pentyl or amyl acetate are examples. Solvent extraction techniques are also applied for recovery of steroids, purification of vitamin B_{12} from microbial sources, and isolation of alkaloids such as morphine and codeine from raw plant material. The simplest equipment for liquid extraction is the separating funnel used for laboratory-scale product recovery. Liquids forming two distinct phases are shaken together in the separating funnel; solute in dilute solution in one solvent transfers to the other solvent to form a more concentrated solution. The two phases are then allowed to separate and the heavy phase is withdrawn from the bottom of the funnel. The phase containing the solute in concentrated form is processed further to purify the product. Whatever apparatus is used for extraction, it is important that contact between the liquid phases is maximised by vigorous mixing and turbulence to facilitate solute transfer.

Extraction with organic solvents is a major separation technique in bioprocessing, particularly for recovery of antibiotics. However, organic solvents are unsuitable for isolation of proteins and other sensitive biopolymers. Techniques are being developed for aqueous two-phase extraction of these molecules. Aqueous solvents which form two distinct phases provide favourable conditions for separation of proteins, cell fragments and organelles with protection of their biological activity. Two-phase aqueous systems are produced when particular polymers or a polymer and salt are dissolved together in

Table 10.2 Examples of aqueous two-phase systems

(From M.R. Kula, 1985, Liquid–liquid extraction of biopolymers. In: M. Moo-Young, Ed, Comprehensive Biotechnology, *vol. 2, pp. 451–471, Pergamon Press, Oxford)*

Component 1	*Component 2*
Polyethylene glycol	Dextran
	Polyvinyl alcohol
	Polyvinylpyrrolidone
	Ficoll
	Potassium phosphate
	Ammonium sulphate
	Magnesium sulphate
	Sodium sulphate
Polypropylene glycol	Polyvinyl alcohol
	Polyvinylpyrrolidone
	Dextran
	Methoxypolyethylene glycol
	Potassium phosphate
Ficoll	Dextran
Methylcellulose	Dextran
	Hydroxypropyldextran
	Polyvinylpyrrolidone

water above certain concentrations. The liquid partitions into two phases, each containing 85–99% water. Some components used to form aqueous two-phase systems are listed in Table 10.2. When added to these mixtures, biomolecules and cell fragments partition between the phases; by selecting appropriate conditions, cell fragments can be confined to one phase as the protein of interest partitions into the other phase. Aqueous two-phase separations are of special interest for extraction of enzymes and recombinant proteins from cell debris produced by cell disruption. After partitioning, product is removed from the extracting phase using other unit operations such as precipitation or crystallisation.

The extent of differential partitioning between phases depends on the equilibrium relationship for the system. The *partition coefficient* K is defined as:

$$K = \frac{C_{Au}}{C_{Al}} \tag{10.24}$$

where C_{Au} is the equilibrium concentration of component A in the upper phase and C_{Al} is the equilibrium concentration of A in the lower phase. If $K > 1$, component A favours the upper phase; if $K < 1$, A is concentrated in the lower phase. In many aqueous systems K is constant over a wide range of concentrations provided the molecular properties of the phases are not changed. Partitioning is influenced by the size, electric charge and hydrophobicity of the particles or solute molecules; biospecific affinity for one of the polymers may also play a role in some systems. Surface free energy of the phase components and ionic composition of the liquids are of paramount importance in determining separation; K is related to both these parameters. Partitioning is also affected by other and sometimes interdependent factors so that it is impossible to predict partition coefficients from molecular properties. For single-stage extraction of enzymes, partition coefficients $\geqslant 3$ are normally required [5].

Even when the partition coefficient is low, good ***product recovery*** or ***yield*** can be achieved by using a large volume of the phase preferred by the solute. Yield of A in the upper phase, Y_u, is defined as:

$$Y_u = \frac{V_u C_{Au}}{V_0 C_{A0}} = \frac{V_u C_{Au}}{V_u C_{Au} + V_l C_{Al}} \tag{10.25}$$

where V_u is volume of the upper phase, V_l is volume of the lower phase, V_0 is the original volume of solution containing the product and C_{A0} is the original product concentration in that liquid. In the lower phase, yield Y_l is defined as:

$$Y_l = \frac{V_l C_{Al}}{V_0 C_{A0}} = \frac{V_l C_{Al}}{V_u C_{Au} + V_l C_{Al}} . \tag{10.26}$$

The maximum possible yield for an ideal extraction stage can be evaluated using Eqs (10.25) and (10.26) and the equilibrium partition coefficient, K. Dividing both numerator and denominator of Eq. (10.25) by C_{Au} and recognising that, at equilibrium, ${}^{C_{Al}}/_{C_{Au}}$ is equal to ${}^{1}/_{K}$:

$$Y_u = \frac{V_u}{V_u + \frac{V_l}{K}} . \tag{10.27}$$

Similarly, dividing the numerator and denominator of Eq. (10.26) by C_{Al} gives:

$$Y_l = \frac{V_l}{V_u K + V_l} . \tag{10.28}$$

Example 10.3 Enzyme recovery using aqueous extraction

Aqueous two-phase extraction is used to recover α-amylase from solution. A polyethylene glycol–dextran mixture is added and the solution separates into two phases. The partition coefficient is 4.2. Calculate the maximum possible enzyme recovery when:

(a) the volume ratio of upper to lower phases is 5.0; and
(b) the volume ratio of upper to lower phases is 0.5.

Solution:
As the partition coefficient is greater than 1, enzyme prefers the upper phase. Yield at equilibrium is therefore calculated for the upper phase. Dividing both numerator and denominator of Eq. (10.27) by V_l gives:

$$Y_u = \frac{\frac{V_u}{V_l}}{\frac{V_u}{V_l} + \frac{1}{K}}.$$

(a) $\frac{V_u}{V_l} = 5.0$. Therefore:

$$Y_u = \frac{5.0}{5.0 + \frac{1}{4.2}} \times 100 = 95\%.$$

(b) $\frac{V_u}{V_l} = 0.5$. Therefore:

$$Y_u = \frac{0.5}{0.5 + \frac{1}{4.2}} \times 100 = 68\%.$$

Increasing the relative volume of the extracting phase enhances recovery.

Another parameter used to characterise two-phase partitioning is the *concentration factor* or *purification factor,* δ_c, defined as the ratio of product concentration in the preferred phase to the initial product concentration:

$$\delta_c = \frac{C_{Al}}{C_{A0}} \text{ (when product partitions to the lower phase)}$$

$$\delta_c = \frac{C_{Au}}{C_{A0}} \text{ (when product partitions to the upper phase).} \tag{10.29}$$

Aqueous extraction in polyethylene glycol–salt mixtures is an effective technique for separating proteins from cell debris. In this system, debris partitions to the lower phase while most of the target proteins are recovered from the upper phase. Extraction can be carried out in a single-stage operation such as that depicted in Figure 10.9 using a polymer mixture which provides a suitable partition coefficient. Equilibrium is approached in extraction operations but rarely reached; for industrial application it is important to consider the time taken for mass transfer and the ease of mechanical separation of the phases. The rate of approach to equilibrium depends on the surface area available for exchange between the phases; this is maximised by rapid mixing. Separation of the phases is sometimes a problem because of the low interfacial tension between aqueous phases; very rapid large-scale extractions can be achieved by combining mixed vessels with centrifugal separators. In many cases, recovery and concentration of product with yields exceeding 90% can be achieved using a single extraction step [5]. When single-stage extraction does not give sufficient recovery, repeated extractions can be carried out in a chain or cascade of contacting and separation units.

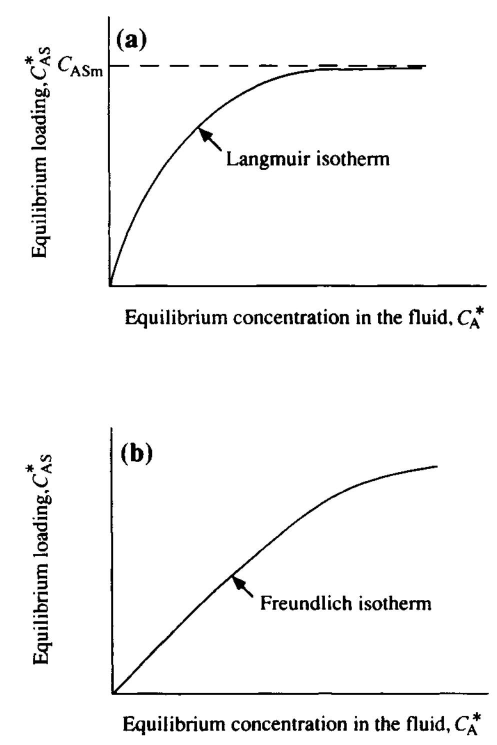
10.6 Adsorption

Adsorption is a surface phenomenon whereby components of a gas or liquid are concentrated on the surface of solid particles or at fluid interfaces. Adsorption is the result of electrostatic, van der Waals, reactive or other binding forces between individual atoms, ions or molecules. Four types of adsorption can be distinguished: exchange, physical, chemical and non-specific.

Adsorption serves the same function as extraction in isolating products from dilute fermentation liquors. Several different adsorption operations are used in bioprocessing, particularly for medical and pharmaceutical products. Ion-exchange adsorption is established practice for recovery of amino acids, proteins, antibiotics and vitamins. Adsorption onto activated charcoal is a method of long standing for purification of citric acid; adsorption of organic chemicals onto charcoal or porous polymeric adsorbents is common in wastewater treatment. Adsorption operations generally have higher selectivity but smaller capacity than extraction. Scale-up procedures for adsorption are less well defined than for extraction; therefore, more experimental data are required for design. Handling of solids in industrial processing is also somewhat more difficult compared with the liquids used in extraction. However, despite these disadvantages, adsorption is gaining increasing application primarily because of its suitability for protein isolation.

In adsorption operations, the substance being concentrated on the surface is called the *adsorbate*; the material to which the adsorbate binds is the *adsorbent*. The ideal adsorbent material has a high surface area per unit volume; this can be achieved if the solid contains a network of fine internal pores which provide an extremely large internal surface area. Carbons and synthetic resins based on styrene, divinylbenzene or acrylamide polymers are commonly used for adsorption of biological molecules. Commercially available adsorbents are porous granular or gel resins with void volumes of 30–50% and pore diameters generally less than 0.01 mm. As an example, the total surface area in particles of activated carbon ranges from 450 to 1800 $m^2\ g^{-1}$. Not all of this area is necessarily available for adsorption; adsorbate molecules only have access to surfaces in pores of appropriate diameter.

Figure 10.10 Adsorption isotherms.



10.6.1 Adsorption Operations

A typical adsorption operation consists of the following stages: a *contacting* or adsorption step which loads solute onto the adsorptive resin, a *washing* step to remove residual unadsorbed material, *desorption* or *elution* of adsorbate with a suitable solvent, *washing* to remove residual eluant, and *regeneration* of the adsorption resin to its original condition. Because adsorbate is bound to the resin by physical or ionic forces, conditions used for desorption must overcome these forces. Desorption is normally accomplished by feeding a stream of different ionic strength or pH; elution with organic solvent or reaction of the sorbed material may be necessary in some applications. Eluant containing stripped solute in concentrated form is processed to recover the adsorbate; operations for final purification include spray drying, precipitation and crystallisation. After elution, the adsorbent undergoes a regenerative treatment to remove any impurities and regain its original adsorptive capacity. Despite regeneration, performance of the resin will decrease with use as complete removal of adsorbed material is impossible. Accordingly, after a few regenerations the adsorbate is replaced.

10.6.2 Equilibrium Relationships For Adsorption

Like extraction, analysis of adsorption depends somewhat on equilibrium relationships which determine the extent to which material can be adsorbed onto a particular surface. When an adsorbate and adsorbent are at equilibrium there is a defined distribution of solute between solid and fluid phases and no further net adsorption occurs. Adsorption equilibrium data are available as *adsorption isotherms.* For adsorbate A, an isotherm gives the concentration of A in the adsorbed phase versus the concentration in the unadsorbed phase at a given temperature. Adsorption isotherms are useful for selecting the most appropriate adsorbent; they also play a crucial role in predicting the performance of adsorption systems.

Several types of equilibrium isotherm have been developed to describe adsorption relationships. However, no single model is universally applicable; all involve assumptions which may or may not be valid in particular cases. One of the simplest adsorption isotherms that accurately describes certain practical systems is the *Langmuir isotherm* shown in Figure 10.10(a). The Langmuir isotherm can be expressed as follows:

$$C^*_{AS} = \frac{C_{ASm}\, K_A\, C^*_A}{1 + K_A\, C^*_A}. \tag{10.30}$$

In Eq. (10.30), C^*_{AS} is the equilibrium concentration or loading of A on the adsorbent in units of, e.g. kg solute kg^{-1} solid or kg solute m^{-3} solid. C_{ASm} is the maximum loading of adsorbate corresponding to complete monolayer coverage of all available adsorption sites, C^*_A is the equilibrium concentration of solute in the fluid phase in units of, e.g. kg m^{-3}, and K_A is a constant. Because of the different units used for fluid- and solid-phase concentrations, K_A usually has units such as m^3 kg^{-1} solid.

Theoretically, Langmuir adsorption is applicable to systems where: (i) adsorbed molecules form no more than a monolayer on the surface; (ii) each site for adsorption is equivalent in terms of adsorption energy; and (iii) there are no interactions between adjacent adsorbed molecules. In many experimental systems at least one of these conditions is not met. For example, many commercial adsorbents possess highly irregular surfaces so that adsorption is favoured at particular points or 'strong sites' on the surface. Accordingly, each site is not equivalent. In addition, interactions between adsorbed molecules exist for almost all real adsorption systems. Recognition of these and other factors has led to application of other adsorption isotherms.

Of particular interest because of its widespread use in liquid–solid systems is the *Freundlich isotherm,* described by the relationship:

$$C^*_{AS} = K_F\, C^{*\,1/n}_A. \tag{10.31}$$

K_F and n are constants characteristic of the particular adsorption system; the dimensions of K_F depend on the dimensions of C^*_{AS} and C^*_A and the value of n. If adsorption is favourable n is > 1; if adsorption is unfavourable n is < 1. The form of the Freundlich isotherm is shown in Figure 10.10(b). Eq. (10.31) applies to adsorption of a wide variety of antibiotics, hormones and steroids.

There are many other forms of adsorption isotherm giving different $C^*_{AS} - C^*_A$ curves [12]. Because the exact mechanisms of adsorption are not well understood, adsorption equilibrium data must be determined experimentally.

Example 10.4 Antibody recovery by adsorption

Cell-free fermentation liquor contains 8×10^{-5} mol l^{-1} immunoglobulin G. It is proposed to recover at least 90% of this antibody by adsorption on synthetic, non-polar resin. Experimental equilibrium data are correlated as follows:

$$C^*_{AS} = 5.5 \times 10^{-5}\, C^{*\,0.35}_A$$

where C^*_{AS} is mol solute adsorbed per cm^3 adsorbent and C^*_A is liquid-phase solute concentration in mol l^{-1}. What minimum quantity of resin is required to treat 2 m^3 fermentation liquor in a single-stage mixed tank?

Solution:

The quantity of resin required is minimum when equilibrium occurs. If 90% of the antibiotic is adsorbed, the residual concentration in the liquid is:

$$\frac{(100-90)\%}{100\%}\,(8\times10^{-5}\,\text{mol l}^{-1}) = 8\times10^{-6}\,\text{mol l}^{-1}.$$

Substituting this value for C_A^* in the isotherm expression gives the equilibrium loading of immunoglobulin:

$$C_{AS}^* = 5.5\times10^{-5}\,(8\times10^{-6})^{0.35} = 9.05\times10^{-7}\,\text{mol cm}^{-3}.$$

The amount of adsorbed antibody is:

$$\frac{90\%}{100\%}\,(8\times10^{-5}\,\text{mol l}^{-1})\,(2\,\text{m}^3)\,.\left|\frac{1000\,\text{l}}{1\,\text{m}^3}\right| = 0.144\,\text{mol}.$$

Therefore, the mass of adsorbent needed is:

$$\frac{0.144\,\text{mol}}{9.05\times10^{-7}\,\text{mol cm}^{-3}} = 1.59\times10^5\,\text{cm}^3.$$

The minimum quantity of resin required is $1.6\times10^5\,\text{cm}^3$, or $0.16\,\text{m}^3$.

Figure 10.11 Movement of the adsorption zone and development of the breakthrough curve for a fixed-bed adsorber.

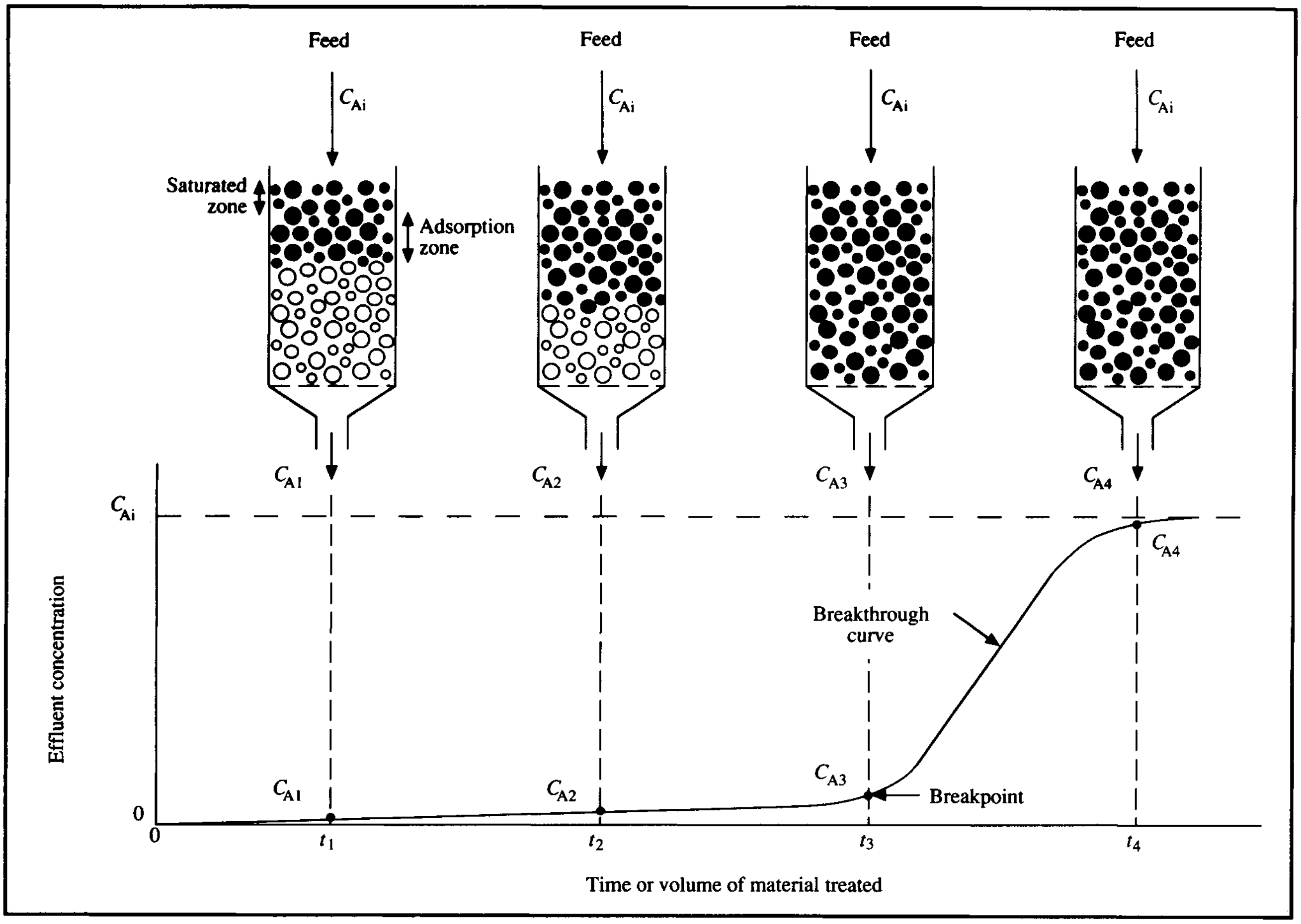
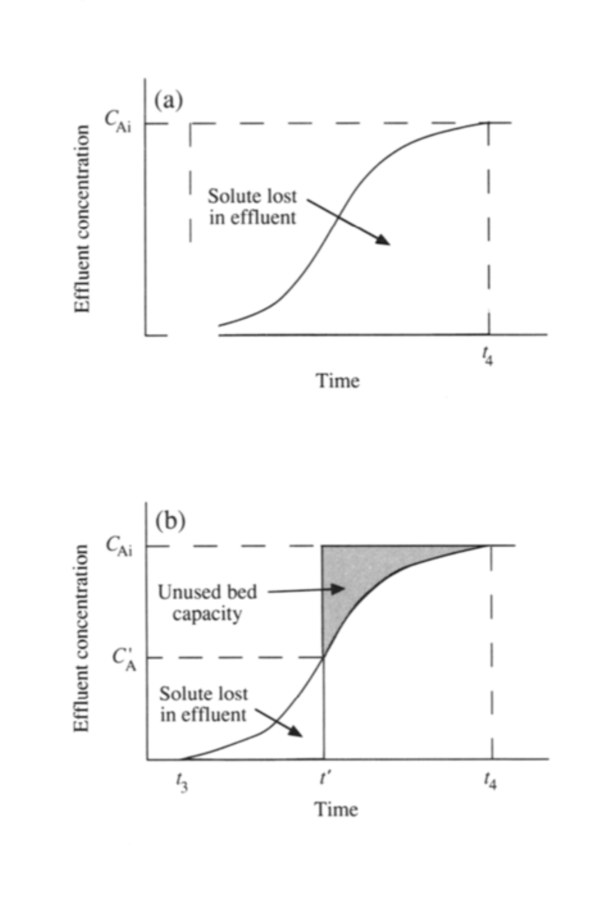


Figure 10.12 Relationship between the breakthrough curve, loss of solute in the effluent, and unused column capacity.



10.6.3 Performance Characteristics of Fixed-Bed Adsorbers

Various types of equipment have been developed for adsorption operations, including fixed beds, moving beds, fluidised beds and stirred-tank contactors. Of these, fixed-bed adsorbers are most commonly applied; the adsorption area available per unit volume is greater in fixed beds than in most other configurations. A fixed-bed adsorber is a vertical column or tube packed with adsorbent particles. Commercial adsorption operations are mostly performed as unsteady-state processes; liquid containing solute is passed through the bed and the loading or amount of product retained in the column increases with time.

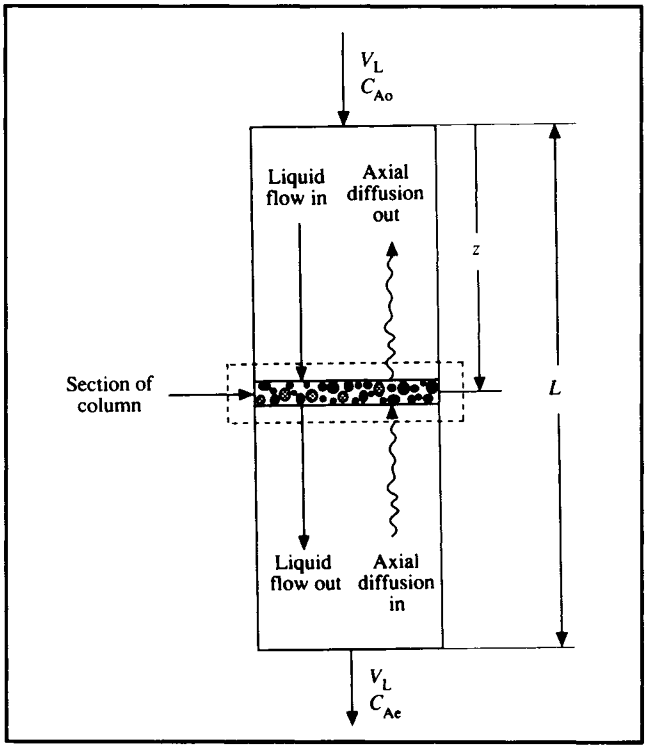
Operation of a downflow fixed-bed adsorber is illustrated in Figure 10.11. Liquid solution containing adsorbate at concentration C_{Ai} is fed at the top of a column which is initially free of adsorbate. At first, adsorbent resin at the top of the column takes up solute rapidly; solution passing through the column becomes depleted of solute and leaves the system with effluent concentration close to zero. As flow of solution continues, the region of the bed where most adsorption occurs, the *adsorption zone*, moves down the column as the top resin becomes saturated with solute in equilibrium with liquid concentration C_{Ai}. Movement of the adsorption zone usually occurs at a speed much lower than the velocity of fluid through the bed, and is called the *adsorption wave*. Eventually the lower edge of the adsorption zone reaches the bottom of the bed, the resin is almost completely saturated, and the concentration of solute in the effluent starts to rise appreciably; this is called the *breakpoint*. As the adsorption zone passes through the bottom of the bed, the resin can no longer adsorb solute and the effluent concentration rises to the inlet value, C_{Ai}. At this time the bed is completely saturated with adsorbate and must be regenerated. The curve in Figure 10.11 showing effluent concentration as a function of time or volume of material processed is known as the *breakthrough curve*.

The shape of the breakthrough curve greatly influences design and operation of fixed-bed adsorbers. Figure 10.12 shows the portion of the breakthrough curve between times t_3 and t_4 when solute appears in the column effluent. The amount of solute lost in the effluent is given by the area under the breakthrough curve. As indicated in Figure 10.12(a), if adsorption continues until the entire bed is saturated and the effluent concentration equals C_{Ai}, a considerable amount of solute is wasted. To avoid this, adsorption operations are usually stopped before the bed is completely saturated. As shown in Figure 10.12(b), if adsorption is halted at time t' when the effluent concentration is C'_A, only a small amount of solute is wasted compared with the process of Figure 10.12(a). The disadvantage is that some portion of the bed capacity is unused, as represented by the shaded area above the breakthrough curve. Because of the importance of the breakthrough curve in determining schedules of operation, much effort has been given to its prediction and to analysis of factors affecting it. This is discussed further in the next section.

10.6.4 Engineering Analysis of Fixed-Bed Adsorbers

In design of fixed-bed adsorbers, the quantity of resin and the time required for adsorption of a given quantity of solute must be estimated. Design procedures involve predicting the shape of the breakthrough curve and the time of appearance of the breakpoint. The form of the breakthrough curve is influenced by factors such as feed rate, concentration of solute in the feed, nature of the adsorption equilibrium and rate of adsorption.

Figure 10.13 Fixed-bed adsorber for mass-balance analysis.



Performance of commercial adsorbers is usually controlled by adsorption rate. This in turn depends on mass-transfer processes within and outside the adsorbent particles. One approach to adsorber design is to conduct extensive pilot studies to examine the effects of major system variables. However, the duration and cost of these experimental studies can be minimised by prior mathematical analysis of the process. Because fixed-bed adsorption is an unsteady-state process and equations for adsorption isotherms are generally non-linear, the calculations involved in engineering analysis are relatively complex compared with many other unit operations. Non-homogeneous packing in adsorption beds and the difficulty of obtaining reproducible results in apparently identical beds add to these problems. It is beyond the scope of this book to consider design procedures in any depth as considerable effort and research is required to establish predictive models for adsorption systems. However, a simplified engineering analysis is presented below.

Let us consider the processes which cause changes in the liquid-phase concentration of adsorbate in a fixed-bed adsorber. The aim of this analysis is to derive an equation for effluent concentration as a function of time (the breakthrough curve). The technique used is the mass balance. Consider the column packed with adsorbent resin shown in Figure 10.13. Liquid containing solute A is fed at the top of the column and flows down the bed. The total length of the bed is L. At distance z from the top is a section of column around which we can perform an unsteady-state mass balance. We will assume that this section is very thin so that z is approximately the same anywhere in the section. The system boundary is indicated in Figure 10.13 by dashed lines; four streams representing flow of material are shown to cross the boundary. The general mass-balance equation given in Chapter 4 can be applied to solute A:

$$\left\{\begin{matrix}\text{mass in}\\\text{through}\\\text{system}\\\text{boundaries}\end{matrix}\right\} - \left\{\begin{matrix}\text{mass out}\\\text{through}\\\text{system}\\\text{boundaries}\end{matrix}\right\} + \left\{\begin{matrix}\text{mass}\\\text{generated}\\\text{within}\\\text{system}\end{matrix}\right\} - \left\{\begin{matrix}\text{mass}\\\text{consumed}\\\text{within}\\\text{system}\end{matrix}\right\} = \left\{\begin{matrix}\text{mass}\\\text{accumulated}\\\text{within}\\\text{system}\end{matrix}\right\}. \tag{4.1}$$

Let us consider each term of Eq. (4.1) to see how it applies to solute A in the designated section of the column. First, because we assume there are no chemical reactions taking place and A can be neither generated nor consumed, the third and fourth terms of Eq. (4.1) are zero. On the left-hand side, this leaves only the input and output terms. What are the mechanisms for input of component A to the section of column? A is brought into the section largely as a result of liquid flow down the column; this is indicated in Figure 10.13 by the solid arrow entering the section. Other mechanisms are related to local mixing and diffusion processes within the interstices or gaps between the resin particles. For example, some A may enter the section from the region just below it by countercurrent diffusion against the direction of flow; this is indicated in Figure 10.13 by the wiggly arrow entering the section from below. Let us now consider movement of A out of the system. The mechanisms for removal of A are the same as for entry: A is carried out of the section by liquid flow and by axial transfer along the length of the tube against the direction of flow. These processes are also indicated in Figure 10.13. The remaining term in Eq. (4.1) is accumulation of A. A will accumulate within the section due to adsorption onto the interior and exterior surfaces of the adsorbent particles. A may also accumulate in liquid trapped within the interstitial spaces or gaps between resin particles.

When appropriate mathematical expressions for rates of flow, axial dispersion and accumulation are substituted into Eq. (4.1), the following equation is obtained:

$$\underset{\text{axial dispersion}}{\mathscr{D}_{Az}\frac{\partial^2 C_A}{\partial z^2}} + \underset{\text{flow}}{u\frac{-\partial C_A}{\partial z}} = \underset{\text{accumulation in the interstices}}{\frac{\partial C_A}{\partial t}} + \underset{\text{accumulation by adsorption}}{\left(\frac{1-\varepsilon}{\varepsilon}\right)\frac{\partial C_{AS}}{\partial t}}. \tag{10.32}$$

In Eq. (10.32), C_A is the concentration of A in the liquid, z is bed depth, t is time and C_{AS} is the average concentration of A in the solid phase. $\mathscr{D}_{Az}$ is the *effective axial dispersion coefficient* for A in the column. In most packed beds $\mathscr{D}_{Az}$ is substantially greater than the molecular diffusion coefficient; the value of $\mathscr{D}_{Az}$ incorporates the effects of axial mixing in the column as the solid particles interrupt smooth liquid flow. ε is the *void fraction* in the bed, defined as:

$$\varepsilon = \frac{V_T - V_s}{V_T} \tag{10.33}$$

where V_T is the total volume of the column and V_s is the volume of the resin particles. u is the interstitial liquid velocity, defined as:

$$u = \frac{F_L}{\varepsilon A_c} \tag{10.34}$$

where F_L is the volumetric liquid flow rate and A_c is the cross-sectional area of the column. In Eq. (10.32), $\partial/_{\partial t}$ denotes the partial differential with respect to time, $\partial/_{\partial z}$ denotes the partial differential with respect to distance, and $\partial^2/_{\partial z^2}$ denotes the second partial differential with respect to distance. Although Eq. (10.32) looks complicated, it is useful to recognise the physical meaning of its components. As indicated below each term of the equation, rates of axial dispersion, flow, and accumulation in the liquid and solid phases are represented; accumulation is equal to the sum of the net rates of axial diffusion and flow into the system.

There are four variables in Eq. (10.32): concentration of A in the liquid, C_A; concentration of A in the solid, C_{AS}; distance from the top of the column, z; and time, t. The other parameters can be considered constant. C_A and C_{AS} vary with time of operation and depth in the column. Theoretically, with the aid of further information about the system, Eq. (10.32) can be solved to provide an equation for the effluent concentration as a function of time: the breakthrough curve. However, solution of Eq. (10.32) is generally very difficult. To assist the analysis, simplifying assumptions are often made. For example, it is normally assumed that dilute solutions are being processed; this results in nearly isothermal operation and eliminates the need for an accompanying energy balance for the system. In many cases the axial-diffusion term can be neglected; axial dispersion is generally significant only at low flow rates. If the interstitial fluid content of the bed is small compared with the total volume of feed, accumulation of A between the particles can also be neglected. With these simplifications, the first and third terms of Eq. (10.32) are eliminated and the design equation is reduced to:

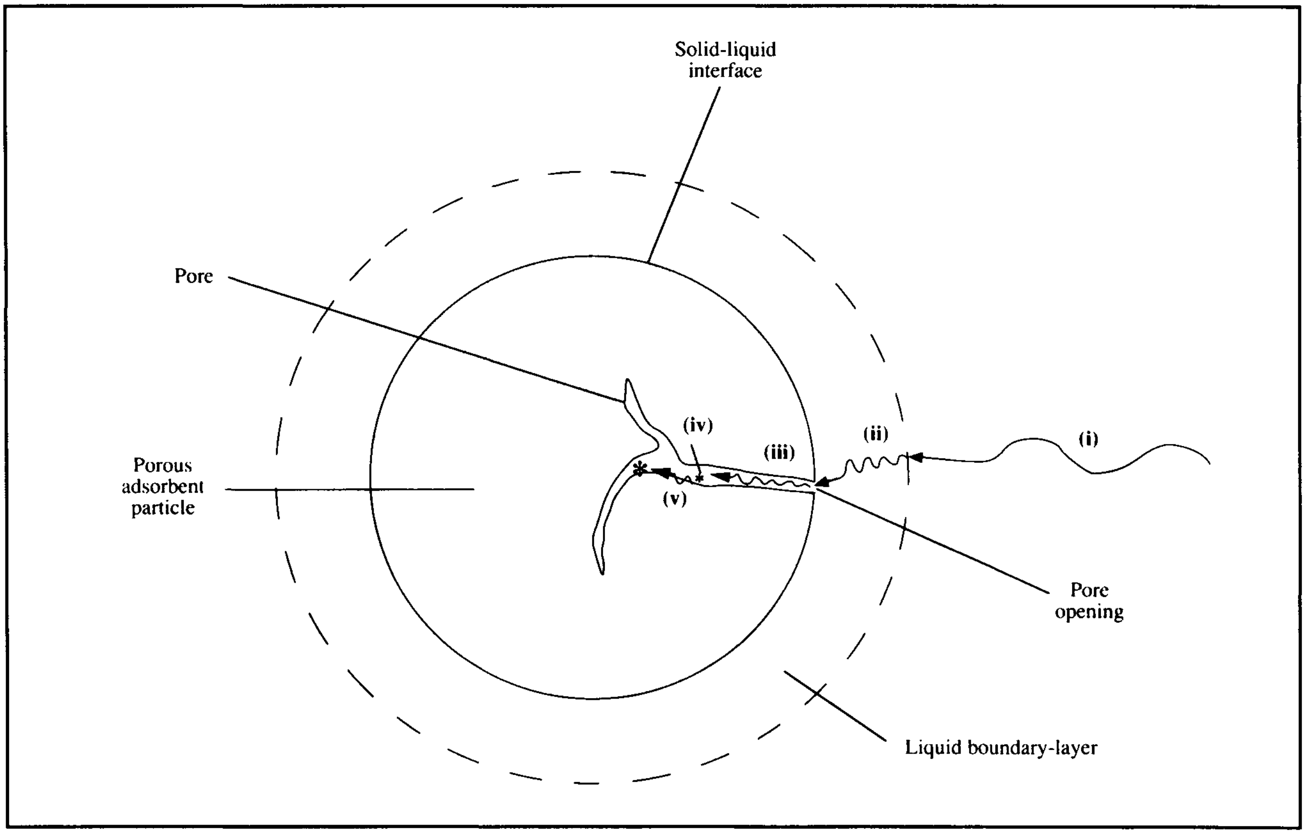
$$u \frac{-\partial C_A}{\partial z} = \left(\frac{1-\varepsilon}{\varepsilon}\right)\frac{\partial C_{AS}}{\partial t}. \tag{10.35}$$

To progress further with this analysis, information about $^{\partial C_{AS}}/_{\partial t}$ is required. This term represents the rate of change of solid-phase adsorbate concentration and depends on the overall rate at which adsorption takes place. Overall rate of adsorption depends on two factors: the rate at which solute is transferred from liquid to solid by mass-transfer mechanisms, and the rate of the actual adsorption or attachment process. The mass-transfer pathway for adsorbate is analogous to that described in Section 9.5.2 for oxygen transfer. There are up to five steps which can pose significant resistance to adsorption as indicated in Figure 10.14. They are:

(i) transfer from the bulk liquid to the liquid boundary layer surrounding the particle;
(ii) diffusion through the relatively stagnant liquid film surrounding the particle;
(iii) transfer through the liquid in the pores of the particle to internal surfaces;
(iv) the actual adsorption process; and
(v) surface diffusion along the internal pore surfaces; i.e. migration of adsorbate molecules within the surface without prior desorption.

Normally only a small amount of adsorption occurs on the outer perimeter of the particle compared to within the pores; accordingly, external adsorption is not shown in Figure 10.14. Bulk transfer of solute is usually rapid because of mixing and convective flow of liquid passing over the solid; the effect of step (i) on overall adsorption rate can therefore be neglected. The adsorption step itself is sometimes very slow and can become the rate-limiting process; however in most cases adsorption occurs relatively quickly so that step (iv) is not rate-controlling. Step (ii) represents the major external resistance to mass transfer, while steps (iii) and (v) represent the major internal resistances. Any or all of these steps can control the overall rate of adsorption depending on the situation. Rate-controlling steps are usually identified experimentally using a small column with packing identical to the industrial-scale system; mass-transfer coefficients can then be measured under appropriate flow conditions.

Figure 10.14 Steps involved in adsorption of solute from liquid to porous adsorbent particle.



Unfortunately however, it is possible that the rate-controlling step changes as the process is scaled up, making rational design difficult.

Greatest simplification of Eq. (10.35) is obtained when the overall rate of mass transfer from liquid to internal surfaces is represented by a single equation. For example, by analogy with Eq. (9.34) or (9.35), we can write:

$$\frac{\partial C_{AS}}{\partial t} = \frac{K_L a}{1-\varepsilon} (C_A - C_A^*) \tag{10.36}$$

where K_L is the overall mass-transfer coefficient, a is the surface area of the solid per unit volume, C_A is the liquid-phase concentration of A, and C_A^* is the liquid-phase concentration of A in equilibrium with C_{AS}. The value of K_L will depend on the properties of the liquid and the flow conditions; C_A^* can be related to C_{AS} by the equilibrium isotherm. In the end, after the differential equations are simplified as much as possible and then integrated with appropriate boundary conditions (usually with a computer), the result is a relationship between effluent concentration (C_A at $z = L$) and time. The height of the column required to achieve a certain recovery of solute can also be evaluated.

As mentioned already in this section, there are many uncertainties associated with design and scale up of adsorption systems. The approach described here will give only an approximate indication of design and operating parameters. Other methods involving various simplifying assumptions can be employed [2]. The above analysis highlights the important role played by mass transfer in practical adsorption operations. *Equilibrium is seldom achieved in commercial adsorption systems; performance is controlled by the overall rate of adsorption.* Most parameters determining the economics of adsorption are those affecting mass transfer. Improvement in adsorber operation can be achieved by enhancing rates of diffusion and reducing mass-transfer resistance.

10.7 Chromatography

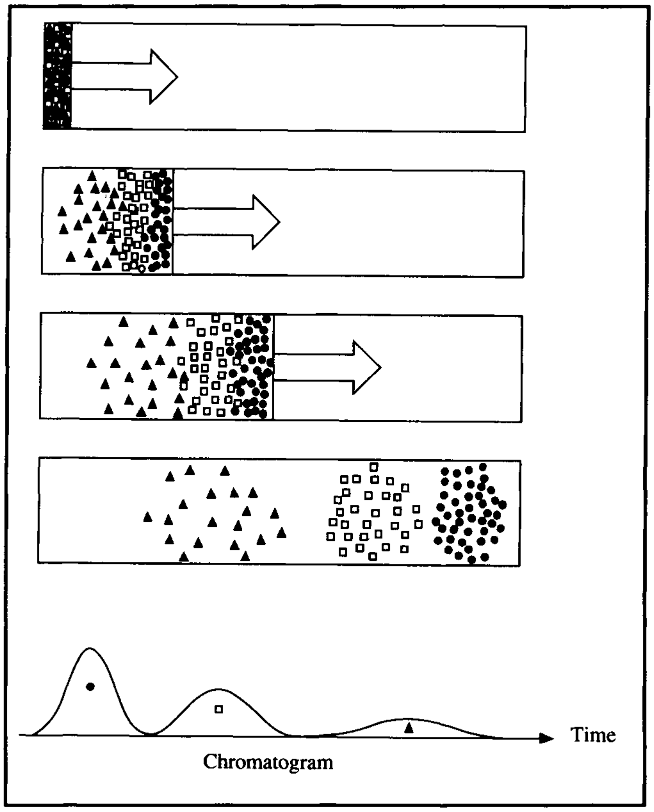
Chromatography is a separation procedure for resolving mixtures and isolating components. Many of the principles described in the previous section on adsorption apply also to chromatography. The basis of chromatography is *differential migration*, i.e. the selective retardation of solute molecules during passage through a bed of resin particles. A schematic description of chromatography is given in Figure 10.15; this diagram shows separation of three solutes from a mixture injected into a column. As solvent flows through the column, the solutes travel at different speeds depending on their relative affinities for the resin particles. As a result, they will be separated and appear for collection at the end of the column at different times. The pattern of solute peaks emerging from a chromatography column is called a *chromatogram*. The fluid carrying solutes through the column or used for elution is known as the *mobile phase*; the material which stays inside the column and effects the separation is called the *stationary phase*.

In *gas chromatography (GC)*, the mobile phase is a gas. Gas chromatography is used widely as an analytical tool for separating relatively volatile components such as alcohols, ketones, aldehydes and many other organic and inorganic compounds. However, of greater relevance to bioprocessing is *liquid chromatography*, which can take a variety of forms. Liquid chromatography finds application both as a laboratory method for sample analysis and as a preparative technique for large-scale purification of biomolecules. In recent years there have been rapid developments in the technology of liquid chromatography aimed at isolation of recombinant products from genetically engineered organisms. Chromatography is a high-resolution technique and therefore suitable for recovery of high-purity therapeutics and pharmaceuticals. Chromatographic methods available for purification of proteins, peptides, amino acids, nucleic acids, alkaloids, vitamins, steroids and many other biological materials include *adsorption chromatography*, *partition chromatography*, *ion-exchange chromatography*, *gel chromatography* and *affinity chromatography*. These methods differ in the principal mechanism by which molecules are retarded in the chromatography column.

(i) *Adsorption chromatography*. Biological molecules have varying tendencies to adsorb onto polar adsorbents such as silica gel, alumina, diatomaceous earth and charcoal. Performance of the adsorbent relies strongly on the chemical composition of the surface, i.e. the types and concentrations of exposed atoms or groups. The order of elution of sample components depends primarily on molecule polarity. Because the mobile phase is in competition with solute for adsorption sites, solvent properties are also important. Polarity scales for solvents are available to aid mobile-phase selection [13].

(ii) *Partition chromatography*. Partition chromatography relies on the unequal distribution of solute between two immiscible solvents. This is achieved by fixing one solvent (the stationary phase) to a support and passing the other solvent containing solute over it. The solvents make intimate contact allowing multiple extractions of solute to occur. Several methods are available to chemically bond the stationary solvent to supports such as silica [14]. When the stationary phase is more polar than the mobile phase, the technique is called *normal-phase chromatography*. When non-polar compounds are being separated it is usual to use a stationary phase which is less polar than the mobile phase; this is called *reverse-phase chromatography*. A common stationary phase for reverse-phase chromatography is hydrocarbon with 8 or 18 carbons bonded to silica gel; these materials are called C_8 and C_{18} packings, respectively. Solvent systems most frequently used are water–acetonitrile and water–methanol; aqueous buffers are also employed to suppress ionisation of sample components. Elution is generally in order of increasing solute hydrophobicity.

Figure 10.15 Chromatographic separation of components in a mixture. Three different solutes are shown schematically as circles, squares and triangles. (From P.A. Belter, E.L. Cussler and W.-S. Hu, 1988, *Bioseparations: Downstream Processing For Biotechnology*, John Wiley, New York.)



(iii) *Ion-exchange chromatography.* The basis of separation in this procedure is electrostatic attraction between the solute and dense clusters of charged groups on the column packing. Ion-exchange chromatography can give high resolution of macromolecules and is used commercially for fractionation of antibiotics and proteins. Column packings for low-molecular-weight compounds include silica, glass and polystyrene; carboxymethyl and diethylaminoethyl groups attached to cellulose, agarose or dextran provide suitable resins for protein chromatography. Solutes are eluted by changing the pH or ionic strength of the liquid phase; salt gradients are the most common way of eluting proteins from ion exchangers. Practical aspects of protein ion-exchange chromatography are described in greater detail elsewhere [15].

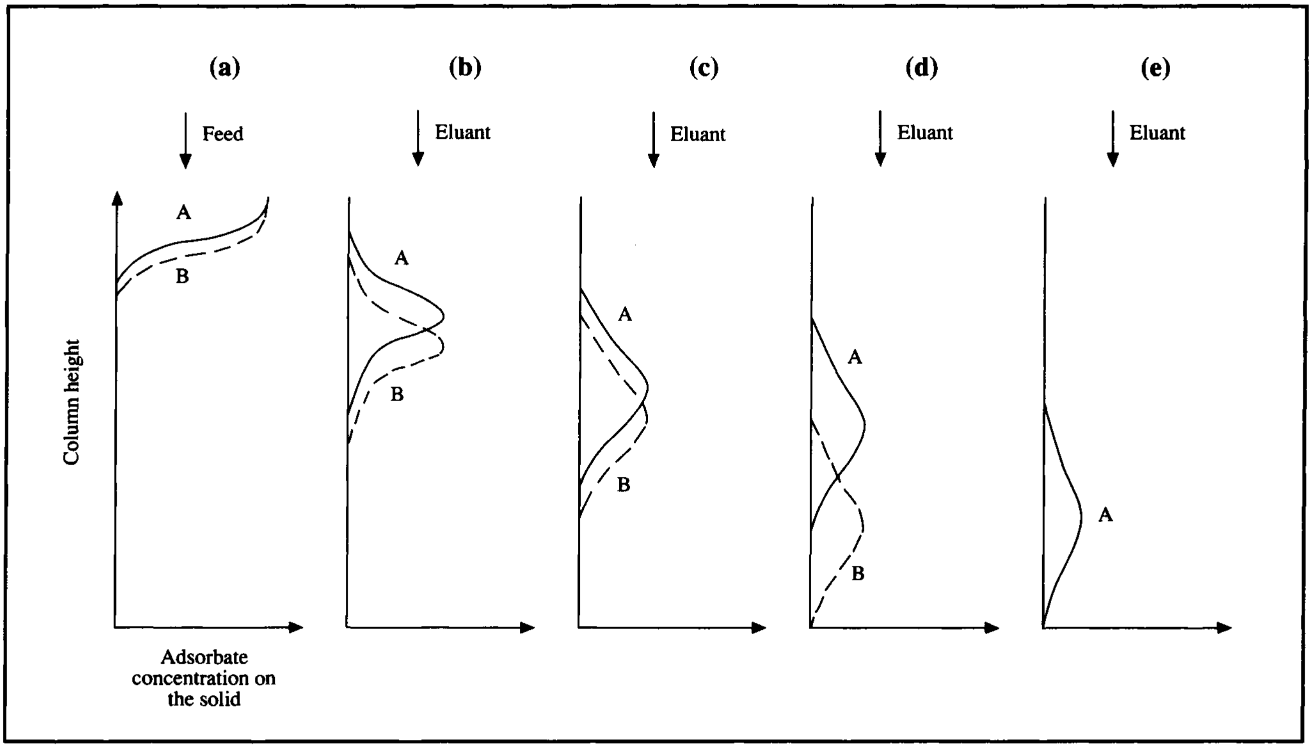
(iv) *Gel chromatography.* This technique is also known as *molecular-sieve chromatography, exclusion chromatography, gel filtration* and *gel-permeation chromatography.* Molecules in solution are separated in a column packed with gel particles of defined porosity. Gels most often used are cross-linked dextrans, agaroses and polyacrylamide gels. The speed with which components travel through the column depends on their effective molecular size. Large molecules are completely excluded from the gel matrix and move rapidly through the column to appear first in the chromatogram. Small molecules are able to penetrate the pores of the packing, traverse the column very slowly, and appear last in the chromatogram. Molecules of intermediate size enter the pores but spend less time there than the small solutes. Gel filtration can be used for separation of proteins and lipophilic compounds. Large-scale gel-filtration columns are operated with upward-flow elution.

(v) *Affinity chromatography.* This separation technique exploits the binding specificity of biomolecules. Enzymes, hormones, receptors, antibodies, antigens, binding proteins, lectins, nucleic acids, vitamins, whole cells and other components capable of specific and reversible binding are amenable to highly selective affinity purification. Column packing is prepared by linking a binding molecule called a *ligand* to an insoluble support; when sample is passed through the column, only solutes with appreciable affinity for the ligand are retained. The ligand must be attached to the support in such a way that its binding properties are not seriously affected; molecules called *spacer arms* are often used to set the ligand away from the support and make it more accessible to the solute. Many ready-made support–ligand preparations are available commercially and are suitable for a wide range of proteins. Conditions for elution depend on the specific binding complex formed: elution usually involves a change in pH, ionic strength or buffer composition. Enzyme proteins can be desorbed using a compound with higher affinity for the enzyme than the ligand, e.g. a substrate or substrate analogue. Affinity chromatography using antibody ligands is called *immuno-affinity chromatography.*

In this section we will consider principles of liquid chromatography for separation of biological molecules such as proteins and amino acids. Choice of stationary phase will depend to a large extent on the type of chromatography employed; however certain basic requirements must be met. For high capacity, the solid support must be porous with high internal surface area; it must also be insoluble and chemically stable during operation and cleaning. Ideally, the particles should exhibit high mechanical strength and show little or no non-specific binding. The low rigidity of many porous gels was initially a problem in industrial-scale chromatography; the weight of the packing material in large columns and the pressures developed during flow tended to compress the packing and impede operation. However, many macroporous gels and composite materials of high rigidity are now available for industrial use.

Two methods for carrying out chromatographic separations are high-performance liquid chromatography (HPLC) and fast protein liquid chromatography (FPLC). In principle, any of the types of chromatography described above can be executed using HPLC and FPLC techniques. Specialised equipment for HPLC and FPLC allows automated injection of sample, rapid flow of material through the column, collection of the separated fractions, and data analysis. Chromatographic separations traditionally performed under atmospheric pressure in vertical columns with manual sample feed and gravity elution are carried out faster and with better resolution using densely-packed columns and high flow rates in HPLC and FPLC systems. The differences between HPLC and FPLC lie in the flow rates and pressures used, the size of the packing material, and the resolution accomplished. In general, HPLC instruments are designed for small-scale, high-resolution analytical applications; FPLC is tailored for large-scale purification. In order to achieve the high resolutions characteristic of HPLC, stationary-phase particles 2–5 μm in diameter are commonly used. Because the particles are so small, HPLC systems are operated under high pressure (5–10 MPa) to achieve flow rates of 1–5 ml min^{-1}. FPLC instruments are not able to develop such high pressures (1–2 MPa), and are therefore operated with column packings of larger size. Resolution is poorer using FPLC compared with

Figure 10.16 Differential migration of two solutes A and B.



HPLC; accordingly, it is common practice to collect only the central peak of the solute pulse emerging from the end of the column and to recycle or discard the leading and trailing edges. FPLC equipment is particularly suited to protein separations; many gels used for gel chromatography and affinity chromatography are compressible and cannot withstand the high pressures exerted in HPLC.

Chromatography is essentially a batch operation; however industrial chromatography systems can be monitored and controlled for easy automation. Cleaning the column in place is generally difficult. Depending on the nature of the impurities contained in the samples, rather harsh treatments with concentrated salt or dilute alkali solutions are required; these may affect swelling of the gel beads and, therefore, liquid flow in the column. Regeneration in place is necessary as re-packing of large columns can be laborious and time-consuming. Repeated use of chromatographic columns is essential because of their high cost.

10.7.1 Differential Migration

Differential migration provides the basis for chromatographic separation and is explained diagrammatically in Figure 10.15. A solution contains two solutes A and B which have different equilibrium affinities for a particular stationary phase. For the sake of brevity, let us say that the solutes are adsorbed onto the stationary phase although they may be adsorbed, bound or entrapped depending on the type of chromatography employed. Assume that A is adsorbed more strongly than B. If a small quantity of solution is passed through the column so that only a limited depth of packing is saturated, both solutes will be retained in the bed as shown in Figure 10.16(a).

A suitable eluant is now passed through the bed. As shown in Figures 10.16(b–e), both solutes will be alternately adsorbed and desorbed at lower positions in the column as flow of eluant continues. Because solute B is more easily desorbed than A, it moves forward more rapidly. Differences in migration velocities of solutes are related to differences in equilibrium distributions between stationary and mobile phases. In Figure 10.16(e), solute B has been separated from A and washed out of the system.

Several parameters are used to characterise differential migration. An important variable is the volume V_e of eluting solvent required to carry the solute through the column until it emerges at its maximum concentration. Each component separated by chromatography has a different elution volume. Another parameter commonly used to characterise elution is the *capacity factor*, k:

$$k = \frac{(V_e - V_o)}{V_o} \tag{10.37}$$

where V_o is the void volume, i.e. the volume of liquid in the column outside of the particles. For two solutes, the ratio of their capacity factors k_1 and k_2 is called the *selectivity* or *relative retention*, δ:

$$\delta = \frac{k_2}{k_1}. \tag{10.38}$$

Eqs (10.37) and (10.38) are normally applied to adsorption, partition, ion-exchange and affinity chromatography. In gel chromatography where separation is a function of effective molecular size, the elution volume is easily related to certain physical properties of the gel column. The total volume of a gel column is:

$$V_T = V_o + V_i + V_s \tag{10.39}$$

where V_T is total volume, V_o is void volume outside the particles, V_i is internal volume of liquid in the pores of the particles, and V_s is volume of the gel itself. The outer volume V_o can be determined by measuring the elution volume of a substance that is completely excluded from the stationary phase; a solute which does not penetrate the gel can be washed from the column using a volume of liquid equal to V_o. V_o is usually about one-third V_T. Solutes which are only partly excluded from the stationary phase elute with a volume described by the following equation:

$$V_e = V_o + K_p V_i \tag{10.40}$$

where K_p is the *gel partition coefficient*, defined as the fraction of internal volume available to the solute. For large molecules which do not penetrate the solid, $K_p = 0$. From Eq. (10.40):

$$K_p = \frac{(V_e - V_o)}{V_i}. \tag{10.41}$$

K_p is a convenient parameter for comparing separation results obtained with different gel-chromatography columns; it is independent of column size and packing density. However experimental determination of K_p depends on knowledge of V_i, which is difficult to measure accurately. V_i is usually calculated using the equation:

$$V_i = a\, W_r \tag{10.42}$$

where a is mass of dry gel and W_r is the *water regain value*, defined as the volume of water taken up per mass of dry gel. The value for W_r is generally specified by the gel manufacturer. If, as is often the case, the gel is supplied already wet and swollen, the value of a is unknown and V_i is determined using the following equation:

$$V_i = \frac{W_r \rho_g}{(1 + W_r \rho_w)} (V_T - V_o) \tag{10.43}$$

where ρ_g is the density of wet gel and ρ_w is the density of water.

Example 10.5 Hormone separation using gel chromatography

A pilot-scale gel-chromatography column packed with Sephacryl resin is used to separate two hormones A and B. The column is 5 cm in diameter and 0.3 m high; the void volume is 1.9×10^{-4} m^3. The water regain value of the gel is 3×10^{-3} m^3 kg^{-1} dry Sephacryl; the density of wet gel is 1.25×10^3 kg m^{-3}. The partition coefficient for hormone A is 0.38; the partition coefficient for hormone B is 0.15. If the eluant flow rate is 0.7 l h^{-1}, what is the retention time for each hormone?

Solution:

The total column volume is:

$$V_T = \pi r^2 h = \pi (2.5 \times 10^{-2}\ \text{m})^2 (0.3\ \text{m}) = 5.89 \times 10^{-4}\ \text{m}^3.$$

$V_o = 1.9 \times 10^{-4}$ m^3; $\rho_w = 1000$ kg m^{-3}. From Eq. (10.43):

$$V_i = \frac{(3 \times 10^{-3}\ \text{m}^3\ \text{kg}^{-1})(1.25 \times 10^3\ \text{kg m}^{-3})}{1 + (3 \times 10^{-3}\ \text{m}^3\ \text{kg}^{-1})(1000\ \text{kg m}^{-3})} (5.89 \times 10^{-4}\ \text{m}^3 - 1.9 \times 10^{-4}\ \text{m}^3)$$

$$= \quad 3.74\times10^{-4}\,\mathrm{m}^3.$$

$K_{pA}=0.38$; $K_{pB}=0.15$. Therefore, from Eq. (10.40):

$$V_{eA}=1.9\times10^{-4}\,\mathrm{m}^3+0.38\,(3.74\times10^{-4}\,\mathrm{m}^3)=3.32\times10^{-4}\,\mathrm{m}^3$$

$$V_{eB}=1.9\times10^{-4}\,\mathrm{m}^3+0.15\,(3.74\times10^{-4}\,\mathrm{m}^3)=2.46\times10^{-4}\,\mathrm{m}^3.$$

The times associated with these elution volumes are:

$$t_A = \frac{3.32\times10^{-4}\,\mathrm{m}^3}{0.71\,\mathrm{h}^{-1}\cdot\left|\dfrac{1\,\mathrm{m}^3}{1000\,\mathrm{l}}\right|\cdot\left|\dfrac{1\,\mathrm{h}}{60\,\mathrm{min}}\right|} = 28\,\mathrm{min}.$$

$$t_B = \frac{2.46\times10^{-4}\,\mathrm{m}^3}{0.71\,\mathrm{h}^{-1}\cdot\left|\dfrac{1\,\mathrm{m}^3}{1000\,\mathrm{l}}\right|\cdot\left|\dfrac{1\,\mathrm{h}}{60\,\mathrm{min}}\right|} = 21\,\mathrm{min}.$$

10.7.2 Zone Spreading

The effectiveness of chromatography depends not only on differential migration but on whether the elution bands for individual solutes remain compact and without overlap. Ideally, each solute should pass out of the column at a different instant in time. In practice, elution bands spread out somewhat so that each solute takes a finite period of time to pass across the end of the column. Zone spreading is not so important when migration rates vary widely because there is little chance that solute peaks will overlap. However if the molecules to be separated have similar structure, migration rates will also be similar and zone spreading must be carefully controlled.

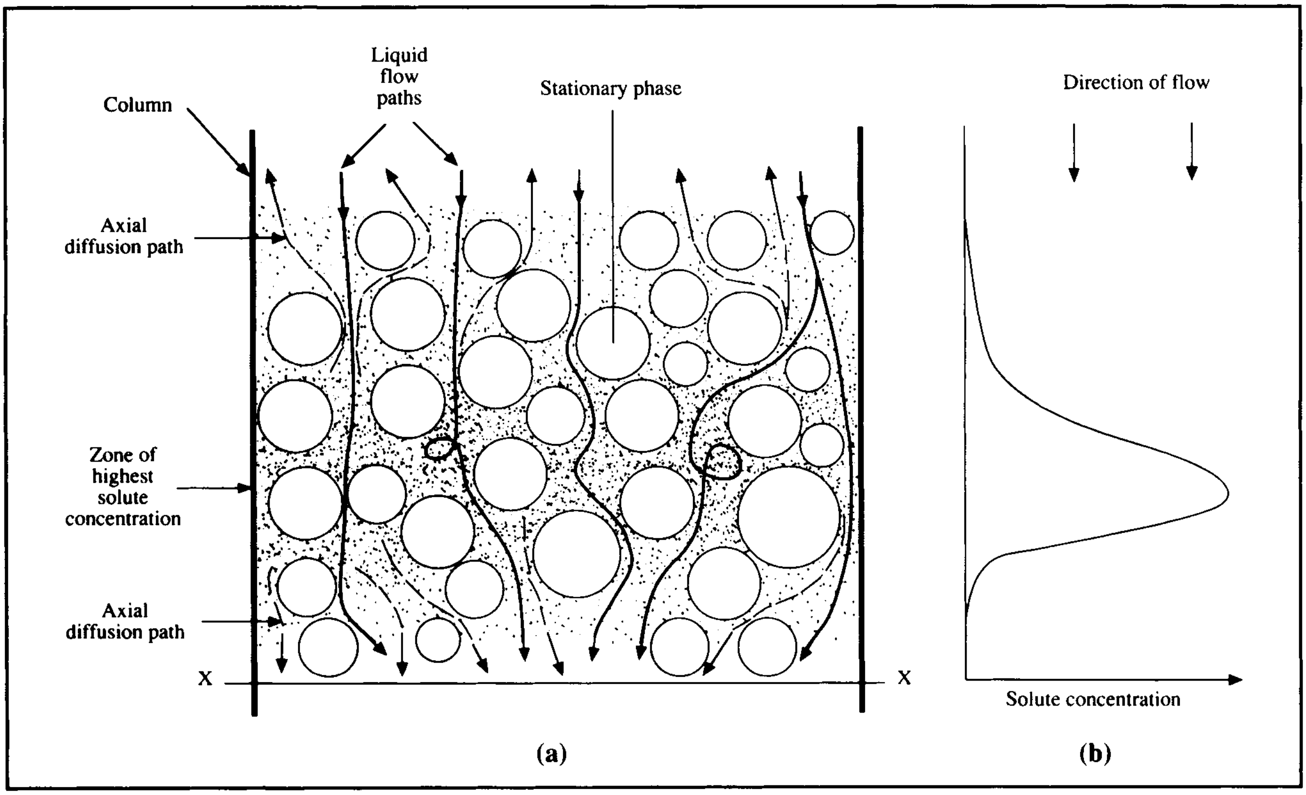
As illustrated in Figure 10.15, typical chromatogram elution bands have a peak of high concentration at or about the centre of the pulse but are of finite width as the concentration trails off to zero before and after the peak. Spreading of the solute peak is caused by several factors represented schematically in Figure 10.17.

(i) *Axial diffusion.* As solute is carried through the column, molecular diffusion of solute will occur from regions of high concentration to regions of low concentration. Diffusion in the axial direction, i.e. along the length of the tube, is indicated in Figure 10.17(a) by broken arrows. Axial diffusion broadens the solute peak by transporting material upstream and downstream away from the region of greatest concentration.

(ii) *Eddy diffusion.* In columns packed with solid particles, actual flow paths of liquid through the bed can be highly variable. As indicated in Figure 10.17(a), some liquid will flow almost directly through the bed while other liquid will take longer and more tortuous paths through the gaps or interstices between the particles. Accordingly, some solute molecules carried in the fluid will move slower than the average rate of progress through the column while others will take shorter paths and move ahead of the average; the result is spreading of the solute band. Differential motion of material due to erratic local variations in flow velocity is known as *eddy diffusion.*

(iii) *Local non-equilibrium effects.* In most columns, lack of equilibrium is the most important factor affecting zone spreading, although perhaps the most difficult to understand. Consider the situation at position X indicated in Figure 10.17(a). A solute pulse is passing through the column; as shown in Figure 10.17(b) concentration within this pulse increases from the front edge to a maximum near the centre and then decreases to zero. As the solute pulse moves down the column, an initial gradual increase in solute concentration will be experienced at X. In response to this increase in mobile-phase solute concentration, solute will bind to the stationary phase and the stationary-phase concentration will start to increase towards an appropriate equilibrium value. Equilibrium is not established immediately however; it takes time for the solute to undergo the mass-transfer steps from liquid to solid as outlined in Section 10.6.4. Indeed, before equilibrium can be established, the mobile-phase concentration increases again as the centre of the solute pulse moves closer to X. Because concentration in the mobile

Figure 10.17 Zone spreading in a chromatography column.



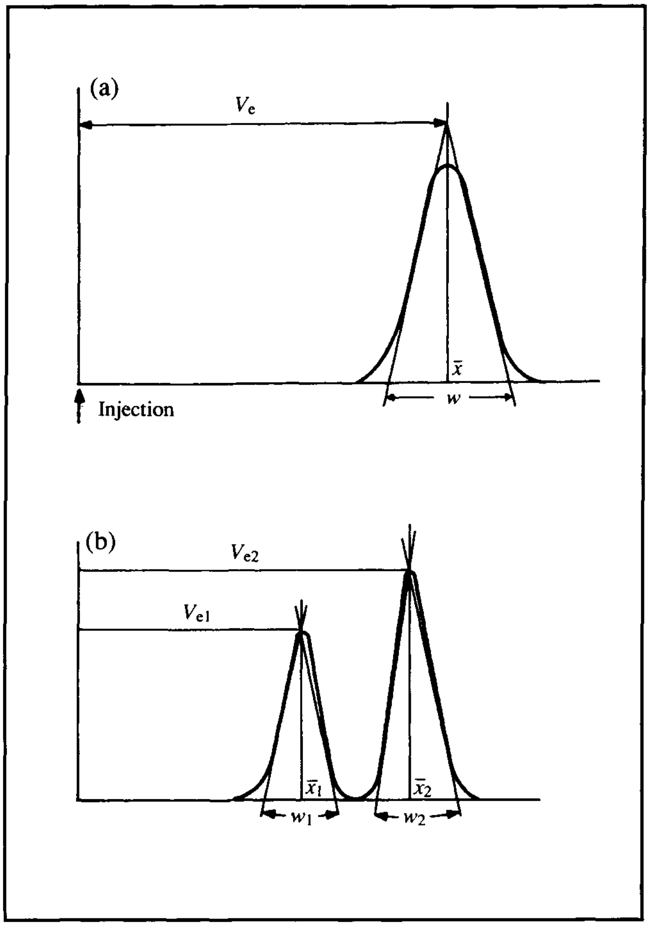
phase is continuously increasing, equilibrium at X remains always out of reach and the stationary-phase concentration lags behind equilibrium values. As a consequence, a higher concentration of solute remains in the liquid than if equilibrium were established, and the front edge of the solute pulse effectively moves ahead faster than the remainder of the pulse. As the peak of the solute pulse passes X, the mobile-phase concentration starts to decrease with time. In response, the solid phase must divest itself of solute to reach equilibrium with the lower liquid-phase concentrations. Again, because of delays due to mass transfer, equilibrium cannot be established with the continuously changing liquid concentration. As the solute pulse passes and the liquid concentration at X falls to zero, the solid phase still contains solute molecules that continue to be released into the liquid. Consequently, the rear of the solute pulse is effectively stretched out until the stationary phase reaches equilibrium with the liquid.

In general, conditions which improve mass transfer will increase the rate at which equilibrium is achieved between the phases and minimise zone spreading. For example, increasing the particle surface area per unit volume facilitates mass transfer and reduces non-equilibrium effects; surface area is usually increased by using smaller particles. On the other hand, increasing the liquid flow rate will exacerbate non-equilibrium effects as the rate of adsorption fails to keep up with concentration changes in the mobile phase. Viscous solutions give rise to considerable zone broadening as a result of slower mass-transfer rates; zone broadening is also more pronounced if the solute molecules are large. Changes in temperature can affect zone broadening in several ways. Because viscosity is reduced at elevated temperatures, heating the column often decreases zone spreading. However, rates of axial diffusion increase at higher temperatures so that the overall effect depends on the system and temperature range tested.

10.7.3 Theoretical Plates in Chromatography

The concept of theoretical plates is often used to analyse zone broadening in chromatography. The idea is essentially the same as that described in Section 10.4 for an ideal equilibrium stage. The chromatography column is considered to be made up of a number of segments or plates of height H; the magnitude of H

Figure 10.18 Parameters for calculation of: (a) number of theoretical plates, and (b) resolution.



is of the same order as the diameter of the resin particles. Within each segment equilibrium is supposed to exist.

As in adsorption operations, equilibrium is not often achieved in chromatography so that the theoretical-plate concept does not accurately reflect conditions in the column. Nevertheless the idea of theoretical plates is applied extensively, mainly because it provides a parameter, the plate height H, which can be used to characterise zone spreading. Use of the plate height, which is also known as the *height equivalent to a theoretical plate* (*HETP*), is acceptable practice in chromatography design even though it is based on a poor model of column operation. HETP is a measure of zone broadening; in general, the lower the HETP value the narrower is the solute peak.

HETP depends on various processes which occur during elution of a chromatography sample. A popular and simple expression for HETP takes the form:

$$H = \frac{A}{u} + Bu + C \tag{10.44}$$

where H is plate height, u is linear liquid velocity, and A, B and C are experimentally-determined kinetic constants. A, B and C include the effects of liquid–solid mass transfer, forward and backward axial dispersion, and non-ideal distribution of liquid around the packing. As outlined in Section 10.6.4, overall rates of solute adsorption and desorption in chromatography depend mainly on mass-transfer steps. Values of A, B and C are reduced by improving mass transfer between liquid and solid phases, resulting in a decrease in HETP and better column performance. Eq. (10.44) and other HETP models are discussed further in other references [16, 17].

HETP for a particular component is related to the elution volume and width of the solute peak as it appears on the chromatogram. If, as shown in Figure 10.18(a), the pulse has the standard symmetrical form of a normal distribution around a mean value $\bar{x}$, the *number of theoretical plates* can be calculated as follows:

$$N = 16\left(\frac{V_e}{w}\right)^2 \tag{10.45}$$

where N is number of theoretical plates, V_e is the distance on the chromatogram corresponding to the elution volume of the solute, and w is the base line width of the peak between lines drawn tangent to the inflection points of the curve. Eq. (10.45) applies if the sample is introduced into the column as a narrow pulse. Number of theoretical plates is related to HETP as follows:

$$N = \frac{L}{H} \tag{10.46}$$

where L is the length of the column. For a given column, the greater the number of theoretical plates the greater is the number of ideal equilibrium stages in the system and the more efficient is the separation. Values of H and N vary for a particular column depending on the component being separated.

10.7.4 Resolution

Resolution is a measure of zone overlap in chromatography and an indicator of column efficiency. For separation of two components, resolution is given by the following equation:

$$R_N = \frac{2(V_{e2} - V_{e1})}{(w_1 + w_2)} \tag{10.47}$$

where R_N is resolution, V_{e1} and V_{e2} are distances on the chromatogram corresponding to elution volumes for components 1 and 2, and w_1 and w_2 the baseline widths of the chromatogram peaks as shown in Figure 10.18(b). Column resolution is a dimensionless quantity; the greater the value of R_N the more separated are the two solute peaks. An R_N value of 1.5 corresponds to a baseline resolution of 99.8% or virtual complete separation; when $R_N = 1.0$ the two peaks overlap by about 2% of the total peak area.

Column resolution can be expressed in terms of HETP. Assuming w_1 and w_2 are approximately equal, Eq. (10.47) becomes:

$$R_N = \frac{(V_{e2} - V_{e1})}{w_2}. \tag{10.48}$$

Substituting for w_2 from Eq. (10.45):

$$R_N = \frac{1}{4}\sqrt{N}\left(\frac{V_{e2} - V_{e1}}{V_{e2}}\right). \tag{10.49}$$

The term

$$\frac{(V_{e2} - V_{e1})}{V_{e2}}$$

can be expressed in terms of k_2 and δ from Eqs (10.37) and (10.38), so that Eq. (10.49) becomes:

$$R_N = \frac{1}{4}\sqrt{N}\left(\frac{\delta - 1}{\delta}\right)\left(\frac{k_2}{k_2 + 1}\right). \tag{10.50}$$

Using the expression for N from Eq. (10.46), the equation for column resolution is:

$$R_N = \frac{1}{4}\sqrt{\frac{L}{H}}\left(\frac{\delta - 1}{\delta}\right)\left(\frac{k_2}{k_2 + 1}\right). \tag{10.51}$$

As is apparent from Eq. (10.51), peak resolution increases as a function of $\sqrt{L}$, where L is the column length. Resolution also increases with decreasing HETP; therefore, any enhancement of mass-transfer conditions reducing H will improve resolution. Derivation of Eq. (10.51) involves Eq. (10.45), which applies to chromatography systems where a relatively small quantity of sample is injected rapidly. Resolution is sensitive to increases in sample size; as the amount of sample increases, resolution declines. In laboratory analytical work it is common to use extremely small sample volumes, of the order of microlitres. However, depending on the type of chromatography used for production-scale purification, sample volumes 5–20% of the column volume and higher are used. Because resolution under these conditions is relatively poor, if the solute peak is collected for isolation of product the central portion of the peak is retained while the leading and trailing edges are recycled back to the feed.

10.7.5 Scaling-Up Chromatography

The aim in scale-up of chromatography is to retain the resolution and solute recovery achieved using a small-scale column while increasing the throughput of material. Strategies for scale up must take into account the dominance of mass-transfer effects in chromatography separations.

The easiest approach to scale-up is to simply increase the flow rate through the column. This gives unsatisfactory results; raising the liquid velocity increases zone spreading and produces high pressures in the column which compress the stationary phase and cause pumping difficulties. The pressure-drop problem can be alleviated by increasing the particle size; however this hinders the overall mass-transfer process and so decreases resolution. Increasing the column length can help regain any resolution lost by increasing the flow rate or particle diameter; however increasing L also has a strong effect in raising the pressure drop through the column.

The solution to scale-up is to keep the same column length, linear flow velocity and particle size as in the small column, but increase the column diameter. The larger capacity of the column is therefore due solely to its greater cross-sectional area. Sample volume and volumetric flow rate are increased in proportion to column volume. In this way, all the important parameters affecting the packing matrix, liquid flow, mass transfer and equilibrium conditions are kept constant; similar column performance can therefore be expected. Because liquid distribution in large-diameter packed columns tends to be poor, care must be taken to ensure liquid is fed evenly over the entire column cross-section.

In practice, variations in column properties and efficiency do occur with scale-up. As an example, compressible solids such as those used in gel chromatography get better support from the column wall in small columns than in large columns; as a result, lower linear flow rates must be used in large-scale systems. An advantage of using gels of high mechanical strength in laboratory systems is that they allow more direct scale-up to commercial operation. The elasticity and compressibility of gels used for fractionation of high-molecular-weight proteins preclude use of long columns in large-scale processes; bed heights in these systems are normally restricted to 0.6–1.0 m.

10.8 Summary of Chapter 10

At the end of Chapter 10 you should:

(i) know what is a *unit operation*;
(ii) be able to describe generally the steps of *downstream processing*;
(iii) understand the theory and practice of *filtration*;
(iv) understand the principles of *centrifugation*, including scale-up considerations;
(v) be familiar with methods used for *cell disruption*;
(vi) understand the concept of an *ideal stage*;
(vii) be able to analyse *aqueous two-phase extractions* in terms of the equilibrium *partition coefficient*, *product yield* and *concentration factor*;
(viii) understand the principles of *adsorption* operations and design of *fixed-bed adsorbers*;
(ix) know the different types of *chromatography* used for separation of biomolecules; and
(x) understand the concepts of *differential migration*, *zone spreading* and *resolution* in chromatography, know what operating conditions enhance chromatography performance, and be able to describe *scale-up procedures* for chromatography columns.

Problems

10.1 Bacterial filtration

A suspension of *Bacillus subtilis* cells is filtered under constant pressure for recovery of protease. A pilot-scale filter is used to measure filtration properties. The filter area is 0.25 m^2, the pressure drop is 360 mmHg, and the filtrate viscosity is 4.0 cP. The cell suspension deposits 22 g cake per litre of filtrate. The following data are measured.

Time (min)	2	3	6	10	15	20
Filtrate volume (l)	10.8	12.1	18.0	21.8	28.4	32.0

(a) Determine the specific cake resistance and filter medium resistance.
(b) What size filter is required to process 4000 l cell suspension in 30 min at a pressure drop of 360 mmHg?

10.2 Filtration of mycelial suspensions

Pelleted and filamentous forms of *Streptomyces griseus* are filtered separately using a small laboratory filter of area 1.8 cm^2. The mass of wet solids per ml of filtrate is 0.25 g ml^{-1} for the pelleted cells and 0.1 g ml^{-1} for the filamentous culture. Viscosity of the filtrate is 1.4 cP. Five filtration experiments at different pressures are carried out with each suspension. The results are as follows:

	Pressure drop (mmHg)				
	100	250	350	550	750
Filtrate volume (ml) *for pelleted suspension*	*Time* (s)				
10	22	12	9	7	5
15	52	26	20	14	12
20	90	49	36	28	22
25	144	75	60	43	34
30	200	110	88	63	51
35	285	149	119	84	70
40	368	193	154	110	90
45	452	240	195	140	113
50	–	301	238	175	141

	Pressure drop (mmHg)				
	100	250	350	550	750
Filtrate volume (ml) *for filamentous suspension*	*Time* (s)				
10	36	22	17	13	11
15	82	47	40	31	25
20	144	85	71	53	46
25	226	132	111	85	70
30	327	194	157	121	100
35	447	262	215	166	139
40	–	341	282	222	180
45	–	434	353	277	229
50	–	–	442	338	283

(a) Evaluate the specific cake resistance as a function of pressure for each culture.
(b) Determine the compressibility for each culture.
(c) A filter press with area 15 m^2 is used to process 20 m^3 filamentous *S. griseus* culture. If the filtration must be completed in one hour, what pressure drop is required?

10.3 Rotary-drum vacuum filtration

Continuous rotary vacuum filtration can be analysed by considering each revolution of the drum as a stationary batch filtration. Per revolution, each cm^2 of filter cloth is used to form cake only for the period of time it spends submerged in the liquid reservoir. A rotary-drum vacuum filter with drum diameter 1.5 m and filter width 1.2 m is used to filter starch from an aqueous slurry. The pressure drop is kept constant at 4.5 psi; the filter operates with 30% of the filter cloth submerged. Resistance due to the filter medium is negligible. Laboratory tests with a 5 cm^2 filter have shown that 500 ml slurry can be filtered in 23.5 min at a pressure drop of 12 psi; the starch cake was also found to be compressible with $s = 0.57$. Use the following steps to determine the drum speed required to produce 20 m^3 filtered liquid per hour.

(a) Evaluate $\mu_f \alpha' c$ from the laboratory test data.
(b) If N is the drum speed in revolutions per hour, what is the cycle time?
(c) From (b), for what period of time per revolution is each cm^2 of filter cloth used for cake formation?
(d) What volume of filtrate must be filtered per revolution to achieve the desired rate of 20 m^3 per hour?
(e) Apply Eq. (10.11) to a single revolution of the drum to evaluate N.
(f) The liquid level is raised so that the fraction of submerged filter area increases from 30% to 50%. What drum speed is required under these conditions?

10.4 Centrifugation of yeast

Yeast cells are to be separated from a fermentation broth. Assume that the cells are spherical with diameter 5 μm and density 1.06 g cm^{-3}. The viscosity of the culture broth is 1.36 $\times 10^{-3}$ N s m^{-2}. At the temperature of separation, the density of the suspending fluid is 0.997 g cm^{-3}. 500 litres broth must be treated every hour.

(a) Specify Σ for a suitably-sized disc-stack centrifuge.
(b) The small size and low density of microbial cells are disadvantages in centrifugation. If instead of yeast, quartz particles of diameter 0.1 mm and specific gravity 2.0 are separated from the culture liquid, by how much is Σ reduced?

10.5 Centrifugation of food particles

Small food particles with diameter 10^{-2} mm and density 1.03 g cm^{-3} are suspended in liquid of density 1.00 g cm^{-3}. The viscosity of the liquid is 1.25 mPa s. A tubular-bowl centrifuge of length 70 cm and radius 11.5 cm is used to separate the particles. If the centrifuge is operated at 10 000 rpm, estimate the feed flow rate at which the food particles are just removed from the suspension.

10.6 Scale-up of disc-stack centrifuge

A pilot-scale disc-stack centrifuge is tested for recovery of bacteria. The centrifuge contains 25 discs with inner and outer diameters 2 cm and 10 cm, respectively. The half-cone angle is 35°. When operated at a speed of 3000 rpm with a feed rate of 3.5 litre min^{-1}, 70% of the cells are recovered. If a bigger centrifuge is to be used for industrial treatment of 80 litres min^{-1}, what operating speed is required to achieve the same sedimentation performance if the larger centrifuge contains 55 discs with outer diameter 15 cm, inner diameter 4.7 cm, and half-cone angle 45°?

10.7 Centrifugation of yeast and cell debris

A tubular-bowl centrifuge is used to concentrate a suspension of genetically-engineered yeast containing a new recombinant protein. At a speed of 12 000 rpm, the centrifuge treats 3 l broth min^{-1} with satisfactory results. It is proposed to use the same centrifuge to separate cell debris from homogenate produced by mechanical disruption of the yeast. If the average size of the debris is one-third that of the yeast and the viscosity of the homogenate is five times greater than the cell suspension, what flow rate can be handled if the centrifuge is operated at the same speed?

10.8 Cell disruption

Micrococcus bacteria are disrupted at 5°C in a Manton-Gaulin homogeniser operated at pressures between 200 and 550 kg_f cm^{-2}. Data for protein release as a function of number of passes through the homogeniser are as follows:

	Pressure drop ($kg_f\ cm^{-2}$)				
	200	300	400	500	550
Number of passes	*% protein release*				
1	5.0	13.5	23.3	36.0	42.0
2	9.5	23.5	40.0	58.5	66.0
3	14.0	33.5	52.5	75.0	83.7
4	18.0	43.0	66.6	82.5	88.5
5	22.0	47.5	73.0	88.5	94.5
6	26.0	55.0	79.5	91.3	–

(a) How many passes are required to achieve 80% protein release at an operating pressure of 460 $kg_f\ cm^{-2}$?
(b) Estimate the pressure required to deliver 70% protein recovery in only two passes?

10.9 Enzyme purification using two-phase aqueous partitioning

Leucine dehydrogenase is recovered from a homogenate of disrupted *Bacillus cereus* cells using an aqueous two-phase polyethylene glycol–salt system. 150 litres of homogenate initially containing 3.2 units enzyme ml^{-1} are processed; a polyethylene glycol–salt mixture is added and two phases form. The enzyme partition coefficient is 3.5.

(a) What volume ratio of upper and lower phases must be chosen to achieve 80% recovery of enzyme in a single extraction step?
(b) If the volume of the lower phase is 100 litres, what is the concentration factor for 80% recovery?

10.10 Recovery of viral particles

Cells of the fall armyworm *Spodoptera frugiperda* are cultured in a fermenter to produce viral particles for insecticide. Viral particles are released into the culture broth after lysis of the host cells. The initial culture volume is 5 litres. An aqueous two-phase polymer solution of volume 2 litres is added to this liquid; the volume of the bottom phase is 1 litre. The virus partition coefficient is 10^{-2}.

(a) What is the yield of virus at equilibrium?
(b) Write a mass balance for viral particles in terms of concentrations and volumes of the phases, equating the amounts of virus present before and after addition of polymer solution.
(c) Derive an equation for the concentration factor in terms of liquid volumes and the partition coefficient only.
(d) Calculate the concentration factor for the viral extraction.

10.11 Gel chromatography scale-up

Gel chromatography is to be used for commercial-scale purification of a proteinaceous diphtheria toxoid from *Corynebacterium diphtheriae* supernatant. In the laboratory, a small column of 1.5 cm inner diameter and height 0.4 m is packed with 10 g dry Sephadex gel; the void volume is measured as 23 ml. A sample containing the toxoid and impurities is injected into the column. At a liquid flow rate of 14 ml min^{-1}, the elution volume for the toxoid is 29 ml; the elution volume for the principal impurity is 45 ml. A column of height 0.6 m and diameter 0.5 m is available for large-scale gel chromatography. The same type of packing is used; the void fraction and ratio of pore volume to total bed volume remain the same as in the bench-scale column. The liquid flow rate in the large column is scaled up in proportion to the column cross-sectional area; the flow patterns in both columns can be assumed identical. The water regain value for the packing is given by the manufacturer as 0.0035 $m^3\ kg^{-1}$ dry gel.

(a) Which is the larger molecule, the diphtheria toxoid or the principal impurity?
(b) Determine the partition coefficients for the toxoid and impurity.
(c) Estimate the elution volumes in the commercial-scale column.
(d) What is the volumetric flow rate in the large column?
(e) Estimate the retention time of toxoid in the large column?

10.12 Protein separation using chromatography

Human insulin A and B from recombinant *Escherichia coli* are separated using pilot-scale affinity chromatography. Laboratory studies have shown that capacity factors for the proteins are 0.85 for the A-chain and 1.05 for the B-chain. The dependence of HETP on liquid velocity satisfies the following type of equation:

$$H = \frac{A}{u} + B u + C$$

where u is linear liquid velocity and A, B and C are constants. Values of A, B and C for the insulin system were found to be $2 \times 10^{-9}\ m^2\ s^{-1}$, 1.5 s and 5.7×10^{-5} m, respectively. Two columns with inner diameter 25 cm are available for the process; one is 1.0 m high, the other is 0.7 m high.

(a) Plot the relationship between H and u from $u = 0.1 \times 10^{-4}$ to $u = 2 \times 10^{-4}$.

(b) What is the minimum HETP? At what liquid velocity is the minimum HETP obtained?
(c) If the larger column is used with a liquid flow rate of 0.31 litres min^{-1}, will the two insulin chains be completely separated?
(d) If the smaller column is used, what is the maximum liquid flow rate that will give complete separation?

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Suggestions for Further Reading

There is an extensive literature on downstream processing of biomolecules. The following is a small selection.

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Chromatography (see also refs 1, 16 and 17)

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Part 4
Reactions and Reactors

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II

Homogeneous Reactions

The heart of a typical bioprocess is the reactor or fermenter. Flanked by unit operations which carry out physical changes for medium preparation and recovery of products, the reactor is where the major chemical and biochemical transformations occur. In many bioprocesses, characteristics of the reaction determine to a large extent the economic feasibility of the project.

Of most interest in biological systems are *catalytic* reactions. By definition, a catalyst is a substance which affects the rate of reaction without altering the reaction equilibrium or undergoing permanent change itself. Enzymes, enzyme complexes, cell organelles and whole cells perform catalytic roles; the latter may be viable or non-viable, growing or non-growing. Biocatalysts can be of microbial, plant or animal origin. Cell growth is an *autocatalytic reaction*: this means that the catalyst is a product of the reaction. The performance of catalytic reactions is characterised by variables such as the reaction rate and yield of product from substrate. These parameters must be taken into account when designing and operating reactors.

In engineering analysis of catalytic reactions, a distinction is made between *homogeneous* and *heterogeneous* reactions. A reaction is homogeneous if the temperature and all concentrations in the system are uniform. Most fermentations and enzyme reactions carried out in mixed vessels fall into this category. In contrast, heterogeneous reactions take place in the presence of concentration or temperature gradients. Analysis of heterogeneous reactions requires application of mass-transfer principles in conjunction with reaction theory. Heterogeneous reactions are treated in Chapter 12.

This chapter covers the basic aspects of reaction theory which allow us to quantify the extent and speed of homogeneous reactions and to identify important factors affecting reaction rate.

11.1 Basic Reaction Theory

Reaction theory has two fundamental parts: *reaction thermodynamics* and *reaction kinetics*. Reaction thermodynamics is concerned with *how far* the reaction can proceed; no matter how fast a reaction is, it cannot continue beyond the point of chemical equilibrium. On the other hand, reaction kinetics is concerned with the *rate* at which reactions proceed.

11.1.1 Reaction Thermodynamics

Consider a reversible reaction represented by the following equation:

$$\mathrm{A} + b\,\mathrm{B} \rightleftharpoons y\,\mathrm{Y} + z\,\mathrm{Z}. \tag{11.1}$$

A, B, Y and Z are chemical species; *b*, *y* and *z* are stoichiometric coefficients. If the components are left in a closed system for an infinite period of time, the reaction proceeds until *thermodynamic equilibrium* is reached. At equilibrium there is no net driving force for further change; the reaction has reached the limit of its capacity for chemical transformation in a closed system. Composition of the equilibrium mixture is determined exclusively by the thermodynamic properties of the reactants and products; it is independent of the way the reaction is executed. Equilibrium concentrations are related by the *equilibrium constant*, *K*. For the reaction of Eq. (11.1):

$$K = \frac{C_{\mathrm{Ye}}{}^{y} C_{\mathrm{Ze}}{}^{z}}{C_{\mathrm{Ae}}\, C_{\mathrm{Be}}{}^{b}} \tag{11.2}$$

where C_{Ae}, C_{Be}, C_{Ye} and C_{Ze} are equilibrium concentrations of A, B, Y and Z, respectively. The value of K depends on temperature as follows:

$$\ln K = \frac{-\Delta G^{\circ}_{\mathrm{rxn}}}{RT} \tag{11.3}$$

where $\Delta G^{\circ}_{\mathrm{rxn}}$ is the *change in standard free energy* per mole of A reacted, R is the ideal gas constant and T is absolute temperature. Values of R are listed in Table 2.5 (p. 20). The superscript ° in $\Delta G^{\circ}_{\mathrm{rxn}}$ indicates standard conditions. Usually, the standard condition for a substance is its most stable form at 1 atm pressure and 25°C; however, for biochemical reactions occurring in solution, other standard conditions may be used [1]. ΔG°_{rxn} is equal to the difference in *standard free energy of formation*, G°, between products and reactants:

$$\Delta G^\circ_{rxn} = y\,G^\circ_Y + z\,G^\circ_Z - G^\circ_A - b\,G^\circ_B. \tag{11.4}$$

Standard free energies of formation are available in handbooks such as those listed in Section 2.6.

Free energy G is related to enthalpy H, entropy S and absolute temperature T as follows:

$$\Delta G = \Delta H - T\Delta S. \tag{11.5}$$

Therefore, from Eq. (11.3):

$$\ln K = \frac{-\Delta H^\circ_{rxn}}{RT} + \frac{\Delta S^\circ_{rxn}}{R}. \tag{11.6}$$

Thus, for exothermic reactions with negative ΔH°_{rxn}, K decreases with increasing temperature. For endothermic reactions and positive ΔH°_{rxn}, K increases with temperature.

Example 11.1 Effect of temperature on glucose isomerisation

Glucose isomerase is used extensively in the USA for production of high-fructose syrup. The reaction is:

$$\text{glucose} \rightleftharpoons \text{fructose}.$$

ΔH°_{rxn} for this reaction is 5.73 kJ gmol^{-1}; ΔS°_{rxn} is 0.0176 kJ gmol^{-1} K^{-1}.

(a) Calculate the equilibrium constants at 50°C and 75°C.
(b) A company aims to develop a sweeter mixture of sugars, i.e. one with a higher concentration of fructose. Considering equilibrium only, would it be more desirable to operate the reaction at 50°C or 75°C?

Solution:
(a) Convert temperatures to degrees Kelvin (K) using the formula of Eq. (2.24):

$$T = 50°\text{C} = 323.15\text{ K}$$
$$T = 75°\text{C} = 348.15\text{ K}.$$

From Table 2.5, $R = 8.3144$ J gmol^{-1} K^{-1} = 8.3144×10^{-3} kJ gmol^{-1} K^{-1}. Using Eq. (11.6)

$$\ln K\,(50°\text{C}) = \frac{-5.73\text{ kJ gmol}^{-1}}{(8.3144 \times 10^{-3}\text{ kJ gmol}^{-1}\text{K}^{-1})\,(323.15\text{K})} + \frac{0.0176\text{ kJ gmol}^{-1}\text{ K}^{-1}}{8.3144 \times 10^{-3}\text{ kJ gmol}^{-1}\text{K}^{-1}}$$

$$K\,(50°\text{C}) = 0.98.$$

Similarly for $T = 75$°C:

$$\ln K\,(75°\text{C}) = \frac{-5.73\text{ kJ gmol}^{-1}}{(8.3144 \times 10^{-3}\text{ kJ gmol}^{-1}\text{K}^{-1})\,(348.15\text{K})} + \frac{0.0176\text{ kJ gmol}^{-1}\text{ K}^{-1}}{8.3144 \times 10^{-3}\text{ kJ gmol}^{-1}\text{K}^{-1}}$$

$$K\,(75°\text{C}) = 1.15.$$

(b) As K increases, the fraction of fructose in the equilibrium mixture increases. Therefore, from an equilibrium point of view, it is more desirable to operate the reactor at 75°C. However, other factors such as enzyme deactivation at high temperatures should also be considered.